

**GEOSPATIAL APPROACH TO OPTIMIZE INCIDENT RESPONSE SYSTEM
WITH DATA ANALYTICS INTEGRATION**

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CHAPTER I

INTRODUCTION

The rapid advancement of geospatial and data analytics technology has transformed decision-making across many sectors, including disaster risk reduction and emergency management. However, disaster risk remains a serious concern in the Philippines, which ranks first in disaster risk among 193 countries, with a WorldRiskIndex score of 46.56 in 2025 (Bündnis Entwicklung Hilft & Ruhr University Bochum, 2025). Timely, location-based information is critical in an emergency, and Web-GIS platforms are increasingly used for decision support, real-time monitoring, early warning, and coordination among stakeholders during hazard events (Daud et al., 2024).

Under Republic Act No. 10121, the NDRRMC is the body that sets policy, coordinates, supervises, monitors, and evaluates disaster risk reduction and management in the country (Republic Act No. 10121, 2010). Even with this law, many municipalities still depend on phone calls, text messages, and two-way radios, which leads to delayed reporting and no real-time location tracking (Consumo et al., 2026). The absence of a centralized geospatial reporting system also makes it difficult for local offices to see where incidents are concentrated and where response is slowest. Studies indicate that geospatial analysis can help by showing incident hotspots and areas with slow response, which supports better use of staff and response teams (Aporto et al., 2026).



The Negros Island Region was created in 2024 and groups Negros Occidental, Negros Oriental, and Siquijor, together with their cities, municipalities, and barangays, into one region (Republic Act No. 12000, 2024). Before this law, Negros Occidental was under Region 6 while Negros Oriental and Siquijor were under Region 7 (Manila Bulletin, 2024). Within this newly formed region, the Talisay City Disaster Risk Reduction and Management Office (DRRMO) and the Bureau of Fire Protection (BFP) Station do not have an existing system for real-time incident recording and coordination, hindering trend analysis and timely response. The proposed Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS) integrates geospatial incident mapping, kernel density hotspot detection, and Prophet-based forecasting to help DRRMO and BFP personnel respond to NDRRMC-catered incidents in the region more efficiently.

In Talisay City, the DRRMO and BFP Station still relied on manual processes such as phone calls, radio, and separate logbooks, which delays reporting, allows errors in location details, and limits the real-time view of ongoing incidents. Geospatial analysis can effectively map incident distribution and highlight high-risk zones, underscoring the need for a digital, map-based response system that supports faster and more accurate decisions (Han & Liu, 2025). Researchers in nearby Oton, Iloilo similarly recommend digitizing and geolocating incident records and using real-time GIS dashboards for more efficient and fair emergency response planning (Aporto et al., 2026).



The purpose of this study was to develop the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration for reporting, verifying, dispatching, and monitoring NDRRMC-catered incidents in the Negros Island Region. The study aimed to improve response efficiency by automating incident intake, verification, and nearest-responder dispatch, which cuts down the time needed to report, confirm, and act on an incident. This helps DRRMO and BFP personnel quickly identify high-risk areas and respond faster, strengthening disaster risk reduction and public safety in the region.

1.1 Background of the Study

The City Disaster Risk Reduction and Management Office (CDRRMO) and the Bureau of Fire Protection (BFP) Station serve as the primary agencies responsible for coordinating disaster preparedness, emergency response, rescue operations, and recovery efforts within the city (Republic Act No. 10121, 2010). As disasters such as floods, landslides, fires, and road accidents continue to affect communities, and with the city's growing population, both offices play a vital role in receiving reports, verifying incidents, dispatching response units, and monitoring cases across their jurisdiction. The offices are staffed by trained personnel who are tasked with recording incident reports, coordinating with response units, and relaying information to city and regional authorities.

Currently, the DRRMO and BFP Station rely on fragmented communication channels, including phone calls, text messages, and two-way radios, for incident reporting (Lagmay et al., 2021). When a resident calls or



radios in a report, personnel manually note the caller's description of the incident type and location in a logbook or dispatch sheet, and each office keeps its own records separately. This process requires responders to manually verify locations and consolidate reports, resulting in delays, incomplete records, and difficulties in monitoring emergency situations (UNDRR, 2023). Another challenge is the lack of a centralized geospatial system for visualizing incident locations and identifying the appropriate response jurisdiction (Esri, 2024). Without spatial analytics, DRRMO and BFP personnel have limited capability to identify high-risk areas, prioritize incidents, and support evidence-based planning for resource allocation (Lagmay et al., 2021).

In response to these challenges, the researchers proposed the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration to address the DRRMO and BFP Station's challenges in managing NDRRMC-catered incidents. The system was designed to let citizens submit geotagged incident reports, verify reports before dispatch, identify and dispatch the nearest available response unit, and generate geospatial hotspot maps and Prophet-based trend forecasts to identify high-risk areas. Through this solution, DRRMO and BFP personnel will be able to access and analyze incident data more efficiently, enabling faster response, more accurate reporting, and more strategic resource allocation across the Negros Island Region.



1.2 Objectives of the Study

The general objective of this study was to design and develop the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration that enables efficient reporting, coordinated response, and monitoring of NDRRMC-catered incidents in the Negros Island Region.

Specifically, this study aimed to:

1. Design and develop the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration with the following features:
 - 1.1. incident report recording and management;
 - 1.2. incident response;
 - 1.3. geospatial mapping interface with identification of the city and congressional district of incidents;
 - 1.4. monitoring and analysis; and
 - 1.5. report generation.
2. Test the functionality of the developed system;
3. Evaluate the quality of the system using the ISO/IEC 25010:2011 criteria and ISO 9241-11:2018 measures.
4. Develop a User's Guide for the developed system.

1.3 Significance of the Study

This study is significant as it develops a Geospatial Approach to Optimize Incident Response System with Data Analytics Integration that enhances the



ability of local government units to report, monitor, and respond to disaster incidents through a centralized mapping platform. By integrating geotagged reporting, geofencing, spatial analytics, and real-time status tracking, the system improves the accuracy, accessibility, and timeliness of disaster information. Its development contributes to more efficient emergency response, better identification of high-risk areas, and informed disaster risk reduction planning at the local government level.

The results of this study will be significant to the following:

City Disaster Risk Reduction and Management Office (CDRRMO). The developed system will help disaster personnel record, verify, and monitor incident reports more efficiently through a centralized platform. It will improve coordination among response units, reduce delays in emergency response, and support better decision-making through organized and location-based incident information.

Local Government Units (LGUs). The system will provide LGUs with reliable geospatial data that can support disaster preparedness, resource allocation, and emergency planning. It will enable local officials to identify high-risk areas and prioritize communities that require immediate intervention during disaster situations.

Citizens and Communities. The study will benefit citizens by providing a convenient platform for submitting geotagged disaster reports and tracking the status of their reported incidents. It will strengthen communication between the



community and emergency responders while promoting faster reporting and improved public safety.

Future Researchers. This study will serve as a valuable reference for future researchers interested in disaster management, geographic information systems, and data analytics. It may provide foundational knowledge for developing more advanced disaster response and decision-support systems.

1.4 Scope and Delimitation of the Study

This study focuses on the design, development, testing, and evaluation of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS) for incidents catered by the NDRRMC within the Negros Island Region (NIR). The system has three user roles: Citizen, who submits incident reports; Response Unit, consisting of BFP and DRRMO personnel who respond to verified incidents; and Administrator, who manages the system, verifies and approves incident reports, and user permissions. Citizen-submitted reports undergo verification before being dispatched to an appropriate response unit. Key features of the system include incident reporting and management, response dispatch, geospatial mapping, geofencing, incident monitoring, Kernel Density Estimation (KDE), and report generation. The system is developed using React, Vite, and Tailwind CSS for the frontend; Node.js and Express for the backend; PostgreSQL for data storage; Leaflet.js for geospatial mapping; Chart.js for data visualization; and Python for KDE analysis. The



system is accessible through a modern web browser and requires a stable internet connection.

While the system provides features for incident reporting, response coordination, monitoring, and spatial analysis, it has several limitations. The system is limited to incidents catered by the NDRRMC within the NIR and does not support coordination or incident sharing between different local government units or jurisdictions. It is intended only for the DRRMO and BFP as response agencies and does not include the police or other government agencies as system users. The system does not predict natural hazards or issue hazard warnings, as these functions remain outside the scope of the study. The KDE feature is limited to identifying spatial concentrations or hotspots of reported incidents and is not intended to predict individual incidents or future disasters. In addition, real-time reporting, mapping, dispatch, and status monitoring require internet connectivity, and system performance may be affected by poor or unstable network connections.

1.5 Definition of Terms

The following key terms have been operationally defined based on their specific usage within the context of this study. These definitions aim to provide clarity and ensure a consistent understanding of the concepts central to the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS), particularly those related to incident reporting, emergency



response, geospatial mapping, spatial analysis, and disaster incident management.

Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS). In this study, this term refers to the web-based system developed to record, manage, map, dispatch, monitor, and analyze disaster incidents reported within the Negros Island Region.

Incident Report. In this study, this term refers to the information submitted by a citizen regarding a disaster or emergency incident, including its location, description, and photo evidence.

Citizen. In this study, this term refers to a registered system user who submits incident reports and monitors the status of their submitted reports.

Response Unit. In this study, this term refers to personnel from the Talisay City Disaster Risk Reduction and Management Office and Bureau of Fire Protection who are responsible for responding to verified incidents assigned to them through the system.

Administrator. In this study, this term refers to an authorized user responsible for managing system operations, accounts, permissions, and other administrative functions within GAOIRS.

Incident Verification. In this study, this term refers to the process of reviewing and validating a citizen-submitted incident report before it is dispatched to an appropriate response unit.



Incident Response. In this study, this term refers to the actions taken by designated response units after a reported incident has been verified and assigned for response.

Geographic Information System (GIS). In this study, this term refers to the mapping component of GAOIRS used to display and manage incident locations and other location-based information.

Geofencing. In this study, this term refers to the use of predefined geographic boundaries in the system to identify the barangay, city, and congressional district where a reported incident is located.

Incident Mapping. In this study, this term refers to the visualization of reported incidents on an interactive map using location data and color-coded markers.

Kernel Density Estimation (KDE). In this study, this term refers to the spatial analysis method used to identify areas with higher concentrations of reported incidents and visualize incident hotspots.

Incident Monitoring. In this study, this term refers to the process of viewing and tracking reported incidents and their response status through the system.

Response Dispatch. In this study, this term refers to the assignment of verified incidents to an appropriate response unit based on the system's incident and location information.



Incident Status Tracking. In this study, this term refers to the monitoring of the progress of a reported incident from verification and response through its subsequent status updates.

Incident Report Generation. In this study, this term refers to the system's capability to compile and export recorded incident information into reports for monitoring and documentation purposes.



CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents a brief review of related literature, studies, and the interrelated system as the basis of making and developing the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration.

2.1 Foreign Literature

Comprehensive Analysis of the Use of Web-GIS for Natural Hazard Management: A Systematic Review

This study reviewed how Web-GIS is used in natural hazard management. Following the PRISMA process, the researchers screened 1,775 articles published from 2014 to 2023 and analyzed 65 of them. The studies were grouped into eight subtopics, including decision support systems, real-time monitoring and early warning, visualization and user interface design, and multi-stakeholder collaboration. The review also noted that Web-GIS studies were fewer in low- and middle-income countries. These findings support the relevance of GAOIRS, which uses a map-based dashboard, real-time monitoring, and shared access for different user roles to manage incidents (Daud et al., 2024).

A Near-Real-Time Global Landslide Incident Reporting Tool Demonstrator Using Social Media and Artificial Intelligence

This study developed a near-real-time tool for reporting landslide incidents. The system continuously monitored social media, collected photographs, and used a trained image classification model to keep only the posts likely to show landslides. In a real-world deployment, it detected landslide



reports with a precision of 76%. The study showed that incident reports can be captured and filtered quickly instead of waiting for manual reporting. These findings support the relevance of GAOIRS, which relies on fast incident intake with photo evidence and includes a verification step before responders are dispatched, since incoming reports may be inaccurate (Pennington et al., 2022).

Designing Multimodal Interactive Dashboard of Disaster Management Systems

This study designed interactive dashboards that present text, image, audio, and video information about disasters in one place. The researchers first studied social media data to identify the requirements, then built the dashboard designs, and had experts evaluate them using heuristics. The evaluators gave positive feedback on the designs. The study highlighted the value of bringing different kinds of incident information into a single screen and of evaluating dashboard usability with experts. This relates to GAOIRS, whose monitoring dashboard also shows incident details, photo evidence, and status updates in one view (AlAbdulaali et al., 2022).

A Study on GIS-Based Spatial Analysis of Emergency Response for Disaster Management: Focusing on Seoul

This study analyzed the operation of fire emergency dispatch in Seoul, examined the impact of the “golden time” on dispatch, and identified factors and improvement measures for effective dispatch within prime time. The results showed that rescue dispatch activities differed depending on the nature of the incident. The number of calls was highest in summer (15,652), compared with spring (13,189), fall (13,128), and winter (14,545), and Saturday was the day of



the week with the most calls. The spatial analysis also showed that rescue incidents were concentrated in some areas. These findings support the hotspot map and time-based reports in GAOIRS, which help show where and when incidents cluster so that response units can be prepared (Park et al., 2024).

Population Access to Hospital Emergency Departments: The Spatial Analysis in Public Health Research

This study examined how spatial analysis can be used to evaluate access to hospital emergency departments (EDs) in Poland. The researchers compared the distribution and travel-time accessibility of EDs in 2011 and 2021 using GIS-based distance measurement across communes. The findings showed that reorganizing the ED network nationally improved overall spatial accessibility, but some areas, including cities that lost nearby facilities, saw accessibility decline. This reinforces that comparing spatial data across two points in time is a useful way to evaluate whether a service network is reaching the people who need it. This supports the practice in GAOIRS of using geospatial mapping and hotspot analysis to evaluate whether incident response coverage is reaching the areas that need it most (Kisiała et al., 2022).

GIS Based Approach of Surface Transport Network Analysis

This study used GIS-based network analysis to determine the best route for healthcare responders to reach an incident location in the Katol metropolitan area. The researchers compared an enhanced GIS route against the standard shortest-path route and found that the optimized route improved travel time by 22%. The study demonstrated that in a densely populated area, an enhanced GIS-based routing approach outperforms a simple shortest-path method when



arriving quickly at an incident matters most. This supports the value of GIS network analysis in GAOIRS for helping response units identify the fastest way to reach a verified incident, not just the shortest path (Gupta & Khode, 2025).

The Role of GIS in Disaster Response: Improving Efficiency and Overcoming Challenges

This review examined how GIS supports disaster response, drawing on studies covering risk assessment, real-time monitoring, evacuation planning, and post-disaster recovery. The researchers found that GIS's integration with technologies such as remote sensing, AI, and machine learning has expanded its ability to support faster and more accurate decisions, but that its use in disaster management still faces challenges related to data quality, system interoperability, and user capabilities. The review highlighted the need for stronger integration between GIS and other disaster management systems. This supports designing GAOIRS around simple role-based access, data verification, and testing with actual DRRMO and BFP users, rather than around its mapping features alone (Han & Liu, 2025).

Emergency Incident Detection from Crowdsourced Waze Data Using Bayesian Information Fusion

This study addressed the delay caused by the traditional reactive approach to emergency reporting, in which a person must call an emergency number, by using crowd-generated reports from the Waze platform to detect incidents. Because the reliability of report sources and the uncertainty about the time and place of reports are major challenges, the researchers built a Bayesian framework that models this uncertainty and integrates reports across space and



time. When tested with Waze data and official incident records from Nashville, Tennessee, the method outperformed strong baselines in F1-score and AUC. This supports the need in GAOIRS to check the reliability of citizen reports and to tag reports that refer to the same incident (Senarath et al., 2020).

Spatiotemporal Clustering and Analysis of Road Accident Hotspots by Exploiting GIS Technology and Kernel Density Estimation

This study applied kernel density estimation (KDE) together with spatial autocorrelation methods to identify road accident hotspots using real crash data from the United Kingdom. The researchers used GIS tools to detect clusters of accidents and to examine how parameter choices, such as bandwidth, affect which areas appear as hotspots. The findings showed that GIS-based KDE, when properly tuned, produces reliable and interpretable hotspot maps that can guide safety interventions. This supports the use of KDE for the heatmap feature in GAOIRS and shows why the bandwidth setting must be chosen carefully to produce a reliable hotspot map (Kazmi et al., 2022).

Forecasting the Human Cost of Disasters under Sustainable Development Goal: A Time Series Analysis Using Facebook Prophet Model

This study used the Facebook Prophet model to forecast disaster-related deaths from 2024 to 2030, using historical data from 2001 to 2023 across 168 countries. The researchers calculated annual mortality rates for each country and used Prophet to project future rates, finding an average global mortality rate of 13.38 deaths per million people with significant variation across countries. The study showed that the Prophet can turn historical disaster records into forward-looking estimates that support prioritization and planning. This is the



basis of the Prophet-based incident forecasting used in GAOIRS to project incident volume from historical DRRMO and BFP records (Ahmad et al., 2025).

2.2 Local Literature

Geospatial Analysis of Emergency Incidents and Disaster Response Management in Oton, Iloilo

This study examined emergency response operations in Oton, Iloilo, to find gaps in service coverage and response efficiency. The researchers digitized 1,428 Patient Care Report forms from July to December 2024 and analyzed incident counts and average response times by barangay using Getis-Ord hotspot analysis and Moran's I. Urbanized barangays had more incidents and faster responses, while remote and rural barangays had fewer incidents and longer responses, and response times were longest in the evening and on weekends. The study recommended digitizing and geolocating incident records and using real-time GIS dashboards. These findings support the heatmap, hotspot, and dashboard features of GAOIRS (Aporto et al., 2026).

A Spatiotemporal Analysis of Fire Incidents in Pampanga Province of Philippines: Inputs for Fire Prevention Programs

This study examined fire incident data of the Bureau of Fire Protection in Pampanga from 2013 to 2020. After cleaning the data, the researchers mapped and charted the incidents to study patterns in space and time. Fires were most likely on Sundays and Mondays in the highly urbanized cities of Angeles, Mabalacat, and San Fernando, mostly in March and December and between 2:00 and 3:00 PM. The study also found that barangays in flood-prone areas were likely to experience fire cases, and it recommended the spatiotemporal



method for planning by the fire bureau. This supports the trend analysis and time-based reports in GAOIRS (de Leon & Miranda, 2022).

Datasets of Fire and Crime Incidents in Pampanga, Philippines

This study collected and analyzed fire and crime incident datasets from the Bureau of Fire Protection and the Philippine National Police for the province of Pampanga. The researchers cleaned, normalized, and mapped both datasets to identify temporal patterns, finding that fires mostly occurred during the dry season and that crime was particularly high in December. The study showed that agency-held incident records, once digitized and cleaned, can reveal clear temporal patterns useful for planning. This supports the practice in GAOIRS of digitizing and cleaning DRRMO and BFP incident records to reveal similar patterns for planning response deployment (Miranda et al., 2023).

Development of Crime Reporting System to Identify Patterns of Crime in Laguna

This study developed a crime reporting system to identify patterns of crime in the province of Laguna, Philippines. The researchers digitized crime records and applied pattern analysis to identify recurring characteristics of reported incidents by location and time. The system demonstrated that a digital incident reporting and pattern-analysis tool can help local authorities move from manually reviewing paper records to systematically identifying trends. This supports the design of GAOIRS as a system that digitizes incident reports and analyzes them for recurring patterns to guide response planning (Villarica et al., 2022).



Black Spot Cluster Analysis of Road Crash Involving Public Utility Vehicles (PUV) Along Commonwealth Avenue Using Kernel Density Estimation

This study identified road crash hotspots involving public utility vehicles along Commonwealth Avenue in Quezon City, which is known as the “killer highway” of Metro Manila, with an average of 17 fatal road crashes per year from 2017 to 2021. The researchers used the crash records of the Metropolitan Manila Development Authority and applied kernel density estimation using free open-source software. The study revealed five fatal crash locations, with high-density crashes from 7:00 to 10:00 AM and from 5:00 to 8:00 PM, specifically on Fridays at Regalado Avenue. The authors said the results can be used to identify black spots and to develop targeted interventions. This supports the use of KDE on agency incident records, as done in GAOIRS (Ballarta et al., 2024).

Visualization and Geo-Mapping of Philippine Fire Incidents

This study explored how data visualization and analysis can extract useful information for planning and decision-making, given the rapid increase in fire incidents recorded in the Philippines from 2013 to 2016. The researchers used the 2012 to 2016 Philippine Nationwide Fire Incident Statistics data of the Bureau of Fire Protection, which was provided through Open Data Philippines. After cleaning and reprocessing the data, they developed a dynamic system called FireStatPH using Flask. This shows that recorded incident data can be turned into visual and map-based outputs for decision-making, which is the same purpose of the analytics and dashboard features of GAOIRS (Oñate, 2022).



Building Model for Crime Pattern Analysis Through Machine Learning Using Predictive Analytics

This study examined how well decision trees and ensemble trees can classify crime through model development. The researchers ran experiments to improve the ensembles in both classification and regression, and they used the synthetic minority oversampling technique (SMOTE) to address the problem of imbalanced data. Using different metrics to evaluate each model, they found that the gradient boosted tree had better classification capability than the decision tree and the random forest in both classification and regression. This supports the practice in GAOIRS of comparing several models using evaluation metrics before choosing the one used for incident forecasting (Asor et al., 2022).

Compliance with the Disaster Risk Reduction and Management Act of 2010 of Local Government Units in the Province of Negros Occidental, Philippines

This study assessed how well the local government units of Negros Occidental comply with the Philippine Disaster Risk Reduction and Management Act, in the areas of risk management practices and institutional arrangements and capacities across the four thematic areas of disaster. The researchers investigated the challenges encountered by the DRRM officers and planning coordinators of 32 LGUs. Using descriptive analysis, they found a high level of compliance in prevention and mitigation, preparedness, and response, and a moderate level in rehabilitation and recovery. LGUs with nine or more hazards were rated high, while 4th class LGUs were rated moderate. The study concluded that failure to provide personnel and financial resources to the DRRM office endangers the safety of the population. This supports the need for practical



tools such as GAOIRS that help DRRM offices in the province manage incidents with the resources they have (Yusay & Caelian, 2022).

An Assessment of the Effectiveness of Government's Risk Management on Disaster Prevention, Mitigation, Response, Recovery, and Rehabilitation in Metro Manila, Philippines

This study assessed the effectiveness of local government units in Metro Manila across four thematic areas of disaster risk reduction and management: preparedness, prevention and mitigation, response, and recovery and rehabilitation. Using a descriptive-evaluative design, the researchers found that all four areas were rated effective overall, with disaster preparedness rated highest. The study noted that consistent implementation across LGUs still depends on adequate personnel and resources at the local level. This supports the need for practical tools such as GAOIRS that help DRRMO and BFP offices in the region carry out NDRRMC-catered response duties with the resources they have (Lagrada & Santiago, 2025).

Acceptance of Mobile Application on Disaster Preparedness: Towards Decision Intelligence in Disaster Management

This study examined user acceptance of a mobile application for disaster preparedness as a step toward decision intelligence in disaster management. The study is relevant because it shows that acceptance by users is an important measure for disaster-related applications, which is why GAOIRS was evaluated for usability with actual end users (Escolano et al., 2023).



2.3 Foreign Systems

Multi-Hazard WebGIS Platform for Coastal Regions (MOSAIC)

The MOSAIC WebGIS platform was developed for coastal regions. It gives access to in-situ and remote sensing data and model predictions in one place, and it includes a database of past hazardous events for the western coast of Portugal, organized by risk indicators. The authors said the platform supports both emergency response and long-term risk planning. This supports the use of a historical incident database together with live monitoring in GAOIRS (Rocha et al., 2021).

Towards a Digital Twin for Supporting Multi-Agency Incident Management in a Smart City

This study identified the challenges faced by emergency services in a multi-agency response, such as scattered data and poor real-time information. Using a systems engineering approach, the researchers built a prototype on Microsoft Azure cloud services that lets stakeholders view available resources, send automatic updates, and add real-time weather and traffic data. The prototype shows how one shared view of an incident can support agencies working together. This supports the shared incident view and the automatic status updates in GAOIRS (Wolf et al., 2022).

The ERMES Chatbot: A Conversational Communication Tool for Improved Emergency Management and Disaster Risk Reduction

The ERMES chatbot is a Telegram-based tool for two-way communication among control centers, field forces, and citizens. The researchers designed, implemented, and validated it using a user-centered approach, and they



co-designed its functions with emergency management practitioners. The chatbot connects to a web dashboard that shows the collected data and supports mission management and early warning. This supports the design of citizen reporting, response unit updates, and a monitoring dashboard built around user needs in GAOIRS (Urbanelli et al., 2024).

Secure Cross-Layer Mobile Sensing Framework for Real-Time Disaster Reporting and Visualisation Using a Mobile Application

This study proposed a secure, scalable mobile system called DEAPP for real-time disaster reporting and visualization. The system connects a native mobile app, a web-based backend, and a cloud database through a cross-layer architecture, and it uses role-based access control, HTTPS encryption, and verified event publication to keep citizen-submitted reports secure and reliable. The researchers also used caching to speed up data access during emergencies. This shows that a citizen-facing mobile reporting app can be built with verification and access control from the start — the same approach GAOIRS takes with its report verification step before dispatch (Mustafa et al., 2025).

Sahana Eden Disaster Management Platform

Sahana Eden is a free and open-source family of web-based applications that addresses common coordination problems during a disaster, such as finding missing people, managing aid and volunteers, and tracking camps, among government groups, civil society, and the victims themselves. Its core features include mapping for situation awareness and geospatial analysis, requests management, volunteer management, and shelter and organization registries. The project began with volunteers in Sri Lanka after the 2004 tsunami and has



been used by dozens of governments and non-governmental organizations. It can also provide a management and ticketing interface for crowdsourced data such as that collected in Ushahidi. This supports the combination of mapping and coordination tools in one platform, as in GAOIRS (Treadgold, 2022).

2.4 Local Systems

iAlerto: A Web and Mobile Alert System for Pasig City DRRMO with Mobile GPS Service Integration

iAlerto was developed to improve incident and emergency reporting by Pasig City residents and to support the Pasig City DRRMO. The developers used Scrum. The mobile application was built with Android Studio, and the web application with CodeIgniter, PHP, and MySQL. The web application manages user profiles, evacuation centers, and incident reports, sends alerts and announcements, and generates reports, while the mobile application supplies the GPS location. Twenty IT professionals and twenty residents and DRRMO employees evaluated it using ISO 9126 functionality criteria, and it was found functionally acceptable. This closely matches the reporting, alert, GPS, and report generation features of GAOIRS (Goh et al., 2023).

Application of Spatiotemporal Analysis and Knowledge Discovery for Databases in the Bureau of Fire Protection as Incident Report System: Tool for Improving Fire Services

This study developed a fire report management system for the Bureau of Fire Protection that analyzes the space and time characteristics of fires and applies Knowledge Discovery in Databases (KDD) methods to identify patterns of fire incidents in Manila. The developers applied SMOTE and one-hot encoding



for data processing and used the Agile method as the software development model. This supports the combination of an incident reporting system and pattern analysis in GAOIRS (Balahadia et al., 2020).

GPS-Based Emergency Response System for Disaster Risk Reduction and Management

This system was developed for the Municipal Disaster Risk Reduction and Management Office of Lambunao, Iloilo, to address delayed emergency reporting, the absence of real-time location tracking, and poor coordination between residents and responders. It has a mobile application for residents and responders and a web-based dashboard for the office. Residents submit emergency requests with details, images, and an automatically captured GPS location, while responders receive alerts, navigate to the incident, and update their status. The dashboard shows incidents, responder locations, and response progress. It was built with Flutter, PHP, and MySQL using the Agile method, and 33 respondents evaluated it using ISO/IEC 25010, giving an overall mean of 4.77 (Excellent). The authors reported that it improved reporting accuracy, reduced response time, and improved coordination. This closely matches the GPS reporting, status updates, and monitoring dashboard of GAOIRS (Consumo et al., 2026).

Project SIGNAL (Strategic Incident Georeferencing for Nationwide Alertness)

Project SIGNAL is an incident reporting and monitoring system of DOST Region 6 for local government units, used for emergencies such as landslides, floods, vehicular accidents, and medical emergencies. Incidents are reported



through a mobile application that automatically geo-tags the location and sends it to the Emergency Operations Center of the LGU, and the recorded data builds a database for planning and risk analysis. The system was implemented in 2022 and had recorded more than 27,000 incident reports as of April 2025, and 132 of the 139 LGUs in Western Visayas had activated their accounts. DOST said it aims to expand the platform, including to the Negros Island Region. This is the closest local example of GAOIRS, which combines geotagged reporting with verification before dispatch, nearest responder dispatch, and trend forecasting for the NIR (Clarín, 2025).

Bandilyo App: A Disaster Risk Reduction Monitoring and Incident Reporting System with Geolocation and SMS Technology

Bandilyo App is a disaster risk reduction and incident reporting system built for the Municipality of Tanauan, Leyte, that combines geolocation and SMS technology in an integrated command-and-control communication system. The developers used Agile methodology to build a mobile application, a web application, and a client-server architecture that supports coordination and humanitarian response activities. The Municipal DRRMO reported that the system supported effective, accurate, and timely information for coordination. This is a close local example of the kind of geolocation-based incident reporting and coordination system GAOIRS is built to provide for DRRMO and BFP personnel in the Negros Island Region (de los Santos et al., 2020).



2.5 Synthesis

Table 1
Synthesis of Related Literature

Related Studies	Features and Functionalities	Developed System
Comprehensive Analysis of the Use of Web-GIS for Natural Hazard Management (Daud et al., 2024)	• Systematic review of 65 studies (2014–2023) • Covers decision support, real-time monitoring, early warning • Covers visualization, UI design, and multi-stakeholder collaboration	• Supported the map-based dashboard, real-time monitoring, and multi-role access in GAOIRS
A Near-Real-Time Global Landslide Incident Reporting Tool Demonstrator (Pennington et al., 2022)	• Social media monitored for landslide reports • AI image classifier filters incoming posts • 76% precision in real-world deployment	• Guided fast incident intake with photo evidence and a verification step before dispatch
Designing Multimodal Interactive Dashboard of Disaster Management Systems (AlAbdulaali et al., 2022)	• Multimodal dashboard (text, image, audio, video) • Requirements drawn from social media study • Expert heuristic evaluation of designs	• Shaped the monitoring dashboard that shows incident details, evidence, and status updates in one view
A Study on GIS-Based Spatial Analysis of Emergency Response: Focusing on Seoul (Park et al., 2024)	• Fire dispatch analyzed for Seoul • Seasonal and weekday call patterns found • Rescue incidents concentrated in specific areas	• Supported hotspot maps and time-based reports showing where and when incidents cluster
Population Access to Hospital Emergency Departments: The Spatial Analysis in Public Health Research (Kisiala et al., 2022)	• GIS-based travel-time accessibility of EDs compared across 2011 and 2021 • National ED network reorganization evaluated across communes	• Supported using geospatial mapping and hotspot analysis to evaluate whether GAOIRS's incident coverage reaches the areas that need it most



GIS Based Approach of Surface Transport Network Analysis (Gupta & Khode, 2025)	• Optimum-route model for healthcare responders (Katol metro area) • Enhanced GIS route vs. shortest-path route compared; 22% faster travel time	• Supported GIS network analysis in GAOIRS for identifying the fastest route to a verified incident, not just the shortest one
The Role of GIS in Disaster Response: Improving Efficiency and Overcoming Challenges (Han & Liu, 2025)	• Review of GIS integration with remote sensing, AI, and machine learning • Data quality, interoperability, and user-capability challenges identified	• Supported designing GAOIRS around simple role-based access, data verification, and testing with actual DRRMO/BFP users
Emergency Incident Detection from Crowdsourced Waze Data (Senarath et al., 2020)	• Bayesian framework for unreliable reports • Crowdsourced Waze data fused with official records • Outperformed baselines in F1-score and AUC	• Supported checking the reliability of citizen reports and tagging same-incident reports
Spatiotemporal Clustering and Analysis of Road Accident Hotspots by Exploiting GIS Technology and KDE (Kazmi et al., 2022)	• KDE with spatial autocorrelation applied to UK crash data • Effect of bandwidth on hotspot detection examined	• Supported the KDE-based heatmap feature in GAOIRS and the need to tune its bandwidth setting carefully
Forecasting the Human Cost of Disasters under Sustainable Development Goal: A Time Series Analysis Using Facebook Prophet Model (Ahmad et al., 2025)	• Prophet model forecasted disaster deaths, 2024–2030, across 168 countries • Average global mortality rate of 13.38 deaths/million found	• Formed the basis of the Prophet-based incident-volume forecasting in GAOIRS
Geospatial Analysis of Emergency Incidents and Disaster Response Management in Oton, Iloilo (Aporto et al., 2026)	• 1,428 Patient Care Report forms digitized • Getis-Ord hotspot analysis and Moran's I used • Response-time gaps found by barangay	• Supported the heatmap, hotspot, and dashboard features



A Spatiotemporal Analysis of Fire Incidents in Pampanga Province (de Leon & Miranda, 2022)	• Fire data mapped and charted, 2013–2020 • Day/month/time patterns identified • Flood-prone barangays linked to fire risk	• Supported trend analysis and time-based reports
Datasets of Fire and Crime Incidents in Pampanga, Philippines (Miranda et al., 2023)	• BFP and PNP fire/crime datasets collected, cleaned, and mapped • Fires linked to dry season; crime highest in December	• Supported digitizing and cleaning DRRMO/BFP incident records to reveal patterns for planning response deployment
Development of Crime Reporting System to Identify Patterns of Crime in Laguna (Villarica et al., 2022)	• Digitized crime records analyzed for recurring patterns • Patterns identified by location and time	• Supported the design of GAOIRS as a system that digitizes reports and analyzes them for recurring patterns
Black Spot Cluster Analysis of Road Crash Involving PUV Along Commonwealth Avenue (Ballarta et al., 2024)	• KDE applied to MMDA crash records • Free open-source GIS tools used • Five fatal crash hotspots identified	• Supported the use of KDE on agency incident records with open-source tools
Visualization and Geo-Mapping of Philippine Fire Incidents (Oñate, 2022)	• Nationwide BFP fire data cleaned (2012–2016) • Dynamic system (FireStatPH) built with Flask	• Supported turning incident records into visual, map-based analytics outputs
Building Model for Crime Pattern Analysis Through Machine Learning (Asor et al., 2022)	• Decision tree, random forest, gradient boosted tree compared • SMOTE used to address data imbalance • Models evaluated with multiple metrics	• Supported comparing several models by evaluation metrics before choosing GAOIRS's forecasting model



Compliance with the DRRM Act of 2010 of LGUs in Negros Occidental (Yusay & Caelian, 2022)	• Compliance assessed across 32 LGUs • High compliance in mitigation/response, moderate in recovery • Personnel and funding gaps identified	• Supported the case for practical tools that help DRRM offices manage incidents with limited resources
An Assessment of the Effectiveness of Government's Risk Management on Disaster Prevention, Mitigation, Response, Recovery, and Rehabilitation in Metro Manila (Lagrada & Santiago, 2025)	• Descriptive-evaluative assessment of LGU DRRM across four thematic areas • All areas rated effective; preparedness rated highest	• Supported the need for practical tools such as GAOIRS to help DRRMO/BFP carry out NDRRMC-catered duties with available resources
Acceptance of Mobile Application on Disaster Preparedness (Escolano et al., 2023)	• User acceptance of a disaster-preparedness app studied • Acceptance framed as key to decision intelligence	• Supported evaluating GAOIRS for usability with actual end users

The reviewed literature shows that effective incident-response platforms combine location-aware reporting, GIS-based visualization, and dashboards that track cases from report to resolution. The foreign studies show that mapping incident, dispatch, and response records reveals where and when cases cluster, that citizen reports need reliability checks before use, and that GIS tools underperform when user access, knowledge, and implementation are neglected. The local studies reinforce this by showing that mapping incidents in space and time exposes high-risk areas and response gaps across different hazards and regions, whether for emergencies in Oton, Iloilo, fires in Pampanga and Manila, or road crashes in Quezon City, and that DRRM offices need both a working tool and adequate personnel and funding to act on what the data shows.



In summary, GAOIRS draws on this literature by combining evidence-based citizen reporting, report verification, nearest-responder dispatch, GIS mapping and hotspot detection, and Prophet-based trend forecasting into a single platform built for the DRRMO and BFP. It aligns with the direction of the reviewed studies by turning separate findings on spatial analysis, dashboard design, and reliability filtering into one system developed specifically for NDRRMC-catered incidents in the Negros Island Region.

Table 2
Synthesis of Related Systems

Related Systems	Features and Functionalities	Developed System
MOSAIC — Multi-Hazard WebGIS Platform for Coastal Regions (Rocha et al., 2021)	• In-situ and remote sensing data in one platform • Model predictions integrated • Database of past hazardous events	• Supported a historical incident database paired with live monitoring
Digital Twin Prototype for Multi-Agency Incident Management (Wolf et al., 2022)	• Azure cloud prototype for a shared resource view • Automatic status updates to stakeholders • Real-time weather and traffic data added	• Supported the shared incident view and automatic status updates
ERMES Chatbot (Urbanelli et al., 2024)	• Telegram-based two-way communication • Connected web dashboard for control centers, field forces, citizens • Co-designed with emergency management practitioners	• Shaped citizen reporting, response-unit updates, and a user-centered monitoring dashboard
Secure Cross-Layer Mobile Sensing Framework — DEAPP (Mustafa et al., 2025)	• Mobile app + web backend + cloud database in a cross-layer architecture • Role-based access control, HTTPS encryption, verified event publication • Caching used to speed up data access	• Supported citizen reporting built with verification and access control from the start, added before dispatch



Sahana Eden Disaster Management Platform (Treadgold, 2022)	• Open-source disaster coordination suite • Mapping, requests, volunteer, and shelter management • Used by governments and NGOs worldwide	• Supported combining mapping and coordination tools in one platform
iAlerto — Web and Mobile Alert System for Pasig City DRRMO (Goh et al., 2023)	• Android app + CodeIgniter/PHP/MySQL web app • Manages profiles, evacuation centers, incident reports • Sends alerts and generates reports	• Matched the reporting, alert, GPS, and report-generation features
Spatiotemporal Analysis and KDD-Based Incident Report System (Balahadia et al., 2020)	• Spatiotemporal and KDD analysis of fire data • SMOTE and one-hot encoding for data processing • Agile development method	• Supported combining incident reporting with pattern analysis
GPS-Based Emergency Response System (Lambunao, Iloilo) (Consumo et al., 2026)	• Resident and responder mobile apps with GPS • Web dashboard for status and response progress • Built with Flutter, PHP, and MySQL	• Matched the GPS reporting, status updates, and monitoring dashboard features
Project SIGNAL (DOST Region 6) (Clarín, 2025)	• Geotagged mobile incident reporting • Database built for LGU planning and risk analysis • 27,000+ reports logged; 132 of 139 LGUs activated	• Closest local example — combined geotagged reporting with verification, dispatch, and forecasting
Bandilyo App — A DRR Monitoring and Incident Reporting System with Geolocation and SMS (de los Santos et al., 2020)	• Geolocation and SMS in an integrated command-and-control system • Mobile app, web app, and client-server architecture built with Agile • Municipal DRRMO reported effective, accurate, timely coordination	• Close local example of geolocation-based incident reporting and coordination for DRRMO/BFP personnel in the NIR

The reviewed systems show that citizen reports paired with location data can be collected and routed to responders through web and mobile platforms, as demonstrated by Ushahidi, Project SIGNAL, the GPS-based system in Lambunao, iReportMo, and iAlerto, while MOSAIC and the digital twin prototype



show that shared dashboards and automatic updates support coordination among multiple agencies. Most of these systems, however, address only one part of the incident-response process, such as reporting, mapping, alerting, or pattern analysis, rather than the process as a whole.

In summary, GAOIRS addresses this gap by bringing evidence-based reporting, verification, nearest-responder dispatch, map and hotspot visualization, trend forecasting, and role-based dashboards into one platform. While the reviewed systems each cover part of what GAOIRS does, few combine all of these functions for the DRRMO and BFP together, which is the gap that GAOIRS is designed to fill.



CHAPTER III

Methodology

This chapter introduces the systematic approach and tools used to develop and evaluate the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. The study adopts the Agile Development Method framework, which provides a structured process for planning, designing, building/developing, testing, deploying, reviewing and launching the system. The chapter outlines the operational framework that guided the project and presents key modeling tools including the Context Diagram and Data Flow Diagrams (DFD) to illustrate system processes, the Entity-Relationship Diagram (ERD) to define data structures, and the Unified Modeling Language (UML) Use Case Diagram and scenarios to describe user interactions. A complete Data Dictionary was provided to define all data elements used in the system. The chapter also specified the necessary hardware and software requirements, presented the project timeline, and discussed the evaluation process, which includes system testing and assessment using the ISO/IEC 25010:2011 Software Quality Model and ISO/IEC 9241-11:2018 Software Quality Model to ensure functionality, reliability, and overall user experience. The chapter concluded with the identification of respondents involved in the system assessment and the ethical considerations observed throughout the study.

3.1 Systems Development Method

Systems Development Method refers to the framework used to structure, plan, and control the information system development process. The study



employed a combination of developmental and descriptive research methods: developmental research focused on the design, development, testing, and refinement of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS), while descriptive research was used to evaluate the system's functionality, usability, and performance through testing procedures and feedback gathered from DRRMO and BFP personnel.

To guide the system development process, the researchers adopted the Agile methodology, specifically an incremental development model, over a plan-driven model such as Waterfall. Agile was chosen because the requirements of the DRRMO and BFP Station became clearer step by step through interviews, feedback sessions, and prototype reviews, and each cycle produced a working part of the system that could be tested and shown to the users. This iterative emphasis on continuous feedback and testing throughout development helped ensure that the system's features matched the actual workflow of its intended users.

Figure 1
Agile Development Method



Figure 1 shows the seven stages of each development cycle: Plan, Launch, Design, Develop, Test, Deploy, and Review. The Review stage feeds its results back into the next Plan stage, so the system improves with every cycle.

Plan

In the Plan stage, the researchers conducted interviews with DRRMO and BFP Station personnel and observed their existing workflows to identify the requirements, challenges, and priorities for each development cycle. Information was gathered on how incidents are currently reported, verified, dispatched, and recorded. These findings were used to define the system's core functions, including incident report recording and management, response dispatch, geospatial mapping, monitoring, and report generation, which became the backlog of features for the cycle.



Design

In the Design stage, the researchers prepared and refined the interface mockups, data flow diagrams, and database design based on feedback from the previous review. Each feature scheduled for the cycle, such as the incident reporting form, the geospatial mapping interface, or the analytics dashboard, was planned in detail before development began.

Develop

In the Develop stage, the modules were coded and integrated in increments: role-based access control, incident report recording and management, incident response and dispatch, geospatial mapping and geofencing, real-time monitoring and heatmap visualization, trend analysis and analytics, and report generation. Partial features could be tested and shown to users before the whole system was assembled.

Test

In the Test stage, the modules were tested individually and as one integrated system. White-box testing checked the internal logic and back-end functions, and black-box testing checked the system from the user's side. Errors were fixed and the affected parts were tested again before moving forward.

Deploy

In the Deploy stage, the tested version of the system was released to a hosted environment so that DRRMO and BFP personnel could try it under conditions similar to actual use.



Review

In the Review stage, the team evaluated the deployed system by gathering feedback from the DRRMO and BFP personnel. The identified issues, user suggestions, and system performance results were documented and analyzed to determine the necessary improvements before the final release.

Launch

In the Launch stage, the researchers reviewed the backlog with DRRMO and BFP Station stakeholders to confirm the scope and priority of the features to be built in the current cycle. The timeline, task assignments, and resources needed to complete the cycle were also set during this stage before development work began.

3.2 Operational Framework

Figure 2

Operational Framework of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

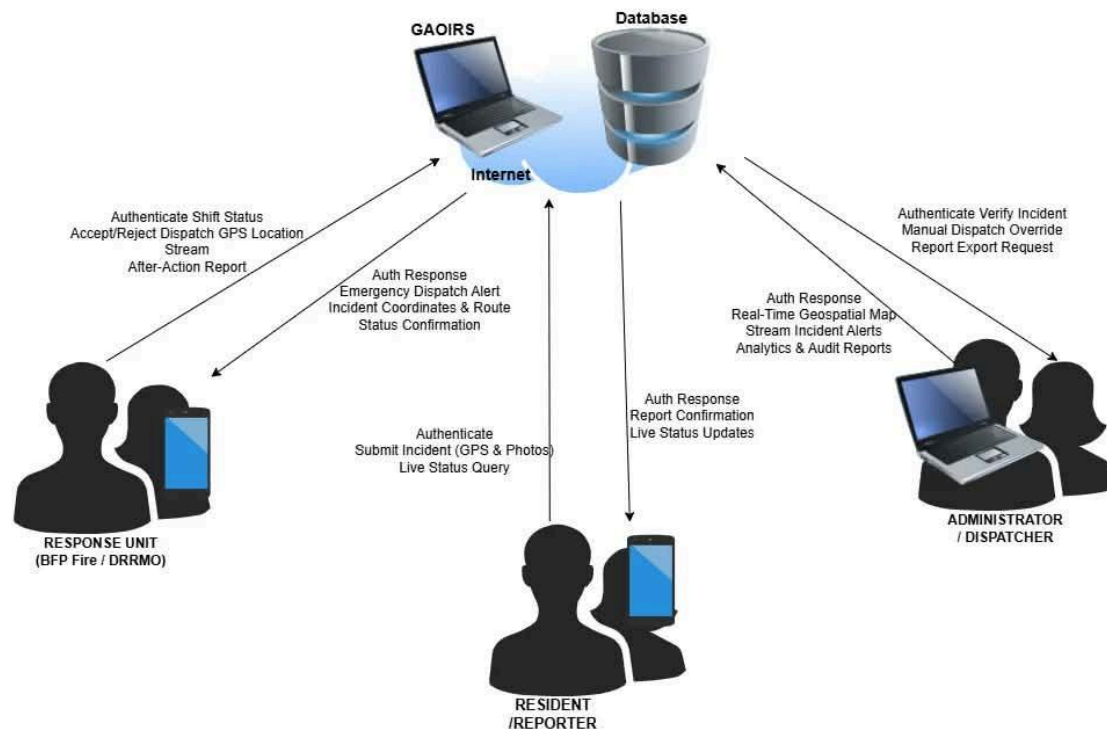


Figure 2 shows how the three system actors interact with the platform. The Citizen (reporter) submits incident details, including GPS location, photo, and incident type, and receives confirmation and status updates. The Response Unit (BFP or DRRMO) sends acknowledgement and status updates, and receives automated alerts with incident data and maps. The Administrator receives the emergency reports, configures role-based access control and system settings, and receives dashboard views, heatmaps, and generated reports. The data are stored in and retrieved from a central database.

3.3 Data Flow Diagram

Figure 3

Context Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

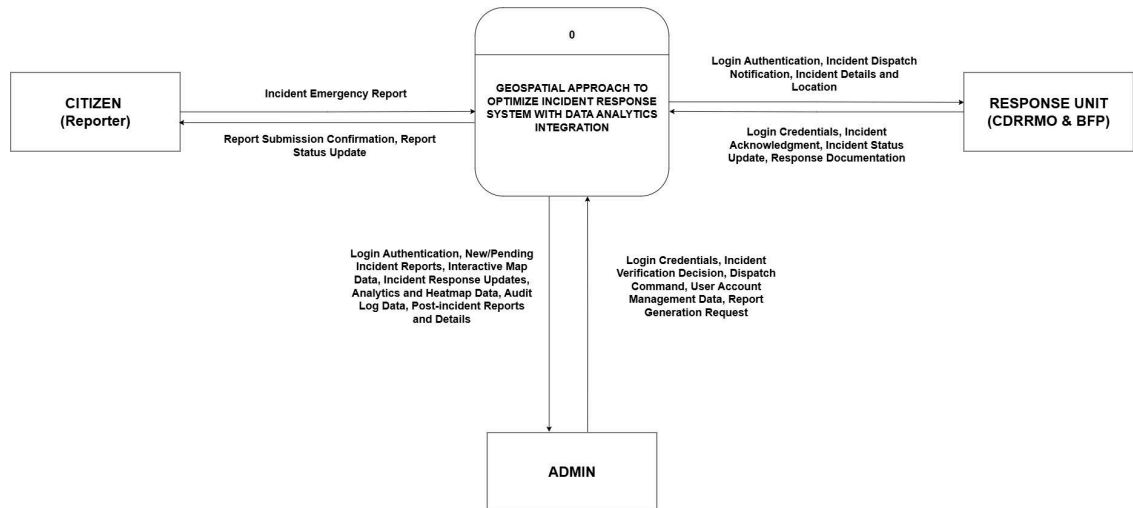


Figure 3 shows the context diagram of the system, highlighting how Citizens, Response Units (CDRRMO and BFP), and Administrators interact with it. Citizens submit incident emergency reports, while Response Units send their login credentials, incident acknowledgments, incident status updates, and response documentation. Administrators send login credentials, incident verification decisions, dispatch commands, user account management data, and report generation requests. The system sends back report submission confirmations and report status updates to Citizens. It sends login authentication, incident dispatch notifications, and incident details and location to Response Units. For Administrators, it returns new and pending incident reports, interactive map data, incident response updates, analytics and heatmap data, audit log

data, and post-incident reports to help each user perform their duties properly and keep incident records organized.

Figure 4

Level 1 Data Flow Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

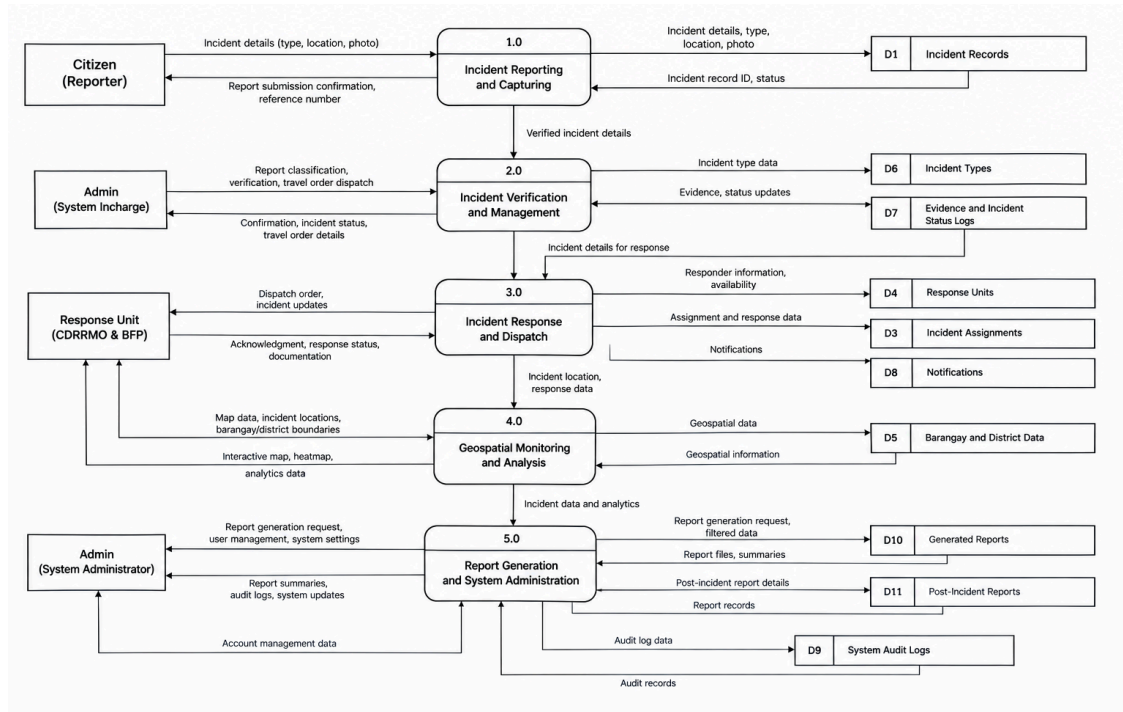


Figure 4 illustrates the Level 1 Data Flow Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. It expands the main process into five interconnected functions that show how data moves throughout the system. Citizens begin the workflow by submitting incident details such as the type, location, and photo to 1.0 Incident Reporting and Capturing. These details were stored in D1 Incident Records, and the Citizen received a report submission confirmation and a reference number. The verified incident details then passed to 2.0 Incident Verification and Management, where the Administrator (System In-charge) classified and verified the report and issued



the travel order for dispatch. This process saved incident type data in D6 Incident Types, and exchanged evidence and status updates with D7 Evidence and Incident Status Logs. The incident details were then forwarded to 3.0 Incident Response and Dispatch, which retrieved responder information and availability from D4 Response Units, recorded assignment and response data in D3 Incident Assignments, and stored notifications in D8 Notifications. Response Units (CDRRMO and BFP) received dispatch orders and incident updates, and sent back acknowledgments, response status, and documentation. The incident location and response data moved on to 4.0 Geospatial Monitoring and Analysis, which used map data, incident locations, and barangay and district boundaries, and stored and retrieved geospatial data in D5 Barangay and District Data. This process gave users the interactive map, heatmap, and analytics data. Finally, the incident data and analytics flowed to 5.0 Report Generation and System Administration. The Administrator (System Administrator) sent report generation requests, user management data, and system settings, and received report summaries, audit logs, and system updates. This process saved generated reports in D10 Generated Reports, post-incident reports in D11 Post-Incident Reports, and audit records in D9 System Audit Logs.

Figure 5

Level 2 Data Flow Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

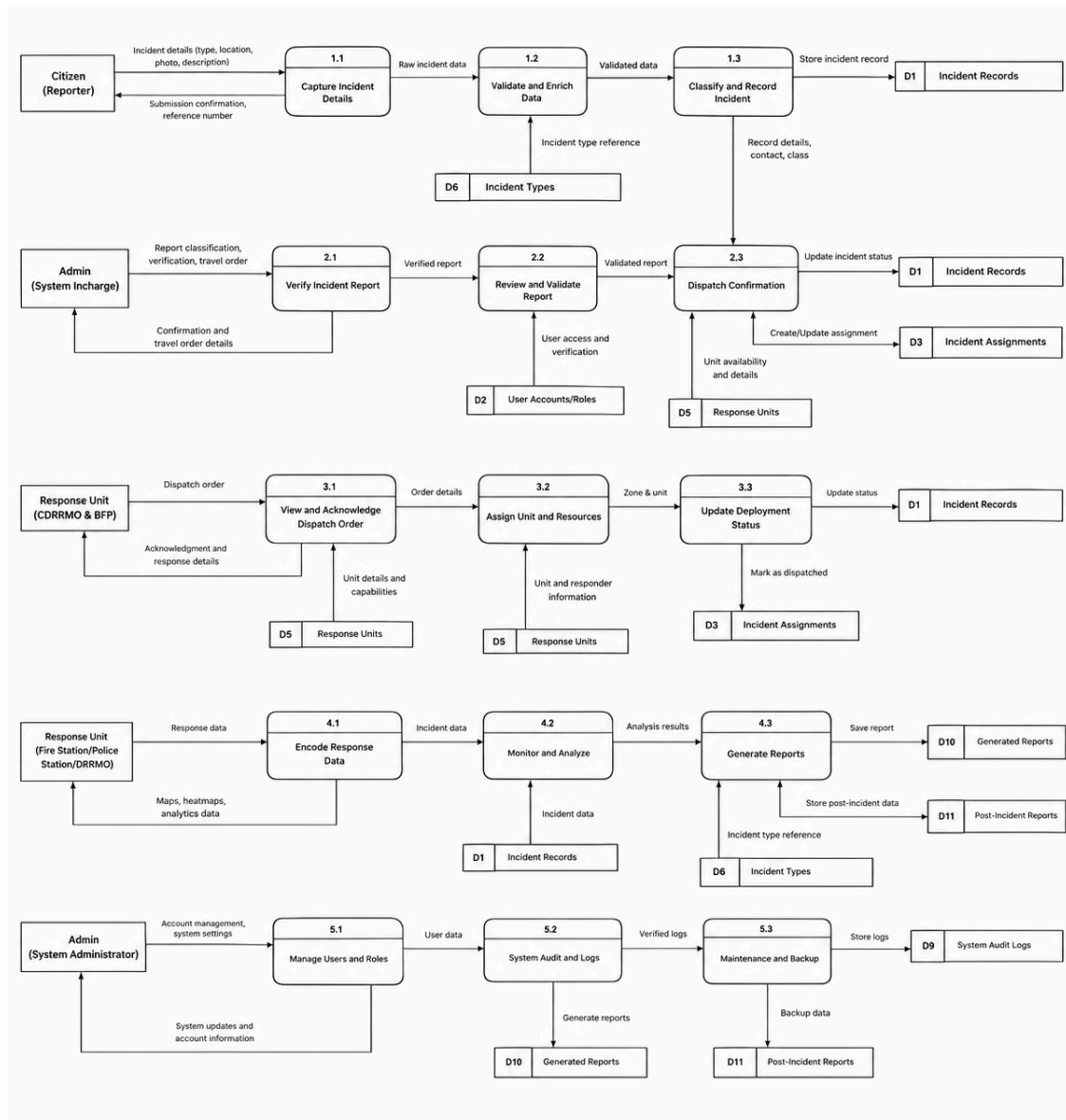


Figure 5 presents the Level 2 Data Flow Diagram of the system, showing how citizens, administrators, and response units report, verify, dispatch, and monitor incidents. The process begins with the Citizen submitting incident details, which are captured, validated against the incident types, and classified before being stored as incident records. The Admin (System In-charge) then verifies and



reviews the report and confirms the dispatch, updating the incident status and creating the assignment based on the availability of response units. The Response Unit (CDRRMO and BFP) views and acknowledges the dispatch order, is assigned the unit and resources, and updates its deployment status. Response data is then encoded and analyzed to produce maps, heatmaps, and analytics data, and the results are used to generate reports and post-incident records. Finally, the Admin (System Administrator) manages users and roles, reviews system audit logs, and performs maintenance and backup. The diagram shows a clear flow of data to support accurate recording, proper verification, coordinated dispatch, and informed monitoring of incidents.

Figure 6

Level 3 Data Flow Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

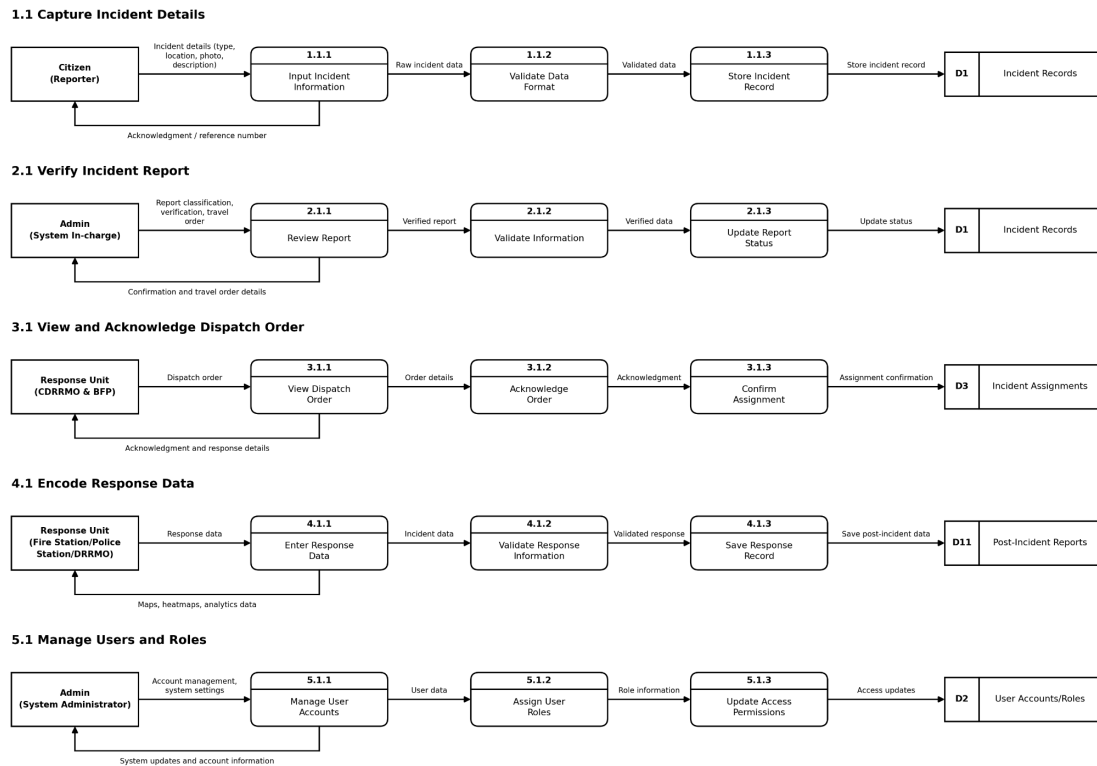


Figure 6 presents the Level 3 Data Flow Diagram of the system, showing selected Level 2 processes broken down into three sub-processes each. Process 1.1 Capture Incident Details begins when the Citizen (Reporter) submits the incident type, location, photo, and description. The system inputs the information, validates the data format, and stores the incident record in D1 Incident Records, then returns an acknowledgment and reference number to the Citizen. In 2.1 Verify Incident Report, the Admin (System In-charge) provides the report classification, verification, and travel order. The report is reviewed, the information is validated, and the report status is updated in D1 Incident Records,

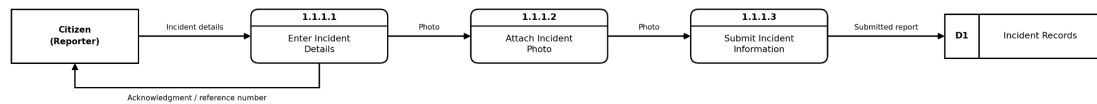


with a confirmation and travel order details sent back to the Admin. In 3.1 View and Acknowledge Dispatch Order, the Response Unit (CDRRMO and BFP) receives the dispatch order, acknowledges it, and confirms the assignment, which is saved in D3 Incident Assignments. In 4.1 Encode Response Data, the Response Unit enters the response data, which is validated and saved as a response record in D11 Post-Incident Reports. The Response Unit then receives maps, heatmaps, and analytics data. Finally, in 5.1 Manage Users and Roles, the Admin (System Administrator) sends the account management data and system settings. The system manages the user accounts, assigns user roles, and updates access permissions in D2 User Accounts/Roles, then returns system updates and account information to the Admin.

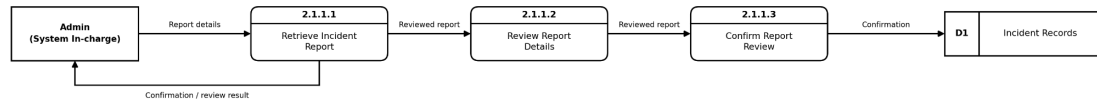
Figure 7

Level 4 Data Flow Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

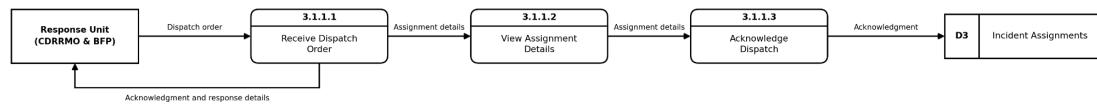
1.1.1 Input Incident Information



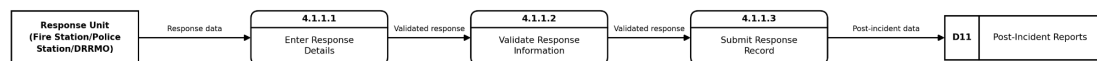
2.1.1 Review Report



3.1.1 View Dispatch Order



4.1.1 Enter Response Data



5.1.1 Manage User Accounts

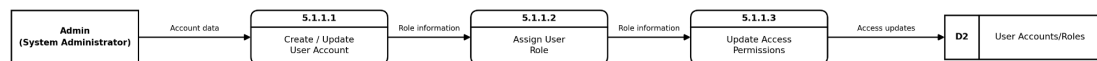


Figure 7 presents the Level 4 Data Flow Diagram of the system, showing the most detailed steps inside five Level 3 processes. In 1.1.1 Input Incident Information, the Citizen (Reporter) enters the incident details, attaches an incident photo, and submits the information. The submitted report is saved in D1 Incident Records, and the Citizen receives an acknowledgment and reference number. In 2.1.1 Review Report, the Admin (System In-charge) sends the report details. The system retrieves the incident report, reviews the report details, and confirms the review, with the confirmation saved in D1 Incident Records and a confirmation or review result returned to the Admin. In 3.1.1 View Dispatch Order, the Response Unit (CDRRMO and BFP) sends the dispatch order, and the



system receives it and displays the assignment details. The Response Unit then acknowledges the dispatch, which is saved in D3 Incident Assignments, and receives an acknowledgment and response details in return. In 4.1.1 Enter Response Data, the Response Unit enters the response details, the system validates the response information, and the response record is submitted to D11 Post-Incident Reports as post-incident data. Lastly, in 5.1.1 Manage User Accounts, the Admin (System Administrator) sends the account data. The system creates or updates the user account, assigns the user role, and updates the access permissions in D2 User Accounts/Roles. Together, the diagrams show a clear flow of data to support accurate recording, proper verification, coordinated dispatch, and secure management of users.

3.4 Entity Relationship Diagram

Figure 8

Entity Relationship Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

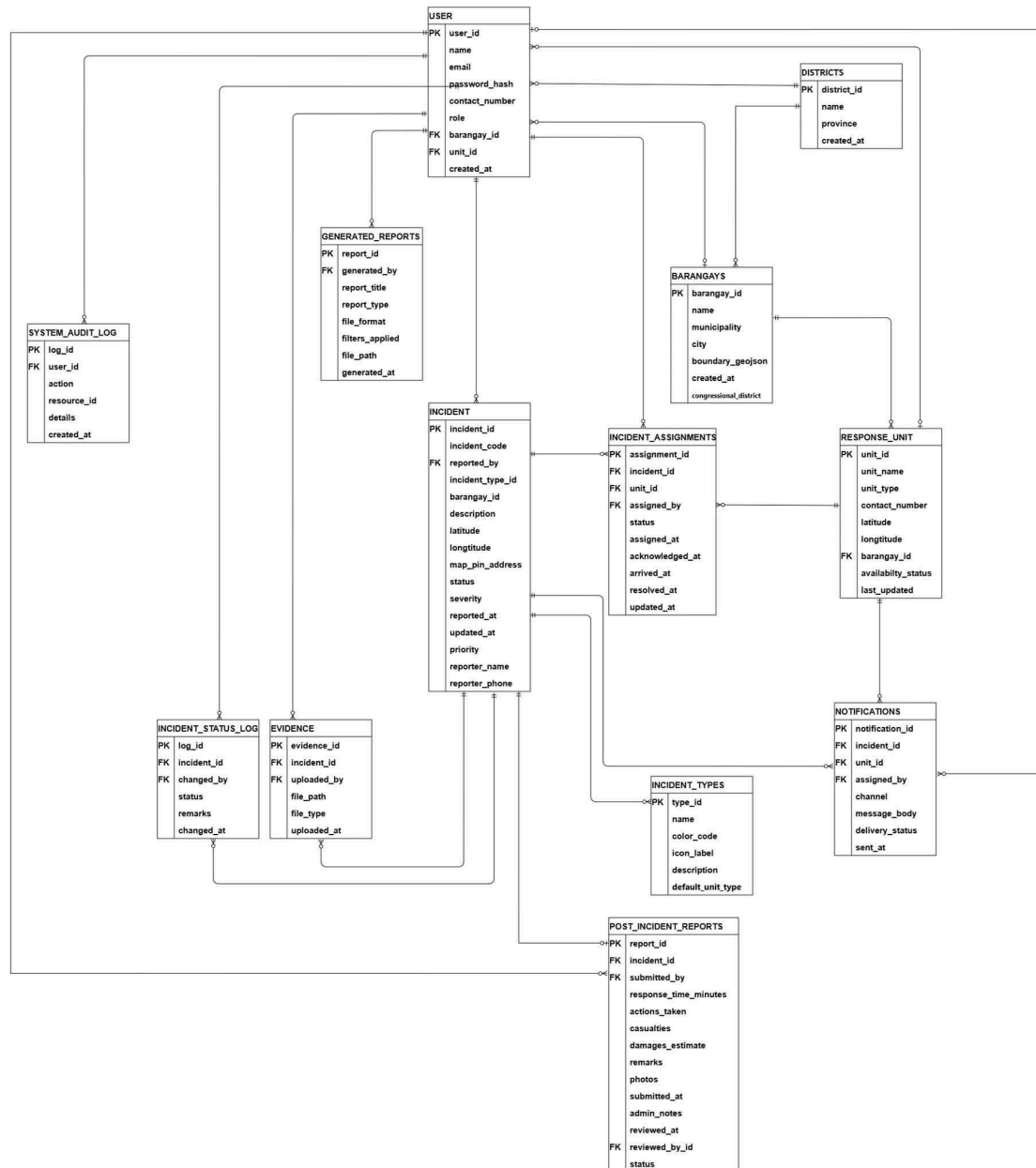


Figure 8 shows the Entity Relationship Diagram (ERD) of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration



(GAOIRS), which includes the main entities such as User, Incident, Response Unit, Incident Assignments, Barangays, Districts, Incident Types, Evidence, Incident Status Log, Notifications, Generated Reports, and Post-Incident Reports. The Incident table serves as the central entity that stores the details of reported incidents, including the incident type, location, description, status, severity, priority, and reporter information. The User table is connected to incidents, assignments, status logs, evidence, generated reports, and post-incident reports to identify the users responsible for submitting, managing, updating, or reviewing system records. The Response Unit table stores information about BFP and DRRMO units and is linked to incident assignments and notifications to support the dispatch and monitoring of response operations. The Incident Assignments table records the assignment of verified incidents to response units, including assignment status and response timestamps. The Barangays and Districts tables provide geographic reference information used to associate users, response units, and incidents with their corresponding jurisdictions. The Incident Types table stores the categories and related attributes of incidents used for classification and map visualization. The Evidence table stores photos and other files uploaded as supporting evidence for reported incidents, while the Incident Status Log records changes in incident status for monitoring and tracking purposes. The Notifications table stores messages sent to assigned response units regarding incident assignments and updates. The Generated Reports table records reports produced by authorized users, while the Post-Incident Reports table stores information about actions taken, response



time, casualties, damages, and other details recorded after an incident has been resolved. The ERD illustrates how these entities are related and how the database supports incident recording, verification, response dispatch, monitoring, documentation, and reporting within the GAOIRS.

3.5 UML Use Case Diagram

Figure 9

Use Case Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

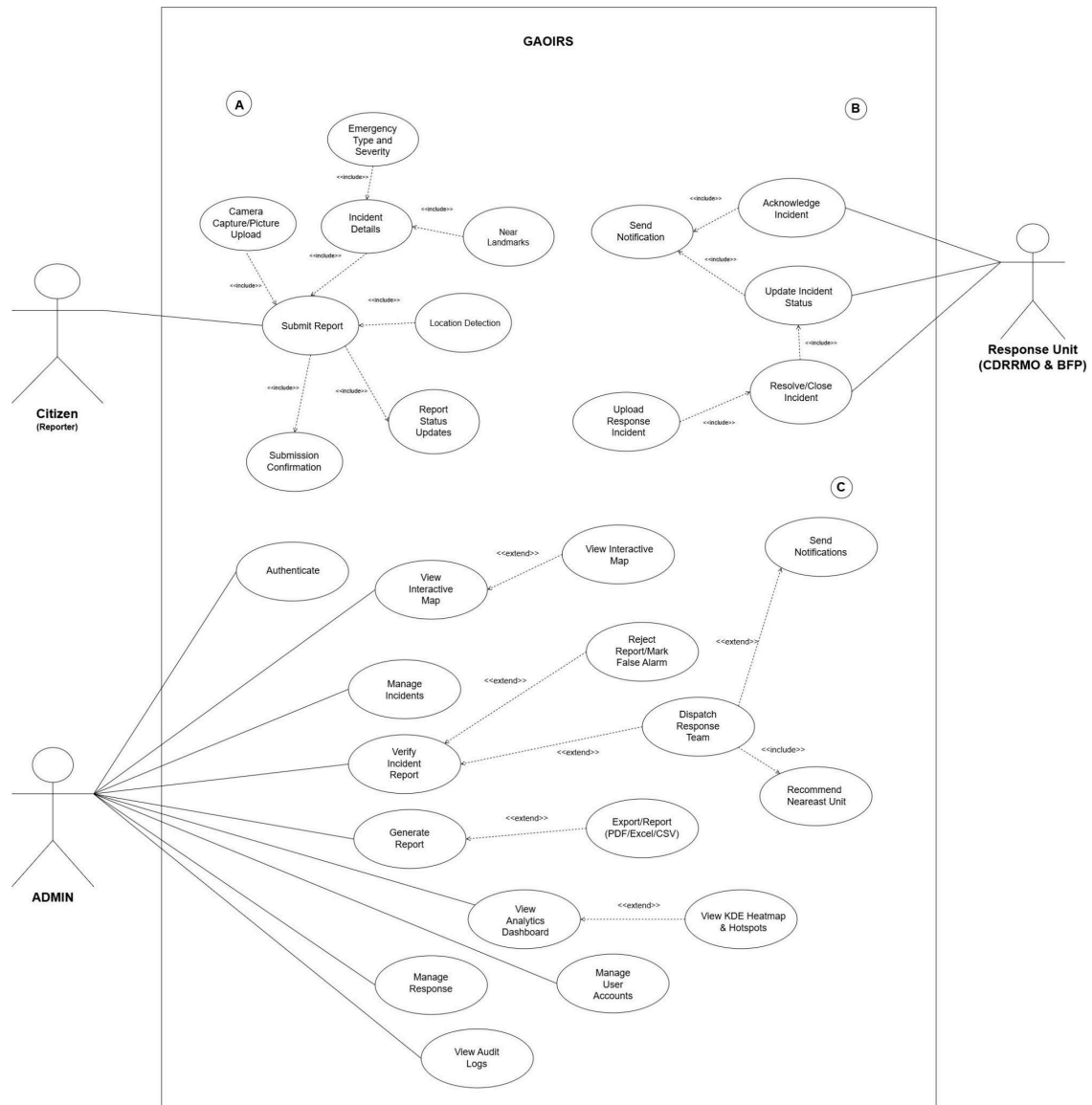


Figure 9 shows what each actor can do in the system. The Citizen can submit an incident report, select the incident type, tag the GPS location, upload photo evidence, track the report status, and receive a submission confirmation. The Response Unit can acknowledge incidents, update and resolve incident



status, view the interactive map, and upload response reports and evidence. The Administrator has the widest access: managing incidents, dispatching response teams, generating reports, viewing the analytics dashboard, managing user accounts, and configuring incident types and access roles. All actors sign in through the Authentication use case (Use Case 1). The diagram sets clear boundaries so that each function is available only to authorized roles.

3.6 Use Case Scenario

Table 3
Submit Report

Use Case 1: Submit Report
Use Case Goal: To allow a citizen to report an incident and have it queued for admin verification.
Scenario 1: Successful submission of an incident report.
Actor: Citizen (Reporter)
Pre-condition: The system is running and accessible. The citizen has the GAOIRS app open.
Main Flow:
Step 1: The citizen selects "Submit Report."
Step 2: The system automatically detects the citizen's current location (Location Detection).
Step 3: The citizen takes or uploads a photo as evidence (Camera Capture/Picture Upload).
Step 4: The citizen selects the emergency type and severity, enters a phone number, and may add the name and near landmarks (Incident Details).
Step 5: The citizen reviews the report details and submits it.
Step 6: The system generates an incident ID, queues the report for admin verification, and displays a confirmation notice (Submission Confirmation).
User Action: Selects "Submit Report," allows location detection, attaches a photo, fills in the incident details, and submits the report.
System Response: Detects the current location, generates an incident ID, queues the report for admin verification, and confirms the submission.

Table 4
Track Report Status

Use Case 2: Track Report Status
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Use Case Goal: To allow the citizen to check the current status of a report they submitted.

Scenario 1: Successful retrieval of a report's status.

Actor: Citizen (Reporter)

Pre-condition: The system is running and accessible. The citizen has submitted at least one report.

Main Flow:

Step 1: The citizen selects "Reports" / "My Report Status Tracking."

Step 2: The system retrieves and displays the current status of the report (e.g., Pending, Verified, Responding, Resolved).

User Action: Selects "Reports" / "My Report Status Tracking" and views the status of a report.

System Response: Retrieves and displays the current status of the selected report.

Table 5

Acknowledge Incident

Use Case 3: Acknowledge Incident

Use Case Goal: To allow a response unit to formally acknowledge a dispatched incident.

Scenario 1: Successful acknowledgment of a dispatched incident.

Actor: Response Unit (CDRRMO & BFP)

Pre-condition: The system is running and accessible. The response unit has been dispatched to an incident.

Main Flow:

Step 1: The response unit receives the dispatch notification with the incident details.

Step 2: The response unit views the incident details and location, and may select "Get Directions."

Step 3: The response unit selects "Acknowledge Incident."

Step 4: The system updates the incident status to "Responding."

Step 5: The system notifies the admin that the unit has acknowledged the incident (Send Notification).

User Action: Opens the dispatched incident and selects "Acknowledge Incident."

System Response: Updates the incident status to "Responding" and notifies the admin.



Table 6
Update Incident Status

Use Case 4: Update Incident Status

Use Case Goal: To allow a response unit to update the progress of an incident it is handling.

Scenario 1: Successful update of an incident's status.

Actor: Response Unit (CDRRMO & BFP)

Pre-condition: The system is running and accessible. The incident has already been acknowledged.

Main Flow:

Step 1: The response unit selects the incident to update.

Step 2: The response unit selects the new status (e.g., En Route, On Scene).

Step 3: The system saves the updated status.

Step 4: The system notifies the relevant parties of the change (Send Notification).

User Action: Selects the incident and chooses its new status.

System Response: Saves the update and notifies the relevant parties.

Table 7
Resolve/Close Incident

Use Case 5: Resolve/Close Incident

Use Case Goal: To allow a response unit to mark an incident as resolved once the response is complete.

Scenario 1: Successful resolution and closure of an incident.

Actor: Response Unit (CDRRMO & BFP)

Pre-condition: The system is running and accessible. The incident has been acknowledged and the response is complete.

Main Flow:

Step 1: The response unit selects "Resolve/Close Incident."

Step 2: The response unit confirms the closure.

Step 3: The system sets the incident status to "Resolved" (Update Incident Status).

Step 4: The system archives the incident record.

User Action: Selects "Resolve/Close Incident" and confirms the closure.

System Response: Sets the status to "Resolved" and archives the incident record.



Table 8
Upload Response Incident

Use Case 6: Upload Response Incident

Use Case Goal: To allow a response unit to upload documentation of how an incident was handled.

Scenario 1: Successful upload of response documentation.

Actor: Response Unit (CDRRMO & BFP)

Pre-condition: The system is running and accessible. The response unit has attended to the incident.

Main Flow:

Step 1: The response unit selects "Upload Response Incident."

Step 2: The response unit attaches the response report, photos, and/or notes.

Step 3: The system saves the response documentation and links it to the incident record.

Step 4: If the incident is not yet marked as resolved, the system closes it (Resolve/Close Incident).

User Action: Selects "Upload Response Incident" and attaches the response report and supporting files.

System Response: Saves and links the documentation to the incident record, and closes the incident if needed.

Table 9
Authenticate

Use Case 7: Authenticate

Use Case Goal: To allow authorized admin users to securely access the GAOIRS dashboard.

Scenario 1: Successful login and access to the dashboard.

Actor: Admin

Pre-condition: The system is running and accessible. The admin has an active registered account.

Main Flow:

Step 1: The admin enters a registered username/email and password.

Step 2: The admin clicks the Login button.

Step 3: The system validates the entered credentials against the user records in the database.

Step 4: The system determines the assigned role of the admin.

Step 5: The system creates a user session and redirects the admin to the dashboard.

User Action: Enters the username/email and password, then clicks "Login."

System Response: Validates the credentials, creates a user session, and redirects the admin to the dashboard.



Scenario 2: Login with invalid credentials.

Actor: Admin

Pre-condition: The system is running and accessible. The admin has an active registered account.

Main Flow:

Step 1: The admin enters a username/email or password that is incorrect.

Step 2: The admin clicks the Login button.

Step 3: The system finds that the credentials do not match and shows an error message.

Step 4: The system keeps the admin on the login page.

User Action: Enters incorrect credentials and clicks "Login."

System Response: Denies access and displays an error message.

Table 10

View Interactive Map

Use Case 8: View Interactive Map

Use Case Goal: To allow the admin to view real-time incident locations and response unit positions on a map.

Scenario 1: Successful viewing of the interactive map.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated.

Main Flow:

Step 1: The admin selects "View Interactive Map."

Step 2: The system retrieves the current incident and response unit locations.

Step 3: The system displays the map with the pinned incidents and units.

User Action: Selects "View Interactive Map."

System Response: Retrieves and displays the current locations of incidents and response units.

Table 11

Manage Incidents

Use Case 9: Manage Incidents

Use Case Goal: To allow the admin to view, filter, and manage all submitted incident reports.

Scenario 1: Successful viewing and filtering of incident records.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated.

Main Flow:

Step 1: The admin selects "Manage Incidents."

Step 2: The system displays the list of incident reports.

Step 3: The admin filters or searches the records as needed.

Step 4: The system displays the filtered results.

Step 5: The admin selects an incident to view its details.



User Action: Selects "Manage Incidents," applies filters or search terms, and opens an incident record.

System Response: Displays the incident reports that match the criteria and the details of the selected incident.

Table 12

Verify Incident Report

Use Case 10: Verify Incident Report

Use Case Goal: To allow the admin to review and validate a citizen-submitted report before it is acted upon.

Scenario 1: Successful verification of a report.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated. A report has been submitted and is pending verification.

Main Flow:

Step 1: The admin opens a pending report.

Step 2: The admin reviews the report details, photo, and pinned location.

Step 3: The admin confirms the report as valid.

Step 4: The system marks the report as verified and makes it ready for dispatch.

User Action: Opens and reviews a pending report, then confirms it as valid.

System Response: Marks the report as verified and makes it ready for dispatch.

Scenario 2: Rejection of a report or marking it as a false alarm (Reject Report/Mark False Alarm).

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated. A report has been submitted and is pending verification.

Main Flow:

Step 1: The admin opens a pending report.

Step 2: The admin reviews the report details, photo, and pinned location.

Step 3: The admin finds the report invalid and selects "Reject Report/Mark False Alarm."

Step 4: The system updates the report status and does not allow it to be dispatched.

User Action: Reviews a pending report and rejects it or marks it as a false alarm.

System Response: Updates the report status to rejected or false alarm.



Table 13
Dispatch Response Team

Use Case 11: Dispatch Response Team

Use Case Goal: To allow the admin to assign a response unit to a verified incident.

Scenario 1: Successful dispatch of a response unit.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated. The incident report has been verified.

Main Flow:

Step 1: The admin selects "Dispatch Response Team" for the verified incident.

Step 2: The system recommends the nearest response unit (Recommend Nearest Unit).

Step 3: The admin confirms the recommended unit or selects another unit.

Step 4: The system sends a notification to alert the assigned unit (Send Notifications).

User Action: Selects "Dispatch Response Team" and confirms the recommended or selected unit.

System Response: Recommends the nearest unit and sends a dispatch notification to the assigned unit.

Table 14
Generate Report

Use Case 12: Generate Report

Use Case Goal: To allow the admin to generate a summary report of incidents and system activity.

Scenario 1: Successful generation and export of a report.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated. Incident data exists in the system.

Main Flow:

Step 1: The admin selects "Generate Report."

Step 2: The admin sets the report parameters (e.g., date range, incident type, area).

Step 3: The system compiles the report.

Step 4: The admin exports the report as PDF, Excel, or CSV (Export/Report).

User Action: Selects "Generate Report," sets the parameters, and chooses an export format.

System Response: Compiles the report and provides the file for download.



Table 15
View Analytics Dashboard

Use Case 13: View Analytics Dashboard

Use Case Goal: To allow the admin to view incident trends and statistics for decision-making.

Scenario 1: Successful viewing of analytics and hotspot data.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated. Incident data exists in the system.

Main Flow:

Step 1: The admin selects "View Analytics Dashboard."

Step 2: The system compiles and displays charts and statistics on incident trends.

Step 3: The admin opens the KDE heatmap and hotspots for spatial analysis (View KDE Heatmap & Hotspots).

User Action: Selects "View Analytics Dashboard" and opens the heatmap view.

System Response: Displays the incident trend statistics and the KDE heatmap of hotspots.

Table 16
Manage User Accounts

Use Case 14: Manage User Accounts

Use Case Goal: To allow the admin to create, edit, or deactivate user accounts.

Scenario 1: Successful update of a user account.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated.

Main Flow:

Step 1: The admin selects "Manage User Accounts."

Step 2: The system displays the list of user accounts.

Step 3: The admin adds, edits, or deactivates an account.

Step 4: The system saves the changes.

User Action: Selects "Manage User Accounts" and makes the needed changes to an account.

System Response: Saves the updated account information.

Table 17
Manage Response

Use Case 15: Manage Response

Use Case Goal: To allow the admin to oversee response units and their assignments.

Scenario 1: Successful update of a response unit's record.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated.



Main Flow:

Step 1: The admin selects "Manage Response."

Step 2: The system displays the list of response units with their status and assignments.

Step 3: The admin updates the unit information or assignments as needed.

Step 4: The system saves the changes.

User Action: Selects "Manage Response" and updates a unit's information or assignment.

System Response: Saves the updated response unit record.

Table 18

View Audit Logs

Use Case 16: View Audit Logs

Use Case Goal: To allow the admin to review a chronological log of system and user actions for accountability.

Scenario 1: Successful viewing of the audit log.

Actor: Admin

Pre-condition: The system is running and accessible. The admin is authenticated.

Main Flow:

Step 1: The admin selects "View Audit Logs."

Step 2: The system displays the log of recorded actions.

Step 3: The admin filters the logs as needed (e.g., by user, date, or action type).

Step 4: The system displays the filtered log results.

User Action: Selects "View Audit Logs" and applies filters as needed.

System Response: Displays the log entries that match the selected criteria.

3.7 Data Dictionary

Data Dictionary furnishes detailed information about the business data, like standard definitions of data elements, their meaning, and allowable values.

Table 19

Users Table

Field Name	Data Type	Data Format	Field Size	Description	Example
user_id (PK)	uuid	Alphanumeric	36	Unique user ID	3f2a9c1e-...
name	varchar	Text	50	Full name	Juan Dela Cruz
email	varchar	Email	30	Email address used to log in	juan@email.com



password_hash	varchar	Encrypted	60	Encrypted password	*****
contact_number	varchar	Text	13	Contact number	+639171234567
role	enum	Text	N/A	User role (admin, responder, citizen)	citizen
barangay_id (FK)	uuid	Alphanumeric	36	Linked barangay (Barangays.barangay_id)	b1e2f3a4-...
unit_id (FK)	uuid	Alphanumeric	36	Linked response unit for responder accounts (Response_Unit.unit_id); empty for other roles	8a6c0f13-...
created_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Account creation date and time	2026-06-01 08:00:00

Table 20
Districts Table

Field Name	Data Type	Data Format	Field Size	Description	Example
district_id (PK)	uuid	Alphanumeric	36	Unique district ID	d4a7c2e9-...
name	varchar	Text	30	District name	District 3
province	varchar	Text	50	Province of the district	Negros Occidental
created_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the record was created	2026-06-01 08:00:00

Table 21
Barangays Table

Field Name	Data Type	Data Format	Field Size	Description	Example
barangay_id (PK)	uuid	Alphanumeric	36	Unique barangay ID	b1e2f3a4-...
name	varchar	Text	50	Barangay name	Zone 2
municipality	varchar	Text	50	Municipality name	Talisay
city	varchar	Text	50	City name	Talisay City
boundary_geojson	json	Text / JSON	N/A	Barangay boundary in GeoJSON format, used for geofence checks and map display	{"type": "Polygon", ...}
created_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the record was created	2026-06-01 08:00:00



congressional_district	varchar	Text	20	Congressional district of the barangay (relates to Districts)	District 3
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Table 22
Response_Unit Table

Field Name	Data Type	Data Format	Field Size	Description	Example
unit_id (PK)	uuid	Alphanumeric	36	Unique response unit ID	8a6c0f13-...
unit_name	varchar	Text	60	Name of the unit	Talisay Fire Station
unit_type	enum	Text	N/A	Type of unit (fire, DRRMO)	fire
contact_number	varchar	Text	13	Unit contact number	+639171234568
latitude	decimal	Numeric	9,6	Base latitude of the unit	10.744000
longitude	decimal	Numeric	9,6	Base longitude of the unit	122.967000
barangay_id (FK)	uuid	Alphanumeric	36	Barangay where the unit is based (Barangays.barangay_id)	b1e2f3a4-...
availability_status	enum	Text	N/A	Availability (available, offline, on standby)	available
last_updated	timestamp	YYYY-MM-DD HH:MM:SS	19	Last location or status update	2026-06-01 14:22:00

Table 23
Incident_Types Table

Field Name	Data Type	Data Format	Field Size	Description	Example
type_id (PK)	uuid	Alphanumeric	36	Unique incident type ID	2b7e5d90-...
name	varchar	Text	50	Incident type name	Fire
color_code	varchar	Text	7	Display color on the map (hex)	#FF4500
icon_label	varchar	Text	30	Map icon label	fire-icon
description	varchar	Text	150	Description of the incident type	Structural fire incidents



default_unit_type	enum	Text	N/A	Type of response unit normally dispatched for this incident type (fire, DRRMO)	fire
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Table 24
Incident Table

Field Name	Data Type	Data Format	Field Size	Description	Example
incident_id (PK)	uuid	Alphanumeric	36	Unique incident ID	9d4e1a77-...
incident_code	varchar	Text	15	Unique incident code	INC-2026-001
reported_by (FK)	uuid	Alphanumeric	36	User who reported the incident (Users.user_id)	3f2a9c1e-...
incident_type_id (FK)	uuid	Alphanumeric	36	Type of incident (Incident_Types.type_id)	2b7e5d90-...
barangay_id (FK)	uuid	Alphanumeric	36	Barangay where the incident occurred (Barangays.barangay_id)	b1e2f3a4-...
description	text	Text	N/A	Details of the incident	Structure fire on the second floor
latitude	decimal	Numeric	9,6	GPS latitude of the incident	10.744000
longitude	decimal	Numeric	9,6	GPS longitude of the incident	122.967000
map_pin_address	varchar	Text	150	Address taken from the map pin	Rizal St., Zone 2, Talisay City
status	enum	Text	N/A	Incident status (Awaiting Verification, Responding, On Scene, Resolved, Closed, False Alarm)	Responding
severity	enum	Text	N/A	Severity level (Low, High, Critical)	High
reported_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the incident was reported	2026-06-01 09:15:00
updated_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time of the last update	2026-06-01 09:40:00



priority	varchar	Text	10	Priority level used to order the incident in the queue	High
reporter_name	varchar	Text	50	Name of the reporter as entered on the report form	Juan Dela Cruz
reporter_phone	varchar	Text	13	Contact number of the reporter as entered on the report form	+639171234567

Table 25
Incident_Assignments Table

Field Name	Data Type	Data Format	Field Size	Description	Example
assignment_id (PK)	uuid	Alphanumeric	36	Unique assignment ID	c5a83e21-...
incident_id (FK)	uuid	Alphanumeric	36	Assigned incident (Incident.incident_id)	9d4e1a77-...
unit_id (FK)	uuid	Alphanumeric	36	Assigned response unit (Response_Unit.unit_id)	8a6c0f13-...
assigned_by (FK)	uuid	Alphanumeric	36	User who made the assignment (Users.user_id)	7c1d4b8a-...
status	enum	Text	N/A	Assignment status (assigned, acknowledged, arrived, resolved)	acknowledged
assigned_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the unit was assigned	2026-06-01 09:20:00
acknowledged_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the unit acknowledged the assignment	2026-06-01 09:22:00
arrived_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the unit arrived at the scene	2026-06-01 09:35:00
resolved_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the unit resolved the incident	2026-06-01 11:00:00
updated_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time of the last update	2026-06-01 11:00:00

Table 26
Notifications Table

Field Name	Data Type	Data Format	Field Size	Description	Example
notification_id (PK)	uuid	Alphanumeric	36	Unique notification ID	e60b7f42-...



incident_id (FK)	uuid	Alphanumeric	36	Related incident (Incident.incident_id)	9d4e1a77-...
unit_id (FK)	uuid	Alphanumeric	36	Response unit that receives the notification (Response_Unit.unit_id)	8a6c0f13-...
assigned_by (FK)	uuid	Alphanumeric	36	User who sent the notification (Users.user_id)	7c1d4b8a-...
channel	varchar	Text	10	Notification channel	sms
message_body	text	Text	N/A	Message content	New incident assigned: INC-2026-001 at Zone 2.
delivery_status	enum	Text	N/A	Delivery status (sent, delivered, failed)	delivered
sent_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the notification was sent	2026-06-01 09:21:00

Table 27
Evidence Table

Field Name	Data Type	Data Format	Field Size	Description	Example
evidence_id (PK)	uuid	Alphanumeric	36	Unique evidence ID	1f8d3c65-...
incident_id (FK)	uuid	Alphanumeric	36	Related incident (Incident.incident_id)	9d4e1a77-...
uploaded_by (FK)	uuid	Alphanumeric	36	User who uploaded the file (Users.user_id)	3f2a9c1e-...
file_path	varchar	Text	255	Location of the stored file	/uploads/incidents/INC-2026-001/photo1.jpg
file_type	varchar	Text	50	File type	image/jpeg
uploaded_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the file was uploaded	2026-06-01 10:05:00

Table 28
Incident_Status_Log Table

Field Name	Data Type	Data Format	Field Size	Description	Example
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log_id (PK)	uuid	Alphanumeric	36	Unique log ID	a92e4b08-...
incident_id (FK)	uuid	Alphanumeric	36	Related incident (Incident.incident_id)	9d4e1a77-...
changed_by (FK)	uuid	Alphanumeric	36	User who changed the status (Users.user_id)	5e9b2d47-...
status	enum	Text	N/A	New status (same as Incident.status)	Resolved
remarks	varchar	Text	150	Notes about the change	Fire extinguished
changed_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the status was changed	2026-06-01 11:00:00

Table 29
Post_Incident_Reports Table

Field Name	Data Type	Data Format	Field Size	Description	Example
report_id (PK)	uuid	Alphanumeric	36	Unique report ID	4c7a9e13-...
incident_id (FK)	uuid	Alphanumeric	36	Related incident, one report per incident (Incident.incident_id)	9d4e1a77-...
submitted_by (FK)	uuid	Alphanumeric	36	Response unit leader who submitted the report (Users.user_id)	5e9b2d47-...
response_time_minutes	int	Numeric	11	Response time in minutes	15
actions_taken	text	Text	N/A	Response actions taken	Deployed 2 fire trucks
casualties	int	Numeric	11	Number of casualties	0
damages_estimate	varchar	Text	50	Estimated cost of damage	PHP 500,000
remarks	text	Text	N/A	Additional remarks	No injuries reported
photos	json	Text / JSON	N/A	List of file paths of photos attached to the report	["/uploads/reports/r1/photo1.jpg"]



submitted_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the report was submitted	2026-06-01 12:00:00
admin_notes	text	Text	N/A	Administrator comments on the report	Verified and approved
reviewed_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the report was reviewed	2026-06-01 13:00:00
reviewed_by_id (FK)	uuid	Alphanumeric	36	Administrator who reviewed the report (Users.user_id)	7c1d4b8a-...
status	enum	Text	N/A	Review status (PENDING, UNDER_REVIEW, APPROVED, REJECTED); default PENDING	APPROVED

Table 30
Generated_Reports Table

Field Name	Data Type	Data Format	Field Size	Description	Example
report_id (PK)	uuid	Alphanumeric	36	Unique report ID	f3b16d84-...
generated_by (FK)	uuid	Alphanumeric	36	User who generated the report (Users.user_id)	7c1d4b8a-...
report_title	varchar	Text	100	Report title	Monthly Incident Summary - June 2026
report_type	enum	Text	N/A	Type of report	monthly_summary
file_format	enum	Text	N/A	File format (PDF, Excel, CSV)	PDF
filters_applied	text	Text	N/A	Filters used when generating the report	{"month":"June","barangay":"All"}
file_path	varchar	Text	255	Location of the generated file	/reports/monthly_june_2026.pdf
generated_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time the report was generated	2026-07-01 08:00:00

Table 31
System_Audit_Log Table

Field Name	Data Type	Data Format	Field Size	Description	Example
log_id (PK)	uuid	Alphanumeric	36	Unique audit log ID	06d5e2ba-...
user_id (FK)	uuid	Alphanumeric	36	User who performed the action (Users.user_id)	7c1d4b8a-...
action	varchar	Text	10	Action performed (CREATE, UPDATE, DELETE, LOGIN, LOGOUT, EXPORT)	UPDATE
resource_id	varchar	Text	36	ID of the affected record (nullable)	9d4e1a77-...
details	text	Text	N/A	Description of the change (nullable)	Status changed from Responding to Resolved
created_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time of the action	2026-06-01 11:01:00

3.8 Hardware and Software Requirements

Hardware and Software Requirements provide detailed minimum requirements needed to support the developed system's functions and executions

Recommended Minimum Hardware Requirements

- Computer: desktop or laptop with internet access
- Processor: multi-core, at least 2.0 GHz
- RAM: at least 8 GB
- Storage: at least 100 GB of free space for system files, records, evidence, and reports
- Network: stable internet, at least 10 Mbps download speed



- GPS-capable smartphone or tablet: required for capturing GPS coordinates when reporting from the field

Recommended Software Requirements

- Operating System: Windows 11 or macOS
- Front end: React.js, Vite, and Tailwind CSS
- Back end: Node.js and Express.js, with RESTful APIs and WebSocket for real-time updates
- Database: PostgreSQL
- Mapping and charts: Leaflet.js for maps, geofencing, and nearest responder detection; Chart.js for charts
- Analytics: Python (KDE)
- Hosting: Cloud hosting (Vercel for the web application)
- Version control: Git

3.9 Project Timeline

The development of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS) officially began on March 16, 2026, and is scheduled for completion on September 22, 2026, as presented in the project Gantt Chart. The project follows an Agile Development Life Cycle, covering the major stages of planning and analysis, system design, development, integration and testing, deployment and review, and documentation and closure. The timeline organizes the development activities into six major phases, with overlapping activities used where appropriate to support continuous system development and evaluation.

The Planning and Analysis Phase ran from March 16 to April 10, 2026. During this period, the proponents established the data-gathering plan and



coordinated with relevant stakeholders. Interviews and workflow observations were conducted with the DRRMO and BFP Station to understand existing incident-response processes and identify operational requirements. The proponents then conducted process gap analysis, developed the feature backlog, and validated the project scope. The completion of this phase provided the requirements and scope needed to proceed with system design.

From April 13 to May 1, 2026, the project proceeded to the System Design Phase. This phase focused on defining the overall structure and technical design of GAOIRS. The proponents prepared the module listing, context diagram, Data Flow Diagram (DFD), and Use Case Diagram to represent the system processes and user interactions. Database design was also carried out through the development of the Entity-Relationship Diagram (ERD) and Data Dictionary. In addition, interface mockups and prototype reviews were conducted to establish the intended user interface and system workflow before development began.

The Development Phase was conducted from May 4 to July 24, 2026, covering seven development sprints. The phase began with environment setup and database creation, followed by the implementation of role-based access control and account management. Subsequent development focused on incident report recording and management, incident verification, response and dispatch, geospatial mapping and geofencing, real-time monitoring and heatmap visualization, and trend analysis and analytics with forecasting. The final development sprint focused on report generation and post-incident reports, completing the major functional components of GAOIRS.



The Integration and Testing Phase ran from July 27 to August 19, 2026. The developed modules were first integrated and the system environment was configured to ensure that the different components could operate together. White-box testing was then conducted to examine the back-end logic and functions, while black-box testing evaluated the system from the user side on a per-module basis. Identified defects were addressed through remediation and regression retesting to verify that corrections did not introduce new problems and that the integrated system continued to function as intended.

The Deployment and Review Phase took place from August 17 to September 11, 2026. During this stage, GAOIRS was deployed to a hosted environment for evaluation. IT expert evaluation was conducted using the ISO/IEC 25010:2011 software quality model, while end-user evaluation was conducted to assess the system from the perspective of its intended users. The results of these evaluations were subjected to statistical analysis, after which the proponents reviewed the feedback and performed the necessary final system revisions.

The final stage, the Documentation and Closure Phase, began on September 1 and continued through September 22, 2026. This phase involved the preparation of the User's Guide, followed by final manuscript revisions and the completion of the required appendices. The documentation activities were scheduled alongside the final stages of system evaluation and revision, allowing the project documentation to incorporate the finalized system and evaluation



results. The project timeline concludes with the completion of the final manuscript and appendices on September 22, 2026.

Overall, the GAOIRS project timeline provides a structured progression from planning and requirements gathering, through system design and development, to testing, deployment, evaluation, and final documentation. The overlapping activities shown in the Gantt Chart support the Agile approach by allowing related development and review activities to progress within the same project period while maintaining defined deliverables for each phase. This timeline serves as the schedule for the systematic development, validation, and completion of the GAOIRS project.

3.10 System Evaluation Tools

White-Box Testing

For the White-Box Testing, Jest with Node.js was used to evaluate the internal logic and backend functionality of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS). This testing approach focused on examining the behavior of individual functions, data processing, input validation, authentication, incident management, and database-related operations by executing unit tests that directly assessed the system's code behavior. Jest was used to test backend functions and verify whether they produced the expected results under valid, invalid, and boundary input conditions. The testing covered core processes such as user authentication, incident creation and management, report verification, incident



assignment, status updates, and other backend operations. Error-handling and validation conditions were also tested to identify logical errors and ensure that the system responded correctly to unexpected or invalid inputs. Overall, White-Box Testing allowed the researchers to examine the internal program logic, identify coding errors, and verify that the backend functions of GAOIRS operated according to the intended system requirements.

Black-Box Testing

For the Black-Box Testing, Playwright and Qase was used to evaluate GAOIRS from the user's perspective without examining the internal source code. Playwright was used to automate browser-based test scenarios and simulate actual user interactions with the system, while Qase was used to organize, document, and manage the test cases and their corresponding results. The testing covered major system functions such as user authentication, citizen incident reporting, report verification, incident assignment, geospatial mapping, incident monitoring, response status updates, report generation, and other role-specific functionalities. The test scenarios were designed to verify whether the system produced the expected results based on different user inputs and actions. Each test case recorded the expected behavior, actual behavior, and test result to determine whether the corresponding function passed or failed. Overall, Black-Box Testing allowed the researchers to assess the functionality, usability of system interactions, and correct behavior of GAOIRS under actual user scenarios.

System Quality and Usability Instruments

Software quality was evaluated with a questionnaire based on the ISO/IEC 25010:2011 model (International Organization for Standardization, 2011), which has eight characteristics: functional suitability (whether the system has the needed functions and gives correct results), performance efficiency (response time and use of resources), compatibility (working with other systems and environments), usability (ease of learning and use), reliability (consistent performance), security (access control and data protection), maintainability (ease of modification), and portability (working on different devices and environments). Usability was evaluated with a questionnaire based on ISO 9241-11:2018 (ISO, 2018), which defines effectiveness (whether users can complete tasks accurately), efficiency (the effort and time needed), and satisfaction (comfort and acceptability of use). Both instruments used a five-point Likert scale. The findings from these evaluations were used to determine whether the system met user expectations and effectively supported the completion of tasks in an efficient manner. (See Appendix A for reference)

Table 32

Likert Scale, Range, and Verbal Interpretation for the ISO/IEC 25010:2011 and ISO 9241-11:2018 Evaluations

Response	Scale	Range	Verbal Interpretation
Strongly Agree	5	4.20 – 5.00	Very High
Agree	4	3.40 – 4.19	High
Undecided	3	2.60 – 3.39	Average
Disagree	2	1.80 – 2.59	Low
Strongly Disagree	1	1.00 – 1.79	Very Low



The ISO/IEC 25010:2011 Software Quality Model was used to evaluate the overall quality of the developed Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS). The model evaluates software quality through characteristics including Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. In this study, the respondents rated the system using a five-point Likert scale ranging from Strongly Disagree to Strongly Agree. The responses were assigned numerical values from 1 to 5, with higher scores indicating a more favorable assessment of the system. The computed mean scores were interpreted using the corresponding verbal interpretation shown in the table above. This evaluation was used to determine the extent to which GAOIRS met the defined software quality criteria.

The ISO 9241-based usability evaluation was used to assess the usability of the developed GAOIRS from the perspective of its intended users. The evaluation focused on how effectively, efficiently, and satisfactorily users were able to interact with and accomplish tasks using the system. In this study, respondents rated the usability statements using a five-point Likert scale ranging from Strongly Disagree to Strongly Agree. Each response was assigned a numerical value from 1 to 5, and the resulting mean scores were interpreted according to the ranges and verbal interpretations presented in the table. The evaluation was used to determine the perceived usability of GAOIRS in terms of ease of use, understanding, interaction, and user satisfaction.



3.11 Respondents of the Study

The respondents were chosen because of their direct involvement in incident response or their knowledge of software. Five IT experts evaluated the quality of the system using ISO/IEC 25010:2011. Twenty-five end users, made up of four Response Unit personnel and twenty-one citizens, evaluated its usability using ISO 9241-11:2018.

Table 33
Respondents of the Study

Respondent	Frequency	Percent
IT Experts	5	16.7%
Response Unit Personnel	4	13.3%
Citizens	21	70.0%
Total	30	100.0%

The study involved a carefully selected group of 30 respondents who participated in the evaluation of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS). The respondents included IT experts who assessed the system's technical quality and functionality, Response Unit personnel composed of two BFP personnel and two DRRMO personnel who evaluated the system's incident monitoring, verification, dispatch, and response-related functions, and citizen respondents who assessed the incident reporting and status-tracking features from the perspective of the intended reporting users. The respondents were selected based on their relevant involvement with information systems, disaster incident response, or citizen



incident reporting, allowing the researchers to obtain feedback from both technical evaluators and actual or intended system users.

3.12 Ethical Consideration

The researchers prepared formal request letters and sought permission from the appropriate offices and intended participants before conducting interviews, workflow observations, system testing, and evaluation activities for the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS). The purpose, scope, objectives, and procedures of the study were clearly explained to the participating personnel and respondents, particularly those involved in the Disaster Risk Reduction and Management Office (DRRMO) and Bureau of Fire Protection (BFP). Participation in the study was conducted voluntarily, and respondents were given sufficient information about their involvement before data collection and system evaluation activities were conducted.

The researchers strictly maintained the confidentiality and privacy of information obtained throughout the study. Since GAOIRS involves incident reports that may contain location information, descriptions, photographs, and other potentially sensitive details, collected information was handled carefully and was accessed only by authorized persons involved in the study. Personal information and respondent-related information obtained during interviews, observations, and system evaluations were not disclosed to unauthorized individuals and were used only for purposes directly related to the academic



research and development of GAOIRS. Appropriate measures were observed to protect collected data from unauthorized access, alteration, disclosure, or misuse. Access to system information and evaluation results was limited according to the researchers' authorized roles. The researchers also ensured that information presented in the study, including incident-related data, system evaluation responses, and other collected materials, was handled in a manner that protects the identity and privacy of the individuals involved.

In compliance with the Data Privacy Act of 2012 (Republic Act No. 10173), informed consent was obtained from participants before collecting information from them. Participants were informed about the purpose of data collection, the manner in which their information would be used, and the measures implemented to protect their information. The researchers respected the participants' right to participate voluntarily and ensured that information collected during the study was used only for legitimate academic and system-development purposes. The researchers also maintained responsible and accurate handling of incident information throughout the development and evaluation of GAOIRS. Incident data were treated as information requiring verification before being used for response-related functions. This is consistent with the system's design, where citizen-submitted incident reports undergo verification before being dispatched to an appropriate response unit.

Finally, the researchers observed ethical standards throughout the planning, development, testing, deployment, and evaluation of GAOIRS. Regular checks were undertaken to ensure that the collection, processing, storage, and



presentation of information remained consistent with the agreed research purpose, confidentiality requirements, and applicable data-protection principles. These measures were implemented to protect the rights and privacy of participants and to promote responsible research conduct throughout the study.



CHAPTER IV

RESULTS AND DISCUSSION

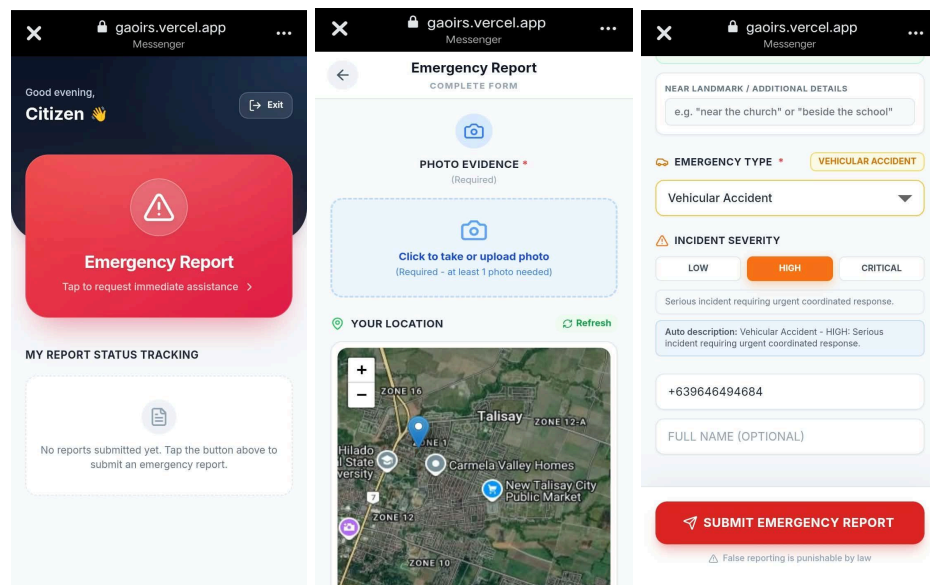
This chapter presents the results of the study based on the objectives of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS). It includes the developed system's modules and interfaces, the results of system testing, and the evaluation results conducted using the ISO/IEC 25010:2011 Software Quality Model and ISO 9241-11:2018 usability measures. The chapter presents the different functionalities of GAOIRS, including incident report recording and management, incident verification and response dispatch, geospatial mapping and geofencing, real-time monitoring and heatmap visualization, trend analysis and forecasting, report generation, and role-based access control. The system evaluation involved five (5) IT experts who assessed the technical quality and functionality of GAOIRS based on the eight quality characteristics of ISO/IEC 25010:2011. Twenty-five (25) end users, consisting of four (4) Response Unit personnel and twenty-one (21) citizens, evaluated the system based on effectiveness, efficiency, and satisfaction using ISO 9241-11:2018. Responses were gathered using a five-point Likert scale, ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). This chapter also discusses the findings obtained from the system testing and evaluation. The results provide an overview of the system's functionality, technical quality, usability, and user experience. Furthermore, the discussion interprets the results in relation to the objectives of the study, identifies the system's observed strengths and areas requiring improvement, and presents the implications of the findings for the use and further enhancement of GAOIRS.

4.1 Results of the Study

The first objective was to design and develop GAOIRS with five features: incident report recording and management, incident response, a geospatial mapping interface, monitoring and analysis, and security and report generation. The result was a working system with all five features. It records and manages incident reports, coordinates and tracks the response, shows incidents on an interactive map, provides trend analysis and a seven-day forecast, and enforces role-based security with report generation. The following figures show each feature.

Figure 10

Incident Report Recording Interface



The figure displays three sequential screenshots of the GAOIRS mobile application interface, illustrating the Incident Report Recording process.

Screenshot 1 (Left): Shows the home screen with a dark blue header. A red button labeled "Emergency Report" with a white warning icon is prominent. Below it, a section titled "MY REPORT STATUS TRACKING" indicates "No reports submitted yet. Tap the button above to submit an emergency report."

Screenshot 2 (Middle): Shows the "Emergency Report" form. It includes a "PHOTO EVIDENCE" section with a camera icon and a note "(Required - at least 1 photo needed)". Below this is a "YOUR LOCATION" section featuring a map of Talisay with a blue location pin and a "Refresh" button.

Screenshot 3 (Right): Shows the "COMPLETE FORM" section. It includes fields for "NEAR LANDMARK / ADDITIONAL DETAILS" (e.g., "near the church" or "beside the school"), "EMERGENCY TYPE" (selected as "VEHICULAR ACCIDENT"), "INCIDENT SEVERITY" (selected as "HIGH"), "Auto description: Vehicular Accident - HIGH: Serious incident requiring urgent coordinated response.", a phone number field (+639646494684), and a "FULL NAME (OPTIONAL)" field. A red "SUBMIT EMERGENCY REPORT" button is at the bottom, with a disclaimer "False reporting is punishable by law" below it.

Figure 10 shows the Incident Report Recording interface, the entry point for reports. The form has fields for photo evidence (up to five photos), the incident location with automatic GPS detection and a map pin that can be



adjusted, the emergency type, the severity (Low, High, or Critical), and the reporter's name and contact number. The description is filled in from the selected type and severity. A warning that false reporting is punishable by law is shown near the Submit button. After submission, the report gets a unique incident code and is saved to the central database for verification, mapping, and monitoring.

Figure 11
Incident Management Interface

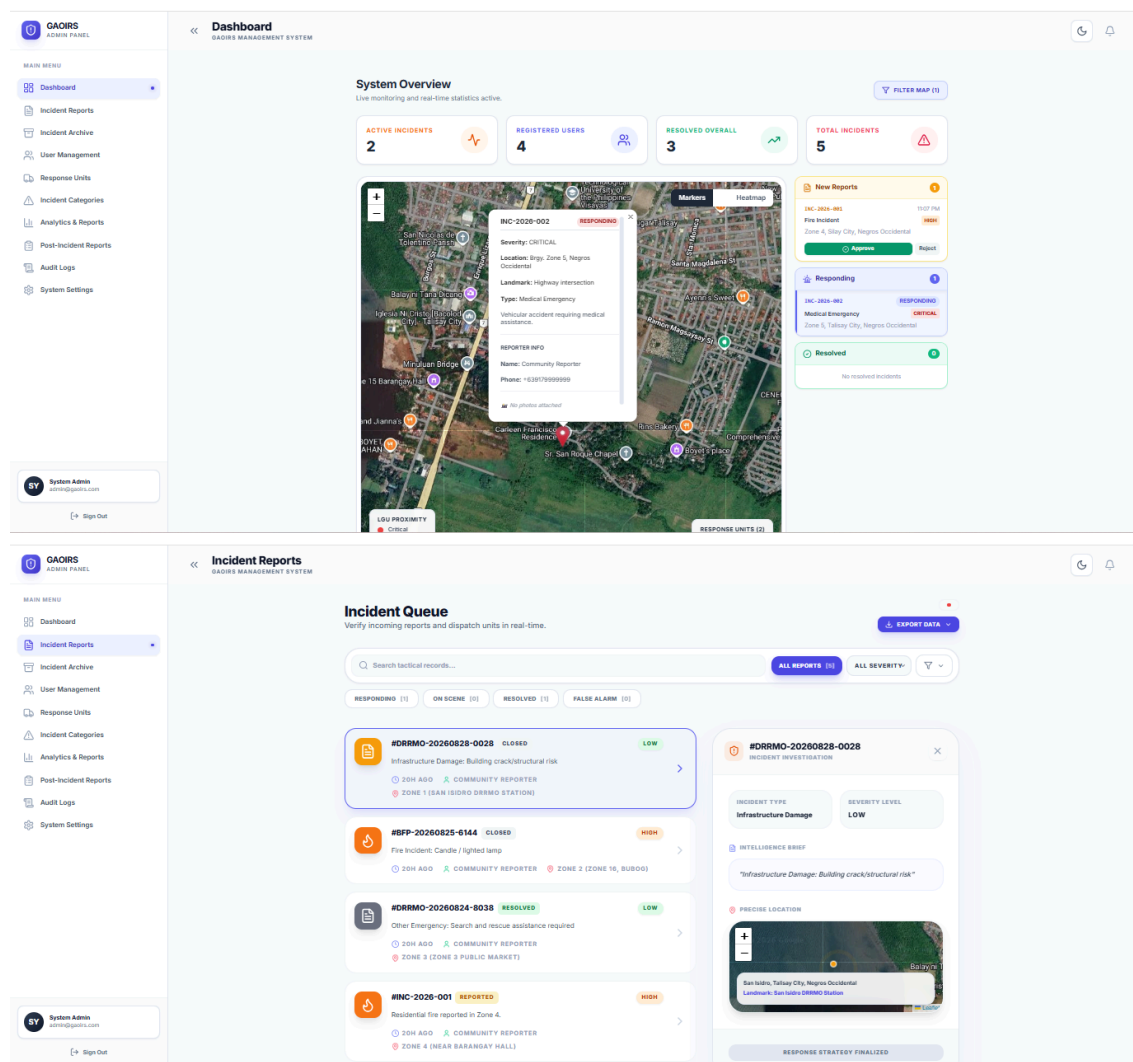
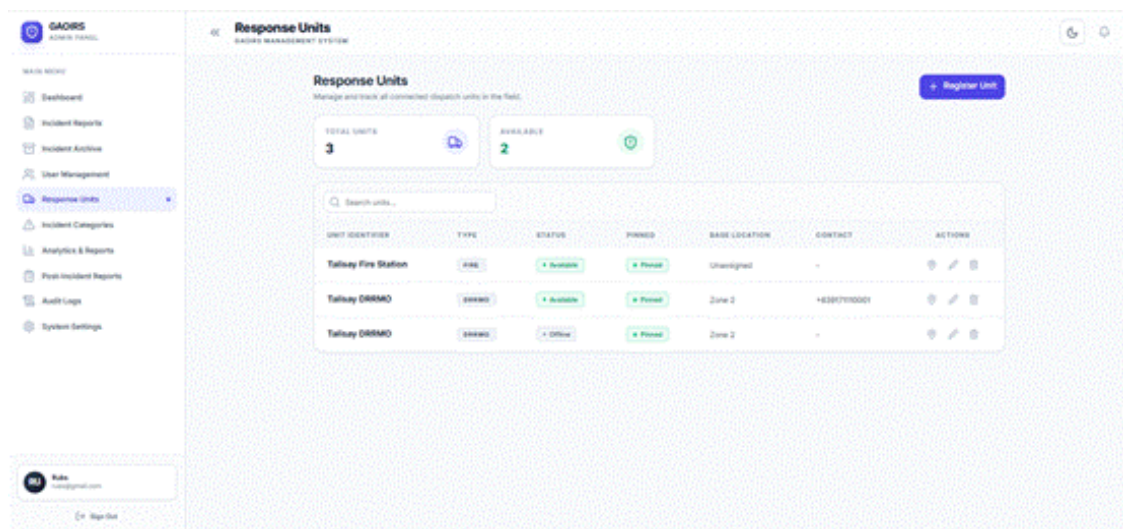


Figure 11 shows the Incident Queue, where administrators review, verify, update, and track reports. The page has a search bar, status tabs (Responding, On Scene, Resolved, and False Alarm), and a severity filter. Each report card shows the incident code, type, severity, and status, and the selected report opens a panel with the incident details, a brief description, and a map of the precise location. An export button lets users download the records.

Figure 12
Incident Response Interface



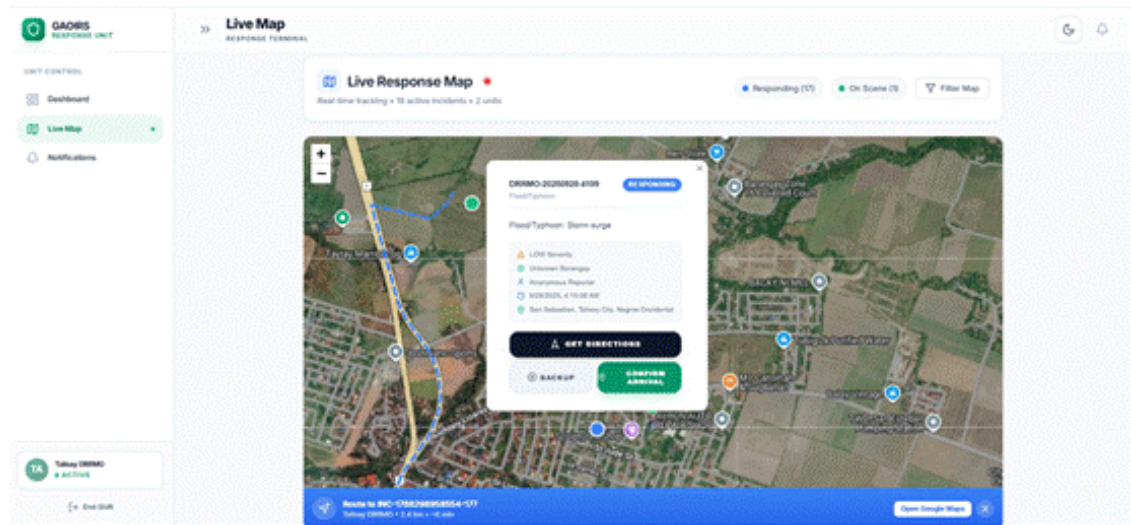


Figure 12 shows the Incident Response module. The upper screen is the Response Units page, where administrators see the registered units, their type, availability, base location, and contact details, and can register, edit, or remove a unit. The lower screen is the Live Response Map used by response units. It shows the units and the active incidents, counts of responding and on-scene incidents, and a pop-up with the incident details and options to get directions, request backup, and confirm arrival. Together, they support assigning responders and tracking the response until the incident is resolved.

Figure 13
Geospatial Mapping Interface

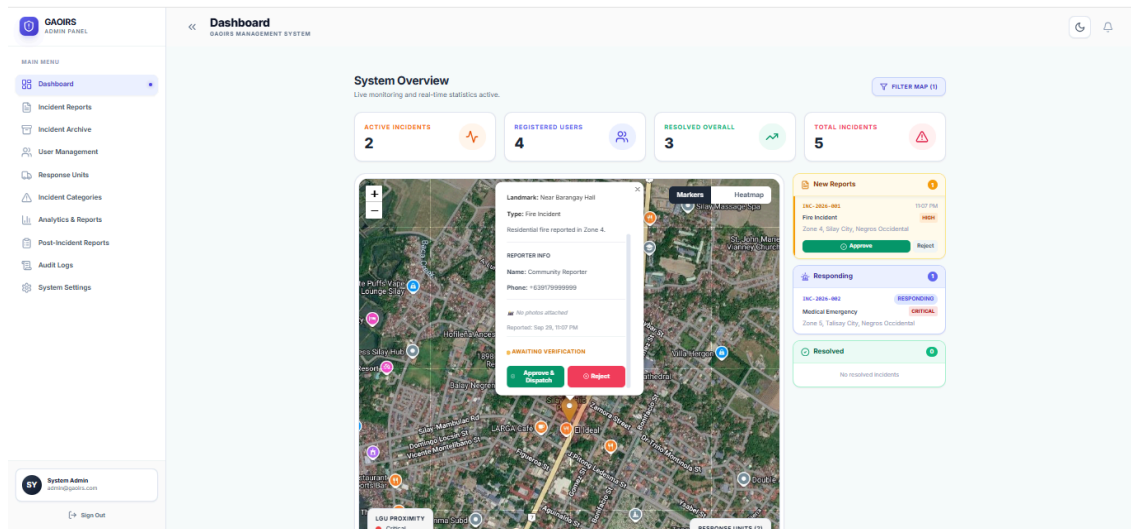


Figure 13 shows the Geospatial Mapping interface on the administrator dashboard. Summary cards show the active incidents, registered users, resolved incidents, and total incidents. The map plots incidents with markers colored by severity and can switch between a marker view and a heatmap view. Selecting a marker shows the incident details, the reporter information, and the evidence photo, with buttons to approve and dispatch or reject the report. The side list groups incidents by new, responding, and reported. The figures on the cards come from the test and demonstration data used in the study.

Figure 14
Monitoring and Analysis Dashboard

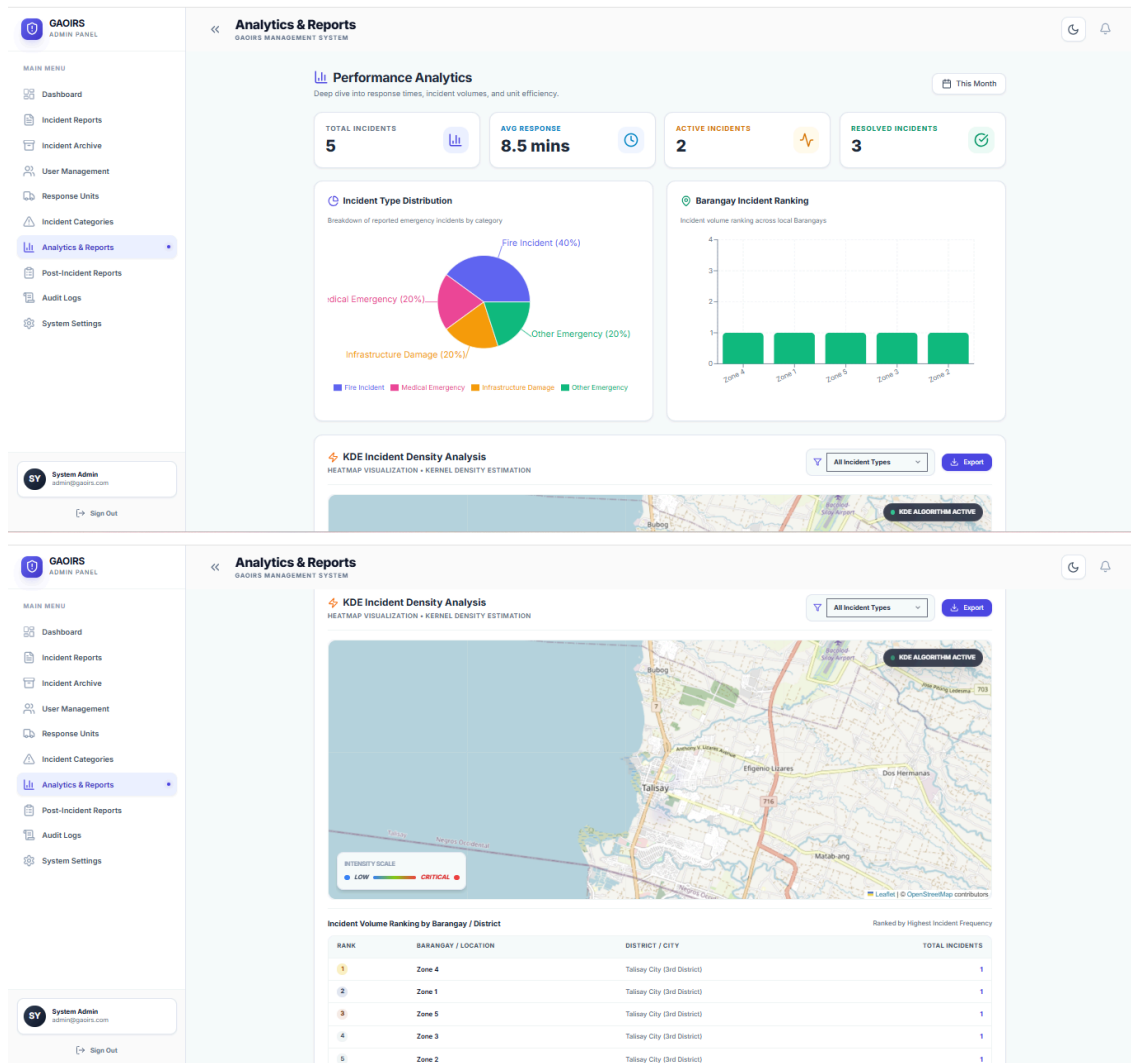
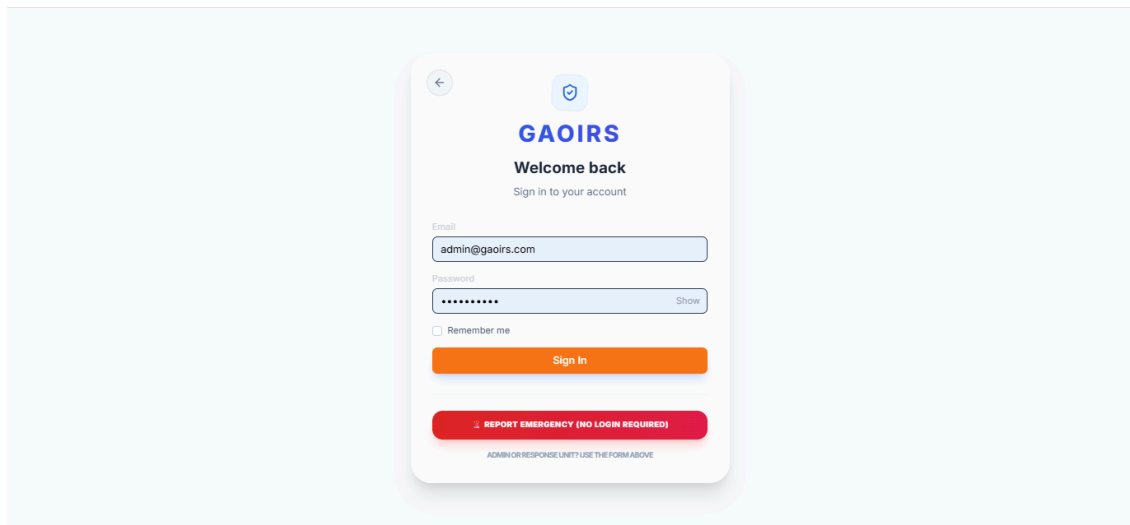


Figure 14 shows the Monitoring and Analysis module. The upper screen shows key indicators (total incidents, average response time, active incidents, and resolved incidents), the distribution of incidents by type, the ranking of barangays by number of incidents, and the seven-day forecast produced by the Prophet model with the average predicted incidents, the trend, and the peak and low days. The lower screen shows the kernel density estimation (KDE) heatmap of incidents, with intensity from low to critical. These help decision-makers see

patterns and plan resources. The values shown come from the synthetic demonstration data described in Section 3.8.

Figure 15

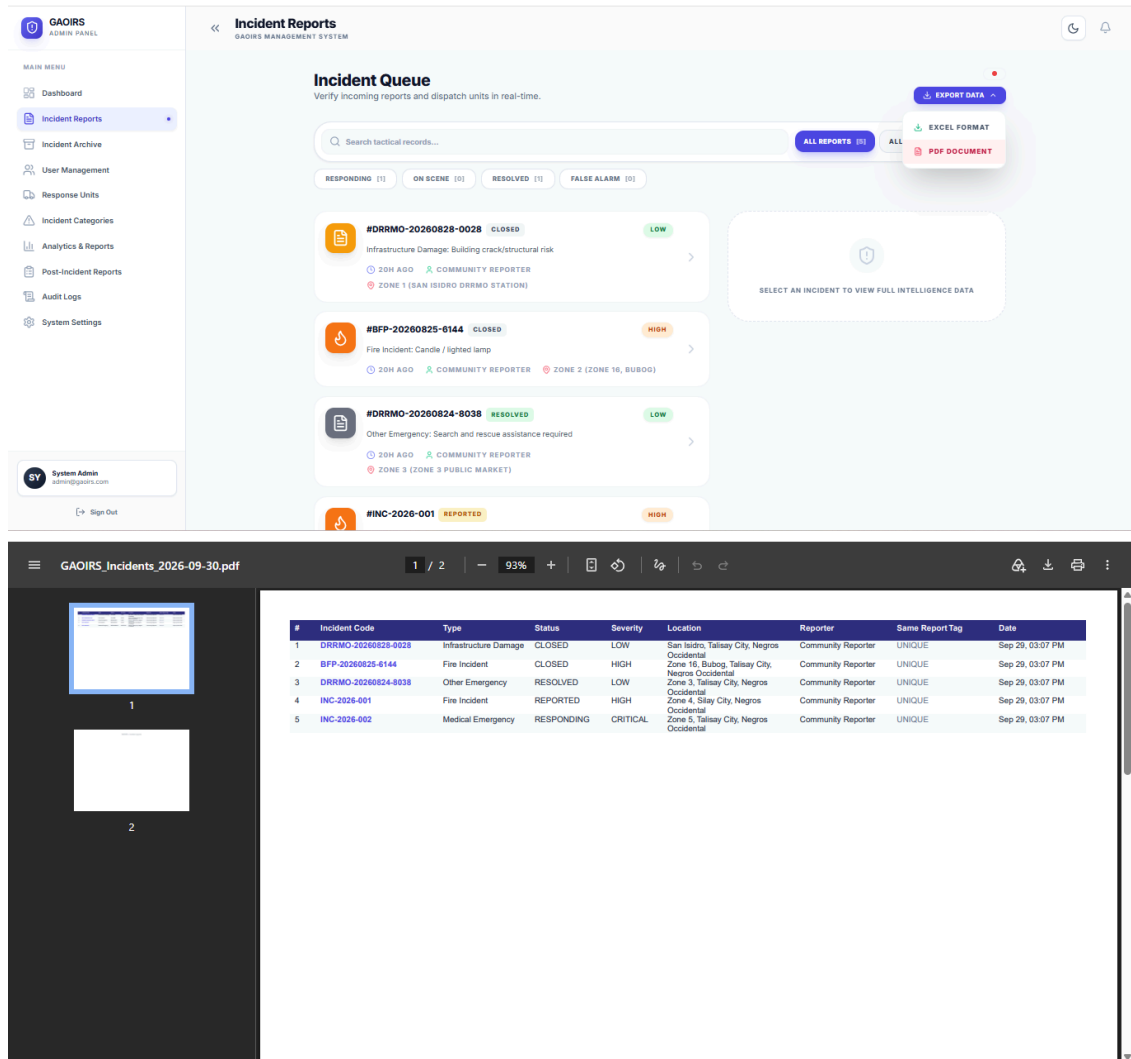
Login and Role-Based Access Interface



The screenshot displays a mobile application login screen for GAOIRS. At the top, there is a back arrow icon and a shield icon. The text "GAOIRS" is prominently displayed in blue, followed by "Welcome back" and "Sign in to your account". Below this, there are input fields for "Email" (containing "admin@gaoirs.com") and "Password" (masked with dots). A "Show" button is next to the password field. There is a checkbox for "Remember me". Below the input fields is an orange "Sign In" button. At the bottom, there is a red button labeled "REPORT EMERGENCY (NO LOGIN REQUIRED)". A small note at the very bottom reads "ADMIN OR RESPONSE UNIT? USE THE FORM ABOVE".

Figure 15 shows the login page of the Security module. Registered users sign in with an email and password, and the system opens the dashboard for their role. Unregistered citizens can use the link for reporting an emergency without logging in. Role-based access control keeps administrator and response unit functions available only to authorized users, which protects the confidentiality and integrity of the incident data.

Figure 16
Report Generation and Export Interface



The screenshot displays the GAOIRS (GADIS Management System) Incident Reports interface. The top section, titled "Incident Queue", shows a list of incidents with filters for "RESPONDING (1)", "ON SCENE (0)", "RESOLVED (1)", and "FALSE ALARM (0)". The incidents listed are:

- #DRRMO-20260828-0028: CLOSED, LOW severity. Infrastructure Damage: Building crack/structural risk. 20H AGO, COMMUNITY REPORTER, ZONE 1 (SAN ISIDRO DRRMO STATION).
- #BFP-20260825-0144: CLOSED, HIGH severity. Fire Incident: Candle / lighted lamp. 20H AGO, COMMUNITY REPORTER, ZONE 2 (ZONE 16, BUBOG).
- #DRRMO-20260824-8038: RESOLVED, LOW severity. Other Emergency: Search and rescue assistance required. 20H AGO, COMMUNITY REPORTER, ZONE 3 (ZONE 3 PUBLIC MARKET).
- #INC-2026-001: REPORTED, HIGH severity.

The bottom section shows a PDF viewer for "GAOIRS_Incidents_2026-09-30.pdf" with a table of archived incidents:

#	Incident Code	Type	Status	Severity	Location	Reporter	Same Report Tag	Date
1	DRRMO-20260828-0028	Infrastructure Damage	CLOSED	LOW	San Isidro, Talisay City, Negros Occidental	Community Reporter	UNIQUE	Sep 29, 03:07 PM
2	BFP-20260825-0144	Fire Incident	CLOSED	HIGH	Zone 16, Bubog, Talisay City, Negros Occidental	Community Reporter	UNIQUE	Sep 29, 03:07 PM
3	DRRMO-20260824-8038	Other Emergency	RESOLVED	LOW	Zone 3, Talisay City, Negros Occidental	Community Reporter	UNIQUE	Sep 29, 03:07 PM
4	INC-2026-001	Fire Incident	REPORTED	HIGH	Zone 4, Silay City, Negros Occidental	Community Reporter	UNIQUE	Sep 29, 03:07 PM
5	INC-2026-002	Medical Emergency	RESPONDING	CRITICAL	Zone 5, Talisay City, Negros Occidental	Community Reporter	UNIQUE	Sep 29, 03:07 PM

Figure 16 shows the Incident Archive page, which supports report generation. It shows the total archived, resolved, closed, and false alarm incidents, a searchable table of archived incidents, and an Export menu with Excel, CSV, and PDF options. Authorized users can use it to produce records for documentation, monitoring, and administrative work.

The second objective was to test the functionality of the features. The system went through alpha testing by the researchers and IT experts, using black-box



and white-box tests, and beta testing by end users, namely Response Unit personnel and citizens, who were able to sign in, submit and track reports, and view mapped incidents. The following tables show the alpha test results. In the tables, Pass means the actual outcome matched the expected outcome, Fail means it did not, and Retest means the outcome could not be fully confirmed.

Black-box testing checked each module from the user's side, without looking at the code.

Table 34

Black-Box Testing of the Alpha Testing of Incident Report Recording and Management

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-SubRep-001	Report form visibility test	All report submission form elements (incident type, location, photo upload, submit button) should be visible on page load.	All expected form elements were visible and properly rendered.	Pass	No issues encountered.
TC-SubRep-002	Incident type selection functionality	Citizen should be able to select an incident type from the list and have the selection reflected in the form.	The selected incident type was captured and reflected correctly in the report.	Pass	Selection worked as intended.
TC-SubRep-003	Automatic location detection accuracy	System should automatically detect and capture the citizen's current location.	The current location was detected and captured accurately.	Pass	Detection worked as intended.



TC-SubRep-004	Photo attachment upload functionality	Citizen should be able to attach a photo as evidence of the incident.	The photo was successfully attached and previewed before submission.	Pass	Upload worked as intended.
TC-SubRep-005	Successful end-to-end report submission	System should save the completed report once all required fields are filled.	The report was saved successfully with all provided details intact.	Pass	Full submission flow worked as intended.
TC-SubRep-006	Submission confirmation and reference number generation	System should display a confirmation message with a unique tracking/reference number.	A confirmation message with a valid, unique reference number was displayed.	Pass	Confirmation flow worked as intended.
TC-SubRep-007	Required field validation	System should prevent submission and display an error when incident type, location, or photo is missing.	The system blocked submission and displayed the correct error prompts for each missing field.	Pass	Validation works as intended; incomplete submissions properly rejected.
TC-SubRep-008	Fallback to manual map pin when GPS fails	System should allow the citizen to manually pin the location when automatic detection is unavailable or inaccurate.	The manual pinning option was correctly offered and the pinned location was captured.	Pass	Fallback flow worked as intended.
TC-TrackStat-001	Status tracking form visibility test	The Track Report Status search field and results area should be visible on page load.	All expected elements were visible and functioning.	Pass	No issues encountered.



TC-TrackStat-002	Successful retrieval by valid reference number	System should display the correct, current status of the selected report.	The system retrieved and displayed the correct status for the tested report.	Pass	Status displayed matched the actual record.
TC-TrackStat-003	Invalid reference number handling	System should display an appropriate message when an invalid or non-existent reference number is entered.	The system correctly displayed a "report not found" message for invalid references.	Pass	Error handling worked as intended.
TC-TrackStat-004	Status accuracy across report lifecycle	The displayed status should correctly reflect each stage of the report's lifecycle (Pending, Verified, Dispatched, Resolved, Rejected).	The status label updated correctly at each stage as the report progressed.	Pass	Status accuracy confirmed across all states.
TC-MgIncId-001	Incident list display test	The list of submitted incident reports should be visible and correctly populated on page load.	All incident records were displayed correctly.	Pass	No issues encountered.
TC-MgIncId-002	Filter by status functionality	System should display only incidents matching the selected status filter.	The filtered results correctly matched the selected status.	Pass	Status filter worked as intended.
TC-MgIncId-003	Filter by date functionality	System should display only incidents within the selected date range.	The filtered results correctly matched the selected date range.	Pass	Date filter worked as intended.



TC-MglIncid-004	Search function accuracy	System should return incidents matching the entered search keyword.	Search results correctly matched the entered keyword.	Pass	Search function worked as intended.
TC-MglIncid-005	Empty result handling	System should display an appropriate message when no records match the applied filter or search.	The system correctly displayed a "no records found" message.	Pass	Handled empty results appropriately.
TC-VerifRpt-001	Pending report review interface test	Report details, photo, and pinned location should be visible when a pending report is opened.	All report details were displayed correctly.	Pass	No issues encountered.
TC-VerifRpt-002	Successful verification and automatic dispatch trigger	System should mark the report as Verified and proceed to Dispatch Response Team.	The report status changed to Verified and the dispatch process was triggered.	Pass	Verification and dispatch trigger worked as intended.
TC-VerifRpt-003	Report rejection / mark false alarm handling	System should allow the admin to reject a report and mark it as a false alarm.	The report status correctly changed to Rejected/False Alarm.	Pass	Rejection flow worked as intended.
TC-VerifRpt-004	Duplicate report flagging	System should flag or group reports that appear to be duplicates of an existing incident.	Duplicate reports submitted for the same incident were correctly flagged for the admin's review.	Pass	Duplicate detection worked as intended.
TC-VerifRpt-005	Verification permission restricted to Admin role	Only users with Admin credentials should be	Non-admin accounts were correctly denied access	Pass	Access control worked as intended.



able to verify
or reject
incident
reports.

to the
verification
function.

The results in Table 34 demonstrate the functionality of the Incident Report Recording and Management module of GAOIRS. The test cases covered the visibility and operation of the incident reporting form, incident type selection, automatic location detection, photo attachment, report submission, confirmation and reference number generation, required field validation, and manual map pinning when GPS detection is unavailable. The testing also covered incident tracking, incident list display, filtering, searching, report verification, and related incident management functions. The results show that the major user-facing functions for recording, submitting, tracking, and managing incident reports were tested according to their expected behavior.

Table 35
Black-Box Testing of the Alpha Testing of Incident Response

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AckIncid-001	Incident notification display test	Dispatch notification and incident details should be visible to the response unit upon assignment.	Notification and details were displayed correctly.	Pass	No issues encountered.
TC-AckIncid-002	Successful acknowledgment and status update	Incident status should update to Acknowledged once the response unit acknowledges it.	The status updated correctly to Acknowledged.	Pass	Acknowledgment flow worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AckIncid-003	Admin notification on acknowledgment	The admin should receive a notification once the response unit acknowledges the incident.	The admin received the acknowledgment notification correctly.	Pass	Notification worked as intended.
TC-AckIncid-004	Duplicate acknowledgment prevention	System should prevent an already-acknowledged incident from being acknowledged again.	The system correctly blocked the duplicate action.	Pass	Validation works as intended.
TC-UpdStat-001	Status options visibility test	Status options (e.g., En Route, On Scene) should be visible for an acknowledged incident.	All status options displayed correctly.	Pass	No issues encountered.
TC-UpdStat-002	Successful status update and real-time reflection	The status should update and reflect in the admin's view in real time.	The status change was saved and reflected on the admin dashboard immediately.	Pass	Update flow worked as intended.
TC-UpdStat-003	Notification sent on status change	Relevant parties should be notified when the incident status changes.	A notification was correctly sent upon each status change.	Pass	Notification worked as intended.
TC-UpdStat-004	Status update blocked before acknowledgment	System should prevent a status update on an incident that has not yet been acknowledged.	The system correctly blocked the update and displayed a message.	Pass	Validation works as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-ResClo-001	Close incident option visibility test	The Resolve/Close Incident option should be visible for an ongoing incident.	Option displayed correctly.	Pass	No issues encountered.
TC-ResClo-002	Successful resolution and archiving	Incident status should change to Resolved and the record archived.	The incident was marked Resolved and archived correctly.	Pass	Closure flow worked as intended.
TC-ResClo-003	Confirmation prompt before closure	System should require confirmation before finalizing closure.	A confirmation prompt was correctly displayed before closure.	Pass	Confirmation step worked as intended.
TC-ResClo-004	Closure blocked without a prior response update	System should prevent closure of an incident that has not been acknowledged or updated.	The system blocked closure and prompted the response unit to update the status first.	Pass	Validation works as intended.
TC-UpldResp-001	Upload form visibility test	File attachment and notes fields should be visible when uploading a response.	All fields displayed correctly.	Pass	No issues encountered.
TC-UpldResp-002	Successful documentation upload and linkage	Documentation should be saved and linked to the correct incident record.	Uploaded files and notes were saved and correctly linked.	Pass	Upload flow worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-UpldResp-003	Automatic closure trigger when incident not yet resolved	System should automatically resolve/close the incident if it is not yet marked closed when documentation is uploaded.	The incident was correctly closed as part of the upload process.	Pass	Automatic closure worked as intended.
TC-UpldResp-004	Unsupported file type rejection	System should reject unsupported file types with an appropriate message.	The system correctly rejected an unsupported file and displayed an error.	Pass	Validation works as intended.
TC-UpldResp-005	File size limit validation	System should reject files that exceed the maximum allowed size.	The system correctly rejected an oversized file and displayed an error.	Pass	File size validation worked as intended.
TC-Dispa-001	Dispatch interface visibility test	The Dispatch Response Team option and unit recommendations should be visible for a verified incident.	All elements displayed correctly.	Pass	No issues encountered.
TC-Dispa-002	Nearest-unit recommendation accuracy	System should recommend the nearest available response unit based on the incident location.	The recommended unit correctly matched the closest available unit to the incident.	Pass	Recommendation accuracy confirmed.
TC-Dispa-003	Successful dispatch and unit assignment	The selected unit should be assigned to the incident once confirmed by the admin.	The unit was correctly assigned to the incident.	Pass	Dispatch flow worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Dispa-004	Dispatch notification delivery	The assigned unit should receive a dispatch notification with the incident details.	The unit received the notification with the correct incident details.	Pass	Notification delivery worked as intended.
TC-Dispa-005	No available unit handling	System should display an appropriate message when no response unit is available.	The system correctly displayed a "no unit available" message.	Pass	Handled correctly.
TC-Dispa-006	Manual override of recommended unit	Admin should be able to manually select a different unit instead of the recommended one.	The admin was able to override the recommendation and assign a different unit.	Pass	Manual override worked as intended.
TC-Avail-001	Availability form visibility test	Availability options (Available, Busy, Off-duty) should be visible to the logged-in response unit.	Options displayed correctly.	Pass	No issues encountered.
TC-Avail-002	Successful availability update	The unit's availability should update immediately upon selection.	Availability was updated and saved correctly.	Pass	Update flow worked as intended.
TC-Avail-003	Status persistence across sessions	The declared status should remain saved after logging out and back in.	The status persisted correctly across sessions.	Pass	Persistence worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Avail-004	Availability reflected in dispatch recommendations	The declared availability should be factored into the nearest-unit recommendation during dispatch.	Only units marked Available were included in dispatch recommendations.	Pass	Integration with dispatch worked as intended.

The results in Table 35 demonstrate the functionality of the Incident Response module of GAOIRS. The test cases covered the display of incident notifications, acknowledgment of assigned incidents, notification of administrators, prevention of duplicate acknowledgments, response status updates, real-time status reflection, status-change notifications, and restrictions on updating incidents before acknowledgment. The testing also included incident resolution and closure, response documentation uploads, file validation, dispatch interface visibility, and responder availability. These results show that the module supports the major response activities from receiving an assigned incident through updating, documenting, and closing the response.

Table 36
Black-Box Testing of the Alpha Testing of Geospatial Interface

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-ViewMap-001	Map load and display test	The interactive map with pinned incidents and units should load and display correctly.	The map loaded and displayed all pins correctly.	Pass	No issues encountered.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-ViewMap-002	City tagging accuracy	Each pin should be correctly tagged with its corresponding city.	All pins displayed the correct city information.	Pass	City tagging worked as intended.
TC-ViewMap-003	Congressional district tagging accuracy	Each pin should be correctly tagged with its corresponding congressional district.	All pins displayed the correct congressional district information.	Pass	District tagging worked as intended.
TC-ViewMap-004	Real-time refresh of incidents/units on map	The map should update in real time as new incidents are submitted or resolved.	The map correctly refreshed to reflect new and resolved incidents.	Pass	Real-time update worked as intended.
TC-ViewMap-005	Map marker click for incident details	Clicking a map marker should display the corresponding incident's details.	Incident details were correctly displayed when a marker was clicked.	Pass	Marker interaction worked as intended.
TC-MapPin-001	Pin tool visibility test	The Pin Location tool should be visible and accessible on the report form.	The tool was visible and accessible.	Pass	No issues encountered.
TC-MapPin-002	Successful manual pin placement and coordinate capture	The system should capture the correct coordinates and city/district and attach them to the report.	Coordinates and location details were correctly captured and attached.	Pass	Pinning flow worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-MapPin-003	Pin outside service area warning	System should display a warning when the pinned location is outside the covered service area.	The system correctly displayed a warning message.	Pass	Validation works as intended.
TC-MapPin-004	Pin adjustment/re-pin functionality	Citizen should be able to move or reset the pin before submitting the report.	The pin was successfully repositioned and the updated coordinates were captured.	Pass	Re-pin functionality worked as intended.

The results in Table 36 demonstrate the functionality of the Geospatial Interface of GAOIRS. The test cases covered map loading and display, city and congressional district tagging, real-time updating of incidents and response units, and viewing incident details through map markers. The testing also examined the Pin Location tool, manual pin placement, coordinate capture, validation of locations outside the service area, and repositioning of map pins before report submission. The results show that the geospatial features provided the expected location-based functions for displaying incidents and capturing accurate incident locations.



Table 37
Black-Box Testing of the Alpha Testing of Monitoring and Analysis

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Analyt-001	Dashboard display test	Charts, statistics, and the KDE heatmap should be visible on page load.	All dashboard elements displayed correctly.	Pass	No issues encountered.
TC-Analyt-002	Trend chart data accuracy	The displayed trend charts should accurately reflect the current incident data.	The chart figures matched the underlying incident records.	Pass	Data accuracy confirmed.
TC-Analyt-003	KDE heatmap and hotspot accuracy	The KDE heatmap should accurately reflect the spatial density and hotspots of incidents.	The heatmap correctly highlighted the known incident clusters.	Pass	Heatmap accuracy confirmed.
TC-Analyt-004	No-data handling	System should display an appropriate message when there is insufficient data to generate analytics.	The system correctly displayed a "no data available" message.	Pass	Handled correctly.
TC-Analyt-005	Forecast/trend analysis display accuracy	The forecast/trend analysis (Prophet, SARIMA, Moving Average) should display projected incident trends based on historical data.	The forecast output was displayed and aligned with the expected trend direction of the test data.	Pass	Forecast display worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AuLog-001	Audit log display test	The chronological log of recorded actions should be visible on page load.	The audit log displayed correctly.	Pass	No issues encountered.
TC-AuLog-002	Filter by user functionality	System should display only log entries related to the selected user.	Filtered results correctly matched the selected user.	Pass	User filter worked as intended.
TC-AuditLog003	Filter by date functionality	System should display only log entries within the selected date range.	Filtered results correctly matched the selected date range.	Pass	Date filter worked as intended.
TC-AuLog-004	Filter by action type functionality	System should display only log entries matching the selected action type.	Filtered results correctly matched the selected action type.	Pass	Action type filter worked as intended.
TC-Au-Log-005	Log entry accuracy check	Each log entry should accurately capture the corresponding system or user action.	All tested log entries matched the actions performed.	Pass	Accuracy confirmed.
TC-MgResp-001	Response unit list display test	The list of response units with their status and assignments should be visible on page load.	All records displayed correctly.	Pass	No issues encountered.
TC-MgResp-002	Successful assignment update	Response unit records should be accurately updated when assignments are changed.	Updates were saved and correctly reflected.	Pass	Update flow worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-MgResp-003	Status/availability visibility accuracy	Each unit's current status and availability should be accurately displayed.	Status and availability displayed matched the actual records.	Pass	Display accuracy confirmed.
TC-MgResp-004	Unauthorized edit prevention	System should prevent non-admin users from editing response unit records.	The system correctly blocked unauthorized edit attempts.	Pass	Access control worked as intended.

The results in Table 37 demonstrate the functionality of the Monitoring and Analysis module of GAOIRS. The test cases covered dashboard display, trend chart accuracy, Kernel Density Estimation (KDE) heatmap and hotspot visualization, handling of insufficient data, and forecast and trend analysis. The testing also included audit log display and filtering, log entry accuracy, response unit information, assignment updates, availability status, and prevention of unauthorized modifications. These results show that the module provides the expected functions for monitoring incident information, analyzing incident patterns, viewing audit records, and managing response unit information.



Table 38

Black-Box Testing of the Alpha Testing of Report Generation

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-GenRpt-001	Report parameters form visibility test	Report parameter fields (date range, incident type, area) should be visible on page load.	All fields displayed correctly.	Pass	No issues encountered.
TC-GenRpt-002	Successful report compilation	System should compile an accurate report based on the selected parameters.	The compiled report data matched the selected parameters.	Pass	Compilation accuracy confirmed.
TC-GenRpt-003	PDF export functionality	System should export the compiled report as a correctly formatted PDF file.	The report downloaded successfully as a properly formatted PDF.	Pass	PDF export worked as intended (Export/Report extension).
TC-GenRpt-004	Excel export functionality	System should export the compiled report as a correctly formatted Excel file.	The report downloaded successfully as a properly formatted Excel file.	Pass	Excel export worked as intended (Export/Report extension).
TC-GenRpt-005	CSV export functionality	System should export the compiled report as a correctly formatted CSV file.	The report downloaded successfully as a properly formatted CSV file.	Pass	CSV export worked as intended (Export/Report extension).
TC-GenRpt-006	Invalid parameter/date range handling	System should display an error when an invalid date range or parameter is entered.	The system correctly displayed an error for invalid parameters.	Pass	Validation works as intended.



The results in Table 38 demonstrate the functionality of the Report Generation module of GAOIRS. The test cases covered the visibility of report parameters, compilation of reports based on selected criteria, and export of generated reports into PDF, Excel, and CSV formats. The testing also verified the system's handling of invalid parameters and date ranges. The results show that the report generation functions were able to compile and export incident information according to the selected parameters while applying appropriate validation to invalid inputs.

Table 39

White-Box Testing of the Alpha Testing of Incident Report Recording and Management

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W012	getIncidents(): paginated incident list query (page 1, limit 10)	Formats the paginated array response.	Returned a data array with total=1 and page=1 pagination object.	Pass	No issues encountered.
TC-W013	getIncidentById(): non-existent incident ID query	Returns 404 Not Found error.	Returned HTTP 404 Not Found status.	Pass	No issues encountered.
TC-W014	verifyIncident(): non-admin verification block (REPORTER, APPROVE)	Blocks non-admin verification with 403.	Returned HTTP 403 Forbidden with "Only admins can verify incidents" message.	Pass	No issues encountered.
TC-W015	verifyIncident(): invalid verification action	Rejects invalid action with 400 error.	Returned HTTP 400 Bad Request with "Must be APPROVE, REJECT, or REQUEST_IN FO" message.	Pass	No issues encountered.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W016	editIncident(): non-admin incident edit guard (RESPONSE_UNIT)	Blocks non-admin editing with 403.	Returned HTTP 403 Forbidden with "Only admins can edit incidents" message.	Pass	No issues encountered.
TC-W018	uploadIncident Photo(): missing file guard validation	Returns 400 Bad Request ("No file uploaded").	Returned 400 Bad Request ("No file uploaded").	Pass	No issues encountered.
TC-W019	uploadIncident Photo(): photo upload URL generator (fire.png)	Returns 200 OK with the Supabase cloud bucket URL.	Returned 200 OK with the Supabase cloud bucket URL.	Pass	No issues encountered.
TC-W020	extractPublicId(): storage path string parser	Extracts the relative storage path string correctly.	Extracted the relative storage path string correctly.	Pass	No issues encountered.
TC-W021	uploadBufferTo Cloudinary(): null image buffer fallback	Returns null safely without crashing the server.	Returned null safely without crashing the server.	Pass	No issues encountered.
TC-W024	generateIncidentCode(): first incident zero-padded code (count = 0)	Generates INC-YYYY-001 format.	Formatted code string "INC-2026-001" with zero-padding.	Pass	No issues encountered.
TC-W025	generateIncidentCode(): incremental code generator (count = 42)	Generates incremented zero-padded code.	Incremented sequence to "INC-2026-043".	Pass	No issues encountered.

The results in Table 39 demonstrate the internal functionality of the Incident Report Recording and Management module of GAOIRS. The test cases examined backend functions for retrieving paginated incident records, accessing incident details, restricting incident verification and editing to authorized users, validating verification actions, handling missing photo uploads, generating photo upload URLs, processing storage paths, and safely handling invalid image buffers. The testing also verified the generation of unique incident codes. The



results show that the tested backend functions and validation mechanisms operated according to their expected behavior.

Table 40
White-Box Testing of the Alpha Testing of Incident Response

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W003	generateToken(): responder unit token signing (unit_id "unit-01")	Encodes response unit identity into the token payload.	Encoded unit_id "unit-01" into the decoded JWT object.	Pass	No issues encountered.
TC-W009	requireResponseUnit(): response unit permission pass (RESPONSE_UNIT role)	Allows request progression via next().	Invoked the next() middleware handler.	Pass	No issues encountered.
TC-W017	updateIncidentStatus(): responder status update (RESPONDING)	Updates incident status to RESPONDING.	Updated incident status to "RESPONDING" with HTTP 200 OK.	Pass	No issues encountered.

The results in Table 40 demonstrate the internal functionality of the Incident Response module of GAOIRS. The test cases examined the generation of response-unit authentication tokens, enforcement of response-unit permissions through middleware, and updating of incident response status. These tests verified whether the backend correctly handled authenticated response-unit requests and processed status changes as intended. The results indicate that the tested internal functions supported the required response-unit authentication, authorization, and incident status operations.

Table 41
White-Box Testing of the Alpha Testing of Geospatial Interface

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W011	createIncident() : NIR geofence boundary check (Manila coordinates)	Rejects the report with a 400 error.	Returned HTTP 400 Bad Request with "Location Out of Coverage Area" message.	Pass	No issues encountered.
TC-W022	getAll(): barangay query pagination (page 1, limit 10)	Calculates total pages accurately.	Returned a 2-record array with totalPages=1 calculation.	Pass	No issues encountered.
TC-W023	create(): barangay spatial record creation (San Jose, Talisay)	Returns 201 Created with the new record.	Returned HTTP 201 Created with generated barangay_id "b3".	Pass	No issues encountered.

The results in Table 41 demonstrate the internal functionality of the Geospatial Interface of GAOIRS. The test cases examined the validation of incident locations against the Negros Island Region coverage boundary, pagination of barangay records, and creation of barangay spatial records. These tests verified that the backend correctly handled geographic coverage validation and the retrieval and creation of spatial information. The results show that the tested geospatial functions produced the expected responses and supported the location-based operations of the system.

Table 42
White-Box Testing of the Alpha Testing of Monitoring and Analysis

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W026	logAuditEvent(): event payload action formatting (LOGIN)	Logs the audit event payload safely.	Stored audit record object with action "LOGIN".	Pass	No issues encountered.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W027	listAuditLogs(): paginated audit logs query (page 1, limit 25)	Returns 200 OK with paginated logs.	Returned HTTP 200 OK with logs array and total=1 metadata.	Pass	No issues encountered.
TC-W028	listDistinctActions(): distinct action strings list	Returns an array of action strings.	Returned 3 distinct action strings including "CREATE_REPORT".	Pass	No issues encountered.

The results in Table 42 demonstrate the internal functionality of the Monitoring and Analysis module of GAOIRS. The test cases examined the recording of audit events, retrieval of paginated audit logs, and generation of distinct action types from the stored logs. These tests verified that system activities could be recorded and retrieved correctly through the backend functions. The results indicate that the tested monitoring and audit-log functions operated according to their expected behavior.

Table 43
White-Box Testing of the Alpha Testing of Report Generation

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W036	getPagination(): default query parameter handling	Returns default page 1, limit 10, skip 0.	Returned page=1, limit=10, skip=0 default values.	Pass	No issues encountered.
TC-W037	getPagination(): skip offset calculation (page 3, limit 15)	Calculates the skip offset accurately.	Calculated skip offset of 30 $((3-1)*15)$ accurately.	Pass	No issues encountered.
TC-W038	getPagination(): negative page boundary guard (page -5)	Enforces minimum page 1 and skip 0.	Enforced minimum page=1 and skip=0.	Pass	No issues encountered.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W039	getPagination() : maximum limit cap protection (limit 500)	Caps the maximum limit to 100 items.	Capped the limit value to the maximum of 100 items.	Pass	Initial run failed (test expected 500); passed on second run after enforcing the 100-item ceiling.
TC-W040	buildPagination Meta(): total pages calculation (page 2, limit 10, total 25)	Calculates totalPages metadata correctly.	Calculated totalPages=3 using Math.ceil(25/10).	Pass	No issues encountered.

The results in Table 43 demonstrate the internal functionality of the Report Generation component of GAOIRS. The test cases examined the handling of default pagination parameters, calculation of pagination offsets, enforcement of page boundaries, maximum-limit protection, and calculation of total pages. The results show that the backend correctly handled pagination values and applied the required limits to ensure that report-related records could be retrieved in an organized and controlled manner. One pagination limit case required an initial correction before passing on the succeeding test.

Table 44

White-Box Testing of the Alpha Testing of System Access, Security, and Utilities

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W001	verifyToken(): missing authorization header	Returns 401 error and blocks request progression.	Returned HTTP 401 Unauthorized with "Missing or invalid authorization header" message.	Pass	No issues encountered.
TC-W002	generateToken(): citizen JWT token generation	Produces a valid signed JWT string containing citizen claims.	Generated a valid signed JWT string with the citizen user payload.	Pass	No issues encountered.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W004	generateToken(): minimal admin payload token	Sets optional unit and barangay fields to null safely.	Assigned null to the optional unit_id and barangay_id fields.	Pass	No issues encountered.
TC-W005	verifyToken(): valid Bearer token verification	Attaches the user object to the request context.	Attached the user payload to req.user and executed next().	Pass	No issues encountered.
TC-W006	requireRole(): undefined user object guard	Returns 401 Unauthorized error.	Returned HTTP 401 Unauthorized status.	Pass	No issues encountered.
TC-W007	requireAdmin(): role mismatch guard check (REPORTER role)	Returns 403 Forbidden error.	Returned HTTP 403 Forbidden status.	Pass	No issues encountered.
TC-W008	requireAdmin(): admin access permission pass (ADMIN role)	Allows request progression via next().	Invoked the next() middleware handler.	Pass	No issues encountered.
TC-W010	requireReporter(): reporter permission pass (REPORTER role)	Allows request progression via next().	Invoked the next() middleware handler.	Pass	No issues encountered.
TC-W029	encrypt(): plain text payload encryption	Produces an encrypted iv:ciphertext string.	Generated a hex iv:ciphertext string with colon separator.	Pass	No issues encountered.
TC-W030	decrypt(): encrypted text payload decryption	Restores the original plaintext string.	Decrypted text back to the exact original string.	Pass	No issues encountered.
TC-W031	decrypt(): invalid format decryption fallback	Returns null safely on invalid format.	Returned null cleanly without throwing an application crash exception.	Pass	No issues encountered.
TC-W032	success(): default success response payload	Formats the default success structure.	Returned { success: true, data: null, message: "Success" }.	Pass	No issues encountered.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-W033	success(): custom data payload formatting	Attaches the custom payload to the success response.	Returned success=true with the custom data payload attached.	Pass	No issues encountered.
TC-W034	error(): default error response payload	Formats the default error structure.	Returned { success: false, error: null, message: "Request failed" }.	Pass	No issues encountered.
TC-W035	error(): custom error payload response	Attaches the error payload to the response.	Returned success=false with the custom error payload attached.	Pass	No issues encountered.

The results in Table 45 demonstrate the internal functionality of the System Access, Security, and Utilities component of GAOIRS. The test cases examined authorization-token verification, JWT generation, access control, security-related functions, and other supporting backend utilities. These tests were conducted to verify that authentication information was properly processed and that unauthorized requests were appropriately restricted. The results show that the tested security and utility functions performed according to their expected behavior.

Table 54

ISO/IEC 25010:2011 Software Quality Evaluation Results (n = 5 IT Experts)

Criteria	Mean Score	Verbal Interpretation
Functional Suitability	4.23	Very High
Performance Efficiency	4.10	High
Compatibility	4.25	Very High
Usability	4.53	Very High



Reliability	4.35	Very High
Security	4.42	Very High
Maintainability	4.20	Very High
Portability	4.67	Very High
Overall Average Mean	4.43	Very High

Table 54 shows that the system obtained an overall average mean of 4.34, interpreted as Very High, based on the evaluation of five (5) IT experts using the ISO/IEC 25010:2011 Software Quality Model. Among the eight evaluated characteristics, Portability received the highest mean score of 4.67, while Performance Efficiency received the lowest mean score of 4.10, interpreted as High. The remaining seven characteristics were interpreted as Very High. Overall, the results indicate that the IT experts gave a Very High assessment of the software quality of GAOIRS.

The third objective also included evaluating usability using ISO 9241-11:2018, which covers effectiveness, efficiency, and satisfaction. The evaluation was answered by twenty-five (25) end users, made up of four (4) Response Unit personnel and twenty-one (21) citizens.

Table 55

ISO 9241-11:2018 Usability Evaluation Results (n = 25 End Users)

Measure	Mean Score	Verbal Interpretation
Effectiveness	4.42	Very High
Efficiency	4.54	Very High
Satisfaction	4.58	Very High



Overall Average Mean	4.51	Very High
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Table 55 shows that the system obtained an overall average mean of 4.51, interpreted as Very High, based on the evaluation of twenty-five (25) end users consisting of four (4) Response Unit personnel and twenty-one (21) citizens using ISO 9241-11:2018. Among the three evaluated measures, Satisfaction received the highest mean score of 4.58, followed by Efficiency with 4.54, while Effectiveness received the lowest mean score of 4.42. All three measures were interpreted as Very High. Overall, the results indicate that the end users gave a Very High assessment of the usability of GAOIRS in terms of effectiveness, efficiency, and satisfaction.

Taken together, the two evaluations show that GAOIRS was well received by IT experts and end users. They also point to performance efficiency and response time as areas for further improvement.

The fourth objective was to develop a User's Guide for the system. The guide was developed and explains how the system works for each user role: Response Unit personnel are guided in recording and responding to incidents, and Administrators in managing user accounts, monitoring system data, and generating reports. The complete guide is in the appendices.

Figure 17

User's Guide Cover Page



Figure 17 shows the cover page of the User's Guide for GAOIRS. The guide serves as a reference for the intended users in understanding and navigating the system's features and functionalities, including incident reporting, incident response, system monitoring, and report generation.



CHAPTER V

CONCLUSION AND RECOMMENDATION

This chapter presents the conclusions drawn from the development, testing, and evaluation of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS). It consolidates the key results of the study and discusses how the developed system addressed the major challenges in incident reporting and response, particularly delayed reporting, uncoordinated response, and limited access to timely spatial and analytical information for NDRRMC-catered incidents in the Negros Island Region. The findings highlight the system's contribution to improving incident reporting and management, coordinating response activities, providing geospatial visualization, and supporting monitoring and data-driven analysis. Furthermore, this chapter includes practical recommendations based on the study's outcomes to sustain and enhance the system's implementation.

5.1 Summary of Findings

The study's overall results showed that all objectives were successfully achieved. The researchers integrated the technical features outlined in the first objective into the system: incident report recording and management, incident response, geospatial mapping with identification of the city and congressional district of incidents, monitoring and analysis, and report generation. They tested its functionality through alpha testing using both Black-box and White-box methods, with all 78 black-box and all 40 white-box test cases passing, and through beta testing by Response Unit personnel and citizens, thereby fulfilling



the second objective. Regarding the third objective, the system's quality was measured through an ISO standard questionnaire assessing the eight characteristics of ISO/IEC 25010:2011. IT experts gave an average score of 4.34, interpreted as 'Very High,' indicating high quality from the experts' perspective. Additionally, following ISO 9241-11:2018, end users gave mean scores of 4.42 in effectiveness, 4.54 in efficiency, and 4.58 in satisfaction, with an overall mean of 4.51, all interpreted as 'Very High.' To help users learn to operate the system effectively, the researchers also developed a User's Guide that clearly explains GAOIRS's capabilities and how it should be properly used by Response Unit personnel and Administrators, thereby fulfilling the fourth objective.

5.2 Conclusions

This study aimed to address the critical challenges faced by the City Disaster Risk Reduction and Management Office (CDRRMO) and the Bureau of Fire Protection (BFP) in handling NDRRMC-catered incidents in the Negros Island Region. These included delays in incident reporting, uncoordinated response among responding units, fragmented records kept through phone calls, text messages, and two-way radios, and limited access to timely spatial and analytical information. To solve these issues, the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration (GAOIRS), developed a system using the Agile methodology, successfully fulfilled all objectives. The study's findings indicate that GAOIRS was fully implemented with all required features, passed all white-box and black-box test cases, received



Very High evaluations from IT experts and end users, and was supported by a comprehensive User's Guide.

The first objective was successfully achieved when GAOIRS was designed and fully developed with all five required features. The system integrated incident report recording and management with photo evidence, automatic GPS detection, and unique incident codes. It also included incident response with response unit registration, assignment, and live status tracking. A geospatial mapping interface identified the city and congressional district of incidents through geofencing and switched between marker and heatmap views. Monitoring and analysis provided incident indicators, barangay rankings, a kernel density estimation heatmap, and a seven-day Prophet forecast. Report generation allowed records to be exported as Excel, and PDF, and role-based access control secured the whole system. These features allowed the DRRMO and BFP to move from fragmented, logbook-based processes to a centralized digital system. The analytics were generated from synthetic data, so they demonstrated that the features worked but did not yet reflect real incident patterns.

The second objective was fully achieved. During the alpha and beta phases, white-box and black-box testing were carried out, which resulted in a 100% pass rate across all test cases. All 40 white-box test cases passed, covering authentication, report generation, access control, incident management, and backend response handling. All 78 black-box test cases also passed across incident reporting, incident response, geospatial mapping, monitoring and



analysis, and report generation. Beta testing showed that Response Unit personnel and citizens could sign in, submit and track reports, and view mapped incidents. This confirmed the system's reliability and its readiness for deployment.

The third objective was successfully achieved, yielding Very High results. An evaluation conducted by five IT experts using ISO/IEC 25010:2011 resulted in an overall mean score of 4.34, categorized as Very High. Among the criteria, Portability scored the highest at 4.67, while Performance Efficiency scored the lowest at 4.10, interpreted as High. Similarly, twenty-five end users, composed of four Response Unit personnel and twenty-one citizens, assessed the system using ISO 9241-11:2018. They produced mean scores of 4.42 for Effectiveness, 4.54 for Efficiency, and 4.58 for Satisfaction, with an overall mean of 4.51, all interpreted as Very High. These findings demonstrated that GAOIRS was highly functional, efficient, and provided a satisfying user experience.

The fourth objective was successfully achieved with the development of a comprehensive User's Guide. The guide explained how the system worked for each user role, guiding Response Unit personnel in recording and responding to incidents and Administrators in managing user accounts, monitoring system data, and generating reports. This empowered users to operate the system confidently and fully utilize its features.



5.3 Recommendations

The City Disaster Risk Reduction and Management Office (CDRRMO) and the Bureau of Fire Protection (BFP) were strongly encouraged to adopt GAOIRS as their official platform for incident reporting and response. They were also encouraged to hold regular training sessions for Response Unit personnel, administrators, and new users, especially on the mapping and analytics features. This approach helped ensure confident and efficient use of the system during emergencies. The DRRMO and BFP were further advised to encode their actual incident records into the system, so that the heatmaps and trend analysis could reflect real incident patterns in each congressional district. The developers were advised to optimize the system's performance and response time as incident and geospatial records grow, since Performance Efficiency was the lowest-rated criterion in the IT experts' evaluation. Local government units and policymakers were urged to support similar digital initiatives by providing funding and by adopting policies that standardize incident reporting tools across the region. For future research and development, improvements such as stronger offline reporting, expanded forecasting with more historical data, and integration with other local government and emergency response platforms were recommended. Additionally, conducting a longitudinal study to evaluate the system's long-term effect on response times, with more Response Unit respondents and tests in other localities of the Negros Island Region, could offer valuable evidence for wider adoption. In conclusion, GAOIRS showed how a centralized, geospatial approach can turn fragmented reporting into faster, better-coordinated response,



and how purposeful, user-centered technology can strengthen the communities it serves.

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APPENDICES

“Appendix A”

ISO/IEC 25010:2011 Software Quality Requirements and Evaluation

Questionnaire

Name(Optional):_____ Date:_____

Instruction:

Please rate the Quality of the **Geospatial Approach to Optimize Incident Response System with Data Analytics** using the scale below by **checking (✓)** the level that corresponds to your answer.

Scale:

1 = Strongly Disagree **2** = Disagree **3** = Undecided **4** = Agree **5** = Strongly Agree

Characteristics	1	2	3	4	5
Functional Suitability					
Functional Completeness The feature is complete that covers the user specified tasks.					
Functional Correctness The system provides correct results with the needed degree of precision.					
Functional Appropriateness The system facilitates the accomplishment of user specified tasks.					
Performance Efficiency	1	2	3	4	5
Time-behavior The system response and processing time when performing its functions, meet the requirements.					
Resource Utilization The amount and type of resources used by a system, when performing its functions, meet the requirements.					
Capacity The maximum limits of the system parameter, meet the requirements					
Compatibility	1	2	3	4	5
Co-existence The system can perform its required functions efficiently while sharing a common environment and resources.					



Interoperability Multiple systems components can exchange information.					
Usability	1	2	3	4	5
Appropriateness Recognisability The users can recognize the appropriateness of the system according to their needs.					
Learnability The system enables the user to learn how to use it with effectiveness, efficiency in emergency situations.					
Operability The system is easy to operate.					
User error protection The system protects users against making errors.					
User Interface Aesthetics The system user interface is pleasing and satisfying for the user.					
Accessibility The system can be used with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
Reliability	1	2	3	4	5
Maturity The system is reliable under normal operation.					
Availability The system is operational and accessible when required for use.					
Fault tolerance The system operates as intended despite the presence of hardware or software faults.					
Recoverability The system can recover the data in the event of an interruption or a failure.					
Security	1	2	3	4	5
Confidentiality The system ensures that data are accessible only to those authorized to have access.					
Integrity The system prevents unauthorized access to, or modification of computer programs or data.					
Non-repudiation					



The system actions or events can be proven to have taken place, cannot be repudiated.					
Accountability The system actions is traceable.					
Authenticity The system can identify a subject or resource.					
Maintainability	1	2	3	4	5
Modularity The system is composed of discrete components such that a change to one component has minimal impact on other components.					
Reusability The system asset can be used in more than one or in building other assets.					
Analyzability The effectiveness and efficiency to assess the impact on a system of an intended change to one or more of its parts, or to diagnose a system for deficiencies or causes of failures.					
Modifiability The system can be effectively and efficiently modified without introducing defects or degrading existing product quality.					
Testability The system effectiveness and efficiency can be performed to determine that test criteria are met.					
Portability	1	2	3	4	5
Adaptability The system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.					
Installability The system effectiveness and efficiency in the installation and/or uninstallation in a specified environment.					
Replaceability The system can replace another specified software product for the same purpose in the same environment.					



ISO/IEC 9241-11:2018

Software Quality Requirements and Evaluation Questionnaire

Name (Optional): _____

Evaluators: ☐ Response Unit (BFP/DRRMO) ☐ Citizen

Software Evaluation: Usability Assessment Based on ISO 9241-11:2018

Instruction:

Please rate the **Geospatial Approach to Optimize Incident Response System with Data Analytics** based on the criteria provided below by checking (✓) the level that best corresponds to your answer.

Scale:

5 – Strongly Agree

4 – Agree

3 – Undecided

2 – Disagree

1 – Strongly Disagree

Effectiveness	1	2	3	4	5
Time to Learn and Use: Task can be performed in a straight forward manner					
Data Entry Time: I can effectively complete my work using this system					
Task Time: Tasks can be performed in a short period of time using this system					
Response Time: I can use it successfully every time					
Time to Install: This system is fast to install					

Efficiency	1	2	3	4	5
Number of Errors: I can efficiently complete my work without spending too much time correcting things with this system					
Completion Rate: I am able to complete my work quickly using this system					

Satisfaction	1	2	3	4	5
I am satisfied with the ease of use of completing the task using this system					



I am satisfied with the amount of time it took to complete the task using this system					
Overall, I am satisfied with this system					



“Appendix B”

LETTER TO BENEFICIARY

**Carlos Hilado Memorial State University**
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March 26, 2026

John Philip Gallego (Chief, Operations and Warning)
Talisay City Disaster Risk Reduction and Management Office
Domingo Lizares St, Talisay City, Negros Occidental

Dear Sir, Gallego

Good day! I am Rubelyn Margaha, a 3rd year student together with my team, Justine Bibanco and Tristan Zane Lim from Carlos Hilado Memorial State University, currently working on our capstone project for our ISCAP1 303. Our project focuses on developing a Geospatial Approach to Optimize Incident Response System with Data Analytics Integration to improve the management of Incident Reporting and Response process.

As part of our data gathering process, we would like to respectfully request a short interview with you to gain insights into your response practice. Your professional experience and perspective would be incredibly valuable in helping me understand the real-world processes and challenges in managing these processes. All information shared will be treated with strict confidentiality and used solely for academic purposes.

We would greatly appreciate the opportunity to speak with you. Please let me know if you are available, and we will coordinate accordingly.

Thank you very much for your time and consideration.

Sincerely,

Margaha, Rubelyn F.
Lim, Tristan Zane C.
Bibanco, Justine M.

Charwin M. Padilla, MIT
Capstone Adviser

Danes P. Tumabing, MIT
Capstone Project Instructor

Danes P. Tumabing, MIT
Chairperson, BSIS Talisay

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(Good governance, Research-oriented, Extension-driven, Education for Sustainable Development, and Nation-building)

March 26, 2026

FO1 Ezra Escamillo B. Osorio (Chief, Personnel and Training)

Talisay City Fire Station

Domingo Lizares St, Talisay City, Negros Occidental

Dear Sir, Osorio

Good day! I am Rubelyn Margaha, a 3rd year student together with my team, Justine Bibanco and Tristan Zane Lim from Carlos Hilado Memorial State University, currently working on our capstone project for our ISCAP1 303. Our project focuses on developing a Geospatial Approach to Optimize Incident Response System with Data Analytics Integration to improve the management of Incident Reporting and Response process.

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We would greatly appreciate the opportunity to speak with you. Please let me know if you are available, and we will coordinate accordingly.

Thank you very much for your time and consideration.

Sincerely,

Margaha, Rubelyn F.

Lim, Tristan Zane C.

Bibanco, Justine M.


Charwin M. Padilla, MIT
Capstone Adviser


Danes P. Tumabing, MIT
Capstone Project 1 Instructor


Danes P. Tumabing, MIT
Chairperson/BSIS Talisay



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“Appendix C”

LETTER TO CAPSTONE PROJECT ADVISER

 **Carlos Hilado Memorial State University**
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CAPSTONE PROJECT ADVISER ACCEPTANCE FORM

Date: March 09, 2026

This is to attest that I, CHARWIN M. PADILLA, have agreed to be the adviser of the following students with the Capstone Project entitled, Geospatial Approach to Optimize Incident Response System with Data Analytics Integration.

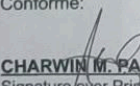
Group Name:
Course/Year/Sec: BSIS 3-A,B,C

Members:

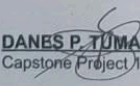
1. Bibanco, Justine M.
2. Lim, Tristan C.
3. Margaha, Rubelyn F.

Furthermore, I agree to set a schedule for the advising and consultation to help the students and ensure the success of the project.

Conforme:


CHARWIN M. PADILLA, MIT
Signature over Printed Name of Adviser

Noted:


DANES P. TUMABING, MIT
Capstone Project Professor

 college.computerstudies@chmsc.edu.ph
(034) 434 8148
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“Appendix D”

BLACK BOX TESTING - ALPHA TESTING

Black-Box Testing of the Alpha Testing of Incident Report Recording and Management

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-SubRep-001	Report form visibility test	All report submission form elements (incident type, location, photo upload, submit button) should be visible on page load.	All expected form elements were visible and properly rendered.	Pass	No issues encountered.
TC-SubRep-002	Incident type selection functionality	Citizen should be able to select an incident type from the list and have the selection reflected in the form.	The selected incident type was captured and reflected correctly in the report.	Pass	Selection worked as intended.
TC-SubRep-003	Automatic location detection accuracy	System should automatically detect and capture the citizen's current location.	The current location was detected and captured accurately.	Pass	Detection worked as intended.
TC-SubRep-004	Photo attachment upload functionality	Citizen should be able to attach a photo as evidence of the incident.	The photo was successfully attached and previewed before submission.	Pass	Upload worked as intended.
TC-SubRep-005	Successful end-to-end report submission	System should save the completed report once all required fields are filled.	The report was saved successfully with all provided details intact.	Pass	Full submission flow worked as intended.



TC-SubRep-006	Submission confirmation and reference number generation	System should display a confirmation message with a unique tracking/reference number.	A confirmation message with a valid, unique reference number was displayed.	Pass	Confirmation flow worked as intended.
TC-SubRep-007	Required field validation	System should prevent submission and display an error when incident type, location, or photo is missing.	The system blocked submission and displayed the correct error prompts for each missing field.	Pass	Validation works as intended; incomplete submissions properly rejected.
TC-SubRep-008	Fallback to manual map pin when GPS fails	System should allow the citizen to manually pin the location when automatic detection is unavailable or inaccurate.	The manual pinning option was correctly offered and the pinned location was captured.	Pass	Fallback flow worked as intended.
TC-TrackStat-001	Status tracking form visibility test	The Track Report Status search field and results area should be visible on page load.	All expected elements were visible and functioning.	Pass	No issues encountered.
TC-TrackStat-002	Successful retrieval by valid reference number	System should display the correct, current status of the selected report.	The system retrieved and displayed the correct status for the tested report.	Pass	Status displayed matched the actual record.
TC-TrackStat-003	Invalid reference number handling	System should display an appropriate message when an invalid or non-existent reference number is entered.	The system correctly displayed a "report not found" message for invalid references.	Pass	Error handling worked as intended.



TC-TrackStat-004	Status accuracy across report lifecycle	The displayed status should correctly reflect each stage of the report's lifecycle (Pending, Verified, Dispatched, Resolved, Rejected).	The status label updated correctly at each stage as the report progressed.	Pass	Status accuracy confirmed across all states.
TC-MgIncId-001	Incident list display test	The list of submitted incident reports should be visible and correctly populated on page load.	All incident records were displayed correctly.	Pass	No issues encountered.
TC-MgIncId-002	Filter by status functionality	System should display only incidents matching the selected status filter.	The filtered results correctly matched the selected status.	Pass	Status filter worked as intended.
TC-MgIncId-003	Filter by date functionality	System should display only incidents within the selected date range.	The filtered results correctly matched the selected date range.	Pass	Date filter worked as intended.
TC-MgIncId-004	Search function accuracy	System should return incidents matching the entered search keyword.	Search results correctly matched the entered keyword.	Pass	Search function worked as intended.
TC-MgIncId-005	Empty result handling	System should display an appropriate message when no records match the applied filter or search.	The system correctly displayed a "no records found" message.	Pass	Handled empty results appropriately.



TC-VerifRpt-001	Pending report review interface test	Report details, photo, and pinned location should be visible when a pending report is opened.	All report details were displayed correctly.	Pass	No issues encountered.
TC-VerifRpt-002	Successful verification and automatic dispatch trigger	System should mark the report as Verified and proceed to Dispatch Response Team.	The report status changed to Verified and the dispatch process was triggered.	Pass	Verification and dispatch trigger worked as intended.
TC-VerifRpt-003	Report rejection / mark false alarm handling	System should allow the admin to reject a report and mark it as a false alarm.	The report status correctly changed to Rejected/False Alarm.	Pass	Rejection flow worked as intended.
TC-VerifRpt-004	Duplicate report flagging	System should flag or group reports that appear to be duplicates of an existing incident.	Duplicate reports submitted for the same incident were correctly flagged for the admin's review.	Pass	Duplicate detection worked as intended.
TC-VerifRpt-005	Verification permission restricted to Admin role	Only users with Admin credentials should be able to verify or reject incident reports.	Non-admin accounts were correctly denied access to the verification function.	Pass	Access control worked as intended.

Black-Box Testing of the Alpha Testing of Incident Response

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AckIncid-001	Incident notification display test	Dispatch notification and incident details should be visible to the response	Notification and details were displayed correctly.	Pass	No issues encountered.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
		unit upon assignment.			
TC-AckIncid-002	Successful acknowledgment and status update	Incident status should update to Acknowledged once the response unit acknowledges it.	The status updated correctly to Acknowledged.	Pass	Acknowledgment flow worked as intended.
TC-AckIncid-003	Admin notification on acknowledgment	The admin should receive a notification once the response unit acknowledges the incident.	The admin received the acknowledgment notification correctly.	Pass	Notification worked as intended.
TC-AckIncid-004	Duplicate acknowledgment prevention	System should prevent an already-acknowledged incident from being acknowledged again.	The system correctly blocked the duplicate action.	Pass	Validation works as intended.
TC-UpdStat-001	Status options visibility test	Status options (e.g., En Route, On Scene) should be visible for an acknowledged incident.	All status options displayed correctly.	Pass	No issues encountered.
TC-UpdStat-002	Successful status update and real-time reflection	The status should update and reflect in the admin's view in real time.	The status change was saved and reflected on the admin dashboard immediately.	Pass	Update flow worked as intended.
TC-UpdStat-003	Notification sent on status change	Relevant parties should be notified when the incident status changes.	A notification was correctly sent upon each status change.	Pass	Notification worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-UpdStat-004	Status update blocked before acknowledgment	System should prevent a status update on an incident that has not yet been acknowledged.	The system correctly blocked the update and displayed a message.	Pass	Validation works as intended.
TC-ResClo-001	Close incident option visibility test	The Resolve/Close Incident option should be visible for an ongoing incident.	Option displayed correctly.	Pass	No issues encountered.
TC-ResClo-002	Successful resolution and archiving	Incident status should change to Resolved and the record archived.	The incident was marked Resolved and archived correctly.	Pass	Closure flow worked as intended.
TC-ResClo-003	Confirmation prompt before closure	System should require confirmation before finalizing closure.	A confirmation prompt was correctly displayed before closure.	Pass	Confirmation step worked as intended.
TC-ResClo-004	Closure blocked without a prior response update	System should prevent closure of an incident that has not been acknowledged or updated.	The system blocked closure and prompted the response unit to update the status first.	Pass	Validation works as intended.
TC-UpldResp-001	Upload form visibility test	File attachment and notes fields should be visible when uploading a response.	All fields displayed correctly.	Pass	No issues encountered.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-UpldResp-002	Successful documentation upload and linkage	Documentation should be saved and linked to the correct incident record.	Uploaded files and notes were saved and correctly linked.	Pass	Upload flow worked as intended.
TC-UpldResp-003	Automatic closure trigger when incident not yet resolved	System should automatically resolve/close the incident if it is not yet marked closed when documentation is uploaded.	The incident was correctly closed as part of the upload process.	Pass	Automatic closure worked as intended.
TC-UpldResp-004	Unsupported file type rejection	System should reject unsupported file types with an appropriate message.	The system correctly rejected an unsupported file and displayed an error.	Pass	Validation works as intended.
TC-UpldResp-005	File size limit validation	System should reject files that exceed the maximum allowed size.	The system correctly rejected an oversized file and displayed an error.	Pass	File size validation worked as intended.
TC-Dispa-001	Dispatch interface visibility test	The Dispatch Response Team option and unit recommendations should be visible for a verified incident.	All elements displayed correctly.	Pass	No issues encountered.
TC-Dispa-002	Nearest-unit recommendation accuracy	System should recommend the nearest available response unit based on the incident location.	The recommended unit correctly matched the closest available unit to the incident.	Pass	Recommendation accuracy confirmed.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Dispa-003	Successful dispatch and unit assignment	The selected unit should be assigned to the incident once confirmed by the admin.	The unit was correctly assigned to the incident.	Pass	Dispatch flow worked as intended.
TC-Dispa-004	Dispatch notification delivery	The assigned unit should receive a dispatch notification with the incident details.	The unit received the notification with the correct incident details.	Pass	Notification delivery worked as intended.
TC-Dispa-005	No available unit handling	System should display an appropriate message when no response unit is available.	The system correctly displayed a "no unit available" message.	Pass	Handled correctly.
TC-Dispa-006	Manual override of recommended unit	Admin should be able to manually select a different unit instead of the recommended one.	The admin was able to override the recommendation and assign a different unit.	Pass	Manual override worked as intended.
TC-Avail-001	Availability form visibility test	Availability options (Available, Busy, Off-duty) should be visible to the logged-in response unit.	Options displayed correctly.	Pass	No issues encountered.
TC-Avail-002	Successful availability update	The unit's availability should update immediately upon selection.	Availability was updated and saved correctly.	Pass	Update flow worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Avail-003	Status persistence across sessions	The declared status should remain saved after logging out and back in.	The status persisted correctly across sessions.	Pass	Persistence worked as intended.
TC-Avail-004	Availability reflected in dispatch recommendations	The declared availability should be factored into the nearest-unit recommendation during dispatch.	Only units marked Available were included in dispatch recommendations.	Pass	Integration with dispatch worked as intended.

Black-Box Testing of the Alpha Testing of Geospatial Interface

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-ViewMap-001	Map load and display test	The interactive map with pinned incidents and units should load and display correctly.	The map loaded and displayed all pins correctly.	Pass	No issues encountered.
TC-ViewMap-002	City tagging accuracy	Each pin should be correctly tagged with its corresponding city.	All pins displayed the correct city information.	Pass	City tagging worked as intended.
TC-ViewMap-003	Congressional district tagging accuracy	Each pin should be correctly tagged with its corresponding congressional district.	All pins displayed the correct congressional district information.	Pass	District tagging worked as intended.
TC-ViewMap-004	Real-time refresh of incidents/units on map	The map should update in real time as new incidents are submitted or resolved.	The map correctly refreshed to reflect new and resolved incidents.	Pass	Real-time update worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-ViewMap-005	Map marker click for incident details	Clicking a map marker should display the corresponding incident's details.	Incident details were correctly displayed when a marker was clicked.	Pass	Marker interaction worked as intended.
TC-MapPin-001	Pin tool visibility test	The Pin Location tool should be visible and accessible on the report form.	The tool was visible and accessible.	Pass	No issues encountered.
TC-MapPin-002	Successful manual pin placement and coordinate capture	The system should capture the correct coordinates and city/district and attach them to the report.	Coordinates and location details were correctly captured and attached.	Pass	Pinning flow worked as intended.
TC-MapPin-003	Pin outside service area warning	System should display a warning when the pinned location is outside the covered service area.	The system correctly displayed a warning message.	Pass	Validation works as intended.
TC-MapPin-004	Pin adjustment/re-pin functionality	Citizen should be able to move or reset the pin before submitting the report.	The pin was successfully repositioned and the updated coordinates were captured.	Pass	Re-pin functionality worked as intended.



Black-Box Testing of the Alpha Testing of Monitoring and Analysis

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Analyt-001	Dashboard display test	Charts, statistics, and the KDE heatmap should be visible on page load.	All dashboard elements displayed correctly.	Pass	No issues encountered.
TC-Analyt-002	Trend chart data accuracy	The displayed trend charts should accurately reflect the current incident data.	The chart figures matched the underlying incident records.	Pass	Data accuracy confirmed.
TC-Analyt-003	KDE heatmap and hotspot accuracy	The KDE heatmap should accurately reflect the spatial density and hotspots of incidents.	The heatmap correctly highlighted the known incident clusters.	Pass	Heatmap accuracy confirmed.
TC-Analyt-004	No-data handling	System should display an appropriate message when there is insufficient data to generate analytics.	The system correctly displayed a "no data available" message.	Pass	Handled correctly.
TC-Analyt-005	Forecast/trend analysis display accuracy	The forecast/trend analysis (Prophet, SARIMA, Moving Average) should display projected incident trends based on historical data.	The forecast output was displayed and aligned with the expected trend direction of the test data.	Pass	Forecast display worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AuLog-001	Audit log display test	The chronological log of recorded actions should be visible on page load.	The audit log displayed correctly.	Pass	No issues encountered.
TC-AuLog-002	Filter by user functionality	System should display only log entries related to the selected user.	Filtered results correctly matched the selected user.	Pass	User filter worked as intended.
TC-AuditLog003	Filter by date functionality	System should display only log entries within the selected date range.	Filtered results correctly matched the selected date range.	Pass	Date filter worked as intended.
TC-AuLog-004	Filter by action type functionality	System should display only log entries matching the selected action type.	Filtered results correctly matched the selected action type.	Pass	Action type filter worked as intended.
TC-Au-Log-005	Log entry accuracy check	Each log entry should accurately capture the corresponding system or user action.	All tested log entries matched the actions performed.	Pass	Accuracy confirmed.
TC-MgResp-001	Response unit list display test	The list of response units with their status and assignments should be visible on page load.	All records displayed correctly.	Pass	No issues encountered.
TC-MgResp-002	Successful assignment update	Response unit records should be accurately updated when assignments are changed.	Updates were saved and correctly reflected.	Pass	Update flow worked as intended.



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-MgResp-003	Status/availability visibility accuracy	Each unit's current status and availability should be accurately displayed.	Status and availability displayed matched the actual records.	Pass	Display accuracy confirmed.
TC-MgResp-004	Unauthorized edit prevention	System should prevent non-admin users from editing response unit records.	The system correctly blocked unauthorized edit attempts.	Pass	Access control worked as intended.

Black-Box Testing of the Alpha Testing of Report Generation

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-GenRpt-001	Report parameters form visibility test	Report parameter fields (date range, incident type, area) should be visible on page load.	All fields displayed correctly.	Pass	No issues encountered.
TC-GenRpt-002	Successful report compilation	System should compile an accurate report based on the selected parameters.	The compiled report data matched the selected parameters.	Pass	Compilation accuracy confirmed.
TC-GenRpt-003	PDF export functionality	System should export the compiled report as a correctly formatted PDF file.	The report downloaded successfully as a properly formatted PDF.	Pass	PDF export worked as intended (Export/Report extension).



Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-GenRpt-004	Excel export functionality	System should export the compiled report as a correctly formatted Excel file.	The report downloaded successfully as a properly formatted Excel file.	Pass	Excel export worked as intended (Export/Report extension).
TC-GenRpt-005	CSV export functionality	System should export the compiled report as a correctly formatted CSV file.	The report downloaded successfully as a properly formatted CSV file.	Pass	CSV export worked as intended (Export/Report extension).
TC-GenRpt-006	Invalid parameter/date range handling	System should display an error when an invalid date range or parameter is entered.	The system correctly displayed an error for invalid parameters.	Pass	Validation works as intended.



“Appendix E”

Sample Black Box Testing – Beta Testing

The first screenshot shows the app's home screen with a greeting 'Good evening, Citizen' and a prominent red 'Emergency Report' button. Below it is a 'MY REPORT STATUS TRACKING' section. The second screenshot shows the 'Emergency Report' form with fields for 'PHOTO EVIDENCE', 'YOUR LOCATION' (with a map), and 'EMERGENCY TYPE' (set to 'VEHICULAR ACCIDENT'). The third screenshot shows the 'INCIDENT SEVERITY' section with 'HIGH' selected, a description of the incident, a phone number, and a 'SUBMIT EMERGENCY REPORT' button.

The screenshot displays the 'Incident Reports' section of the GAOIRS Admin Panel. It features a sidebar with navigation options like Dashboard, Incident Reports, Incident Archive, User Management, Response Units, Incident Categories, Analytics & Reports, Post-Incident Reports, Audit Logs, and System Settings. The main area shows an 'Incident Queue' with a list of reports, including details like incident ID, type, severity, and location. A detailed view of a specific incident is shown on the right, including a map and response strategy.



GAOBS ADMIN PANEL

Response Units
DISPATCH MANAGEMENT SYSTEM

Manage and track all connected dispatch units in the field.

Response Units

TOTAL UNITS: 3 AVAILABLE: 2

Search units...

UNIT IDENTIFIER	TYPE	STATUS	PINNED	BASE LOCATION	CONTACT	ACTIONS
Talabay Fire Station	FIRE	Available	Phone	Unassigned	-	
Talabay DRBMO	DRBMO	Available	Phone	Zone 2	+6397700001	
Talabay DRBMO	DRBMO	Offline	Phone	Zone 2	-	

Admin
admin@gmail.com
Sign Out

GAOBS RESPONSE UNIT

Live Map
RESPONSE TERMINAL

UNIT CONTROL

Dashboard
Live Map
Notifications

Live Response Map
Real-time tracking + 18 active incidents + 2 units

Responding (10) On Scene (8) Filter Map

DRBMO-20250208-8109
Fire/Dispatch

Fire/Dispatch: Storm surge

- LOW Severity
- Unknown Reporter
- Anonymous Report
- WDR 0025, 4:10:48 AM
- San Sebastian, Talabay City, Negros Occidental

GET DIRECTIONS
BACKUP
CAUTION
ARRIVAL

Route to: 9W-178K794358554-177
Talabay DRBMO: 1/2, 4 km + 10 min

Open Google Maps

1A Talabay DRBMO
Available

Sign Out

GAOBS ADMIN PANEL

Dashboard
DISPATCH MANAGEMENT SYSTEM

MAIN MENU

Dashboard
Incident Reports
Incident Archive
User Management
Response Units
Incident Categories
Analytics & Reports
Post-Incident Reports
Audit Logs
System Settings

System Admin
admin@gmail.com
Sign Out

System Overview
Live monitoring and real-time statistics active.

ACTIVE INCIDENTS: 2 REGISTERED USERS: 4 RESOLVED OVERALL: 3 TOTAL INCIDENTS: 5

Landmarks Near Barangay Hall
Type: Fire Incident
Residential fire reported in Zone 4.

REPORTER INFO
Name: Community Reporter
Phone: +63977000009
No photos attached
Reported: Sep 25, 10:07 PM

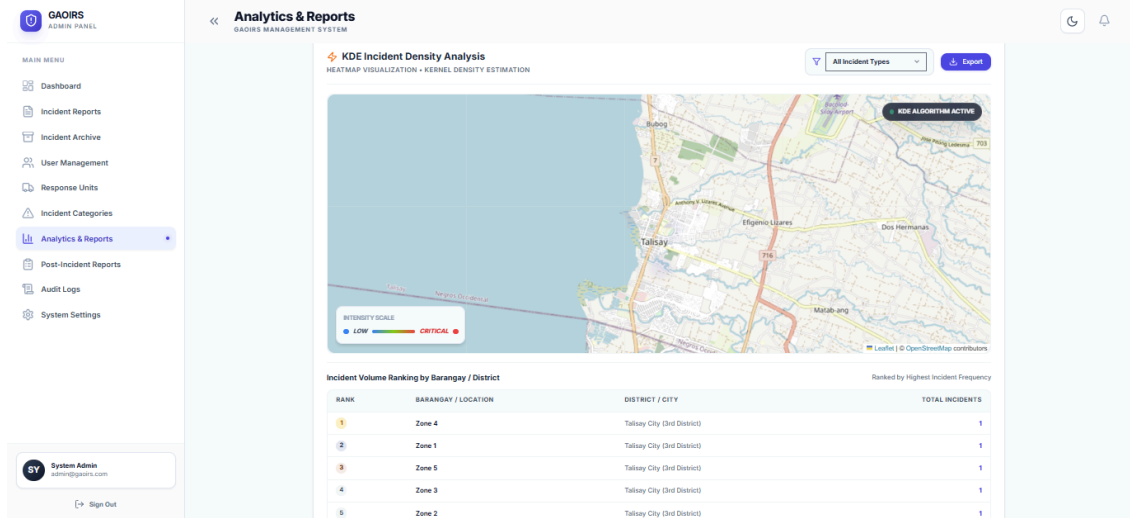
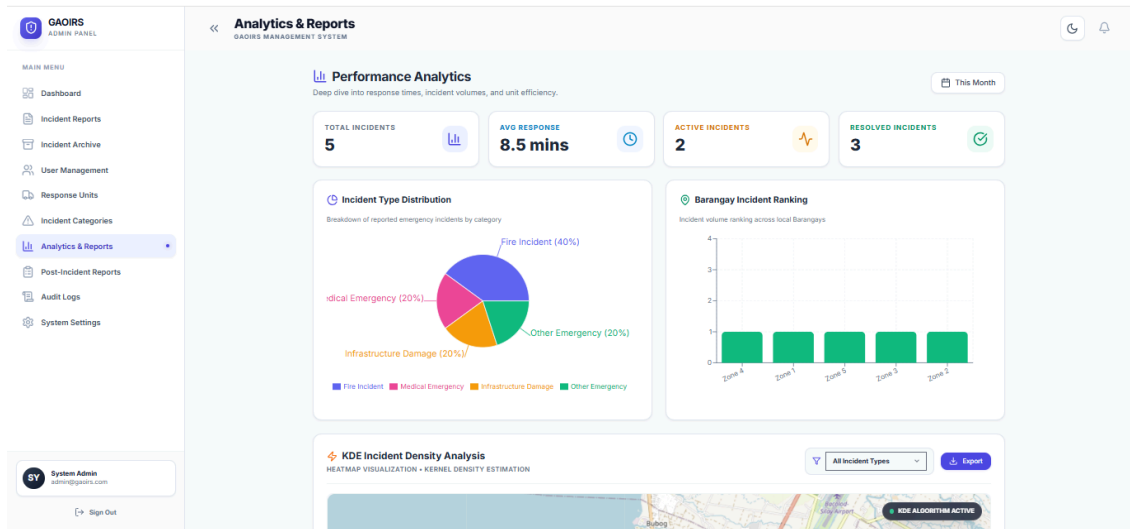
AWAITING VERIFICATION
Approve & Dispatch Report

New Reports
JAC-000-460
Fire Incident
Zone 5, Talabay City, Negros Occidental
10:07 PM
Approve Report

Responding
JAC-000-460
Fire Incident
Zone 5, Talabay City, Negros Occidental
CRITICAL
RESPONDING

Resolved
No resolved incidents

RESPONSE UNITS (2)





“Appendix F”

WHITE BOX TESTING - ALPHA TESTING

White-Box Testing of Authentication Middleware

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W001	verifyToken()	Header missing validation	headers: {}	Returned HTTP 401 Unauthorized with "Missing or invalid authorization header" message	Returns 401 error and blocks request progression	Pass
TC-W002	generateToken()	Citizen JWT token generation	user_id: "usr-101", role: "citizen"	Generated valid signed JWT string with citizen user payload	Produces valid signed JWT string containing citizen claims	Pass
TC-W003	generateToken()	Responder unit token signing	user_id: "usr-202", unit_id: "unit-01"	Encoded unit_id "unit-01" into decoded JWT object	Encodes response unit identity into token payload	Pass
TC-W004	generateToken()	Minimal admin payload token	user_id: "usr-303", role: "admin"	Assigned null to optional unit_id and barangay_id fields	Sets optional unit/barangay fields to null safely	Pass
TC-W005	verifyToken()	Valid Bearer token verification	headers: { authorization: "Bearer " }	Attached user payload to req.user and executed next()	Attaches user object to request context	Pass



White-Box Testing of Role-Based Access Control (RBAC) Middleware Guards

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W006	requireRole()	Undefined user object guard	req.user = undefined	Returned HTTP 401 Unauthorized status	Returns 401 Unauthorized error	Pass
TC-W007	requireAdmin()	Role mismatch guard check	req.user = { role: "REPORTER" }	Returned HTTP 403 Forbidden status	Returns 403 Forbidden error	Pass
TC-W008	requireAdmin()	Admin access permission pass	req.user = { role: "ADMIN" }	Invoked next() middleware handler	Allows request progression via next()	Pass
TC-W009	requireResponseUnit()	Response unit permission pass	req.user = { role: "RESPONSE_UNIT" }	Invoked next() middleware handler	Allows request progression via next()	Pass
TC-W010	requireReporter()	Reporter permission pass	req.user = { role: "REPORTER" }	Invoked next() middleware handler	Allows request progression via next()	Pass

White-Box Testing of Incident Management & Geofencing Controller

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W011	createIncident()	NIR geofence boundary check	lat: 14.5995, lng: 120.9842 (Manila)	Returned HTTP 400 Bad Request with "Location Out of Coverage Area" message	Rejects report with 400 error	Pass
TC-W012	getIncidents()	Paginated incident list query	page: "1", limit: "10"	Formatted data array with total=1 and page=1 pagination object	Formats paginated array response	Pass



TC-W013	getIncidentById()	Non-existent incident ID query	params: { id: "invalid-id" }	Returned HTTP 404 Not Found status	Returns 404 Not Found error	Pass
TC-W014	verifyIncident()	Non-ADMIN verification block	user.role: "REPORTER", action: "APPROVE"	Returned HTTP 403 Forbidden with "Only admins can verify incidents" message	Blocks non-admin verification with 403	Pass
TC-W015	verifyIncident()	Invalid verification action	action: "INVALID_TYPE"	Returned HTTP 400 Bad Request with message "Must be APPROVE, REJECT, or REQUEST_INFO"	Rejects invalid action with 400 error	Pass
TC-W016	editIncident()	Non-ADMIN incident edit guard	user.role: "RESPONSE_UNIT"	Returned HTTP 403 Forbidden with "Only admins can edit incidents" message	Blocks non-admin editing with 403	Pass
TC-W017	updateIncidentStatus()	Responder status update	status: "RESPONDING", remarks: "En route"	Updated incident status to "RESPONDING" with HTTP 200 OK	Updates status to RESPONDING	Pass

White-Box Testing of Photo Evidence Upload & Cloud Storage Services

ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Result	Result
TC-W018	uploadIncidentPhoto()	Missing file guard	req.file = undefined	Returns 400 Bad Request ("No file	Matches expected output	Pass



		validation		uploaded").		
TC-W019	uploadIncidentPhoto()	Photo upload URL generator	originalName: "fire.png"	Returns 200 OK with Supabase cloud bucket URL.	Matches expected output	Pass
TC-W020	extractPublicId()	Storage path string parser	url: "https://.../incident.jpg"	Extracts relative storage path string correctly.	Matches expected output	Pass
TC-W021	uploadBufferToCloudinary()	Null image buffer fallback	buffer = null	Returns null safely without crashing server.	Matches expected output	Pass

White-Box Testing of Barangay Spatial Records Controller

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W022	getAll()	Barangay query pagination	page: "1", limit: "10"	Returned 2 records array with totalPages= 1 calculation	Calculates total pages accurately	Pass
TC-W023	create()	Barangay spatial record creation	name: "San Jose", city: "Talisay"	Returned HTTP 201 Created with generated barangay_id "b3"	Returns 201 Created with new record	Pass

White-Box Testing of Incident Code Generator Algorithm (INC-YYYY-XXX)

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W024	generateIncidentCode()	First incident zero-padded code	incidentCount = 0	Formatted code string "INC-2026-001" with zero-padding	Generates INC-YYYY-001 format	Pass



TC-W025	generateIncidentCode()	Incremental code generator	incidentCount = 42	Incremented sequence to format "INC-2026-043"	Generates incremented zero-padded code	Pass
----------------	------------------------	----------------------------	--------------------	---	--	------

White-Box Testing of System Audit Logging Controller

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W026	logAuditEvent()	Event payload action formatting	user_id: "usr-101", action: "LOGIN"	Stored audit record object with action "LOGIN"	Logs audit event payload safely	Pass
TC-W027	listAuditLogs()	Paginated audit logs query	page: "1", limit: "25"	Returned HTTP 200 OK with logs array and total=1 metadata	Returns 200 OK with paginated logs	Pass
TC-W028	listDistinctActions()	Distinct action strings list	req = {}	Returned 3 distinct action strings array containing "CREATE_REPORT"	Returns array of action strings	Pass

White-Box Testing of Sensitive Data Encryption Utility

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W029	encrypt()	Plain text payload encryption	"Confidential Contact 09171234567"	Generated hex iv:ciphertext string formatted with colon separator	Produces encrypted iv:ciphertext string	Pass
TC-W030	decrypt()	Encrypted text payload decryption	encryptedText String	Decrypted text back to exact original string "Emergency Responder"	Restores original plaintext string	Pass



				Location Payload"		
TC-W031	decrypt()	Invalid format decryption fallback	"invalid_for mat_string"	Returned null cleanly without throwing application crash exception	Returns null safely on invalid format	Pass

White-Box Testing of API Response Helper Utility

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W032	success()	Default success response payload	empty parameters	Returned object { success: true, data: null, message: "Success" }	Formats default success structure	Pass
TC-W033	success()	Custom data payload formatting	data: { id: "INC-101" }	Returned object with success=true and custom data payload attached	Attaches custom payload to success response	Pass
TC-W034	error()	Default error response payload	empty parameters	Returned object { success: false, error: null, message: "Request failed" }	Formats default error structure	Pass
TC-W035	error()	Custom error payload response	error: { code: "UNAUTHO RIZED" }	Returned object with success=false and custom error payload attached	Attaches error payload to response	Pass



White-Box Testing of Database Query Pagination Utility

ID	Tested Code Segment	Test Description	Input Values	Actual Result	Expected Behavior	Result
TC-W036	getPagination()	Default query parameter handling	query: {}	Returned object with page=1, limit=10, skip=0 default values	Returns default page: 1, limit: 10, skip: 0	Pass
TC-W037	getPagination()	Skip offset calculation	page: 3, limit: 15	Calculated skip offset value 30 $((3-1)*15)$ accurately	Calculates skip offset accurately	Pass
TC-W038	getPagination()	Negative page boundary guard	page: -5, limit: 10	Enforced minimum page=1 and skip=0 boundary protection	Enforces minimum page 1 and skip 0	Pass
TC-W039	getPagination()	Maximum limit cap protection	page: 1, limit: 500	Capped limit value to maximum ceiling of 100 items	Caps maximum limit to 100 items	Pass
TC-W040	buildPaginationMeta()	Total pages calculation	page: 2, limit: 10, total: 25	Calculated totalPages= 3 using $\text{Math.ceil}(25/10)$	Calculates totalPages metadata correctly	Pass



“Appendix G”

Sample White Box Testing – Beta Testing

White-box testing using the JEST Results:

```
PASS src/middleware/authMiddleware.test.js
White-Box Unit Testing - Module 1: Authentication Middleware & Session Authorization
  ✓ TC-W001: verify_header - should return 401 error if Authorization header is missing (619 ms)
  ✓ TC-W002: citizen_token - should generate valid JWT signature for citizen payload (647 ms)
  ✓ TC-W003: responder_token - should encode unit_id into responder JWT token (615 ms)
  ✓ TC-W004: admin_token - should set optional unit/barangay fields to null safely (611 ms)
  ✓ TC-W005: login_verification - should attach decoded payload to req.user for valid token (656 ms)

Test Suites: 1 passed, 1 total
Tests:       5 passed, 5 total
Snapshots:   0 total
Time:        3.868 s, estimated 5 s
Ran all test suites matching src/middleware/authMiddleware.test.js.
```

```
PASS src/middleware/roleMiddleware.test.js
White-Box Unit Testing - Module 2: Role-Based Access Control (RBAC) Middleware Guards
  ✓ TC-W006: undefined_user_guard - should return 401 error if req.user is undefined (610 ms)
  ✓ TC-W007: role_mismatch_guard - should return 403 error if user role does not match required role (611 ms)
  ✓ TC-W008: admin_access_pass - should call next() if user possesses ADMIN role (612 ms)
  ✓ TC-W009: responder_access_pass - should call next() if user possesses RESPONSE_UNIT role (613 ms)
  ✓ TC-W010: reporter_access_pass - should call next() if user possesses REPORTER role (607 ms)

Test Suites: 1 passed, 1 total
Tests:       5 passed, 5 total
Snapshots:   0 total
Time:        3.559 s, estimated 4 s
Ran all test suites matching src/middleware/roleMiddleware.test.js.
PS C:\Users\Tristan Zane\OneDrive\Desktop\CAPSTONE\backend>
```

```
export const createIncident = async (req, res) => {
  try {
    const { incident_type_id, description, latitude, longitude, map_pin_address, landmark, severity, force_create, reporter_name, reporter_phone } = req.body;

    // --- Duplicate Detection ---
    // Check for similar incidents (same type + nearby location within 5 minutes)
    const fiveMinutesAgo = new Date(Date.now() - 5 * 60 * 1000);
    const recentSimilar = await prisma.incident.findFirst({
      where: {
        incident_type_id,
        status: { not: 'FALSE_ALARM' },
        reported_at: { gte: fiveMinutesAgo },
        latitude: { gte: latitude - 0.001, lte: latitude + 0.001 }, // ~100m radius
        longitude: { gte: longitude - 0.001, lte: longitude + 0.001 }
      }
    });

    if (recentSimilar && !force_create) {
      return res.status(400).json({
        success: false,
        message: 'Similar incident already reported nearby within last 5 minutes',
        warning: true,
        existing_incident: {
          incident_id: recentSimilar.incident_id,
          incident_code: recentSimilar.incident_code,
          reported_at: recentSimilar.reported_at
        },
        shouldAllow: true
      });
    }
  }
}
```



```
PASS src/controllers/incidentController.test.js (8.987 s)
White-Box Unit Testing - Module 3: Incident Management & Geofencing Controller
  ✓ TC-W011: coverage_boundary - should reject reporting outside NIR coverage area (e.g. Manila location)
3748 ms)
  ✓ TC-W012: get_incidents_pagination - should format paginated incident query response accurately (602 ms)
  ✓ TC-W013: get_incident_by_id - should return 404 when incident ID does not exist (609 ms)
  ✓ TC-W014: verify_non_admin_block - should block non-ADMIN users from verifying reports with 403 Forbidden
n (611 ms)
  ✓ TC-W015: verify_invalid_action - should reject invalid verification actions with 400 Bad Request (614
s)
  ✓ TC-W016: edit_non_admin_block - should block non-ADMIN users from editing incidents with 403 Forbidden
(611 ms)
  ✓ TC-W017: update_dispatch_status - should update incident status to RESPONDING and return updated recor
(666 ms)

Test Suites: 1 passed, 1 total
Tests:       7 passed, 7 total
Snapshots:   0 total
Time:        9.107 s, estimated 11 s
Ran all test suites matching src/controllers/incidentController.test.js.
```

```
PASS src/controllers/uploadController.test.js
White-Box Unit Testing - Module 4: Photo Evidence Upload & Cloud Storage Services
  ✓ TC-W018: missing_file_guard - should return 400 Bad Request if req.file is missing (638 ms)
  ✓ TC-W019: process_photo_upload - should process image file upload successfully (669 ms)
  ✓ TC-W020: extract_public_id - should extract storage path from full Supabase URL correctly (608 ms)
  ✓ TC-W021: null_buffer_fallback - should return null when image buffer is missing or undefined (608 ms)

Test Suites: 1 passed, 1 total
Tests:       4 passed, 4 total
Snapshots:   0 total
Time:        4.949 s, estimated 6 s
Ran all test suites matching src/controllers/uploadController.test.js.
```

```
PASS src/controllers/barangayController.test.js
White-Box Unit Testing - Module 5: Barangay Spatial Records Controller
  ✓ TC-W022: pagination_accuracy - should calculate pagination accurately for page 1 and limit 10 (611 ms)
  ✓ TC-W023: create_record - should return 201 Created upon successful creation of barangay (668 ms)

Test Suites: 1 passed, 1 total
Tests:       2 passed, 2 total
Snapshots:   0 total
Time:        1.981 s, estimated 5 s
Ran all test suites matching src/controllers/barangayController.test.js.
```

```
PASS src/utills/incidentCode.test.js
White-Box Unit Testing - Module 6: Incident Code Generator Algorithm (INC-YYYY-XXX)
  ✓ TC-W024: first_incident_code - should generate zero-padded code format INC-YYYY-001 for first incident
(621 ms)
  ✓ TC-W025: incremented_code - should generate incremented zero-padded code INC-YYYY-043 for 42 existing i
ncidents (656 ms)

Test Suites: 1 passed, 1 total
Tests:       2 passed, 2 total
Snapshots:   0 total
Time:        1.809 s, estimated 4 s
Ran all test suites matching src/utills/incidentCode.test.js.
```



```
(use node --trace-warnings flag to show where the warning was created)
PASS src/controllers/auditController.test.js
White-Box Unit Testing - Module 7: System Audit Logging Controller
  ✓ TC-W026: log_event_action - should handle missing parameter gracefully (620 ms)
  ✓ TC-W027: list_audit_logs - should format paginated audit logs cleanly with 200 OK (605 ms)
  ✓ TC-W028: distinct_actions - should map distinct audit action strings into array (604 ms)

Test Suites: 1 passed, 1 total
Tests:      3 passed, 3 total
Snapshots:  0 total
Time:       2.36 s, estimated 3 s
Ran all test suites matching src/controllers/auditController.test.js.
```

```
PASS src/utis/encryptionUtil.test.js
White-Box Unit Testing - Module 8: Sensitive Data Encryption Utility
  ✓ TC-W029: encrypt_payload - should successfully encrypt plain text sensitive data (654 ms)
  ✓ TC-W030: decrypt_payload - should successfully decrypt encrypted text back to original string (618 ms)
  ✓ TC-W031: invalid_format_fallback - should return null when trying to decrypt invalid format text (656 ms)

Test Suites: 1 passed, 1 total
Tests:      3 passed, 3 total
Snapshots:  0 total
Time:       2.374 s, estimated 3 s
Ran all test suites matching src/utis/encryptionUtil.test.js.
```

```
PASS src/utis/apiResponse.test.js
White-Box Unit Testing - Module 9: Standardized API Response Helper Utility
  ✓ TC-W032: default_success_format - should format default success response correctly (617 ms)
  ✓ TC-W033: custom_payload_success - should format custom payload success response accurately (617 ms)
  ✓ TC-W034: default_error_format - should format default error response correctly (611 ms)
  ✓ TC-W035: custom_error_payload - should format custom error response with error payload (608 ms)

Test Suites: 1 passed, 1 total
Tests:      4 passed, 4 total
Snapshots:  0 total
Time:       2.934 s, estimated 3 s
Ran all test suites matching src/utis/apiResponse.test.js.
```

```
(use node --trace-warnings flag to show where the warning was created)
PASS src/utis/pagination.test.js
White-Box Unit Testing - Module 10: Database Query Pagination Utility
  ✓ TC-W036: default_pagination_params - should return default page 1, limit 10, skip 0 when query is empty (604 ms)
  ✓ TC-W037: skip_offset_calculation - should calculate skip accurately for page 3 with limit 15 (604 ms)
  ✓ TC-W038: negative_page_guard - should enforce minimum page 1 when page is zero or negative (616 ms)
  ✓ TC-W039: max_limit_cap_guard - should cap limit to 100 when query requests 500 items (613 ms)
  ✓ TC-W040: total_pages_calculation - should calculate total pages correctly in metadata (667 ms)

Test Suites: 1 passed, 1 total
Tests:      5 passed, 5 total
Snapshots:  0 total
Time:       3.585 s, estimated 4 s
Ran all test suites matching src/utis/pagination.test.js.
```

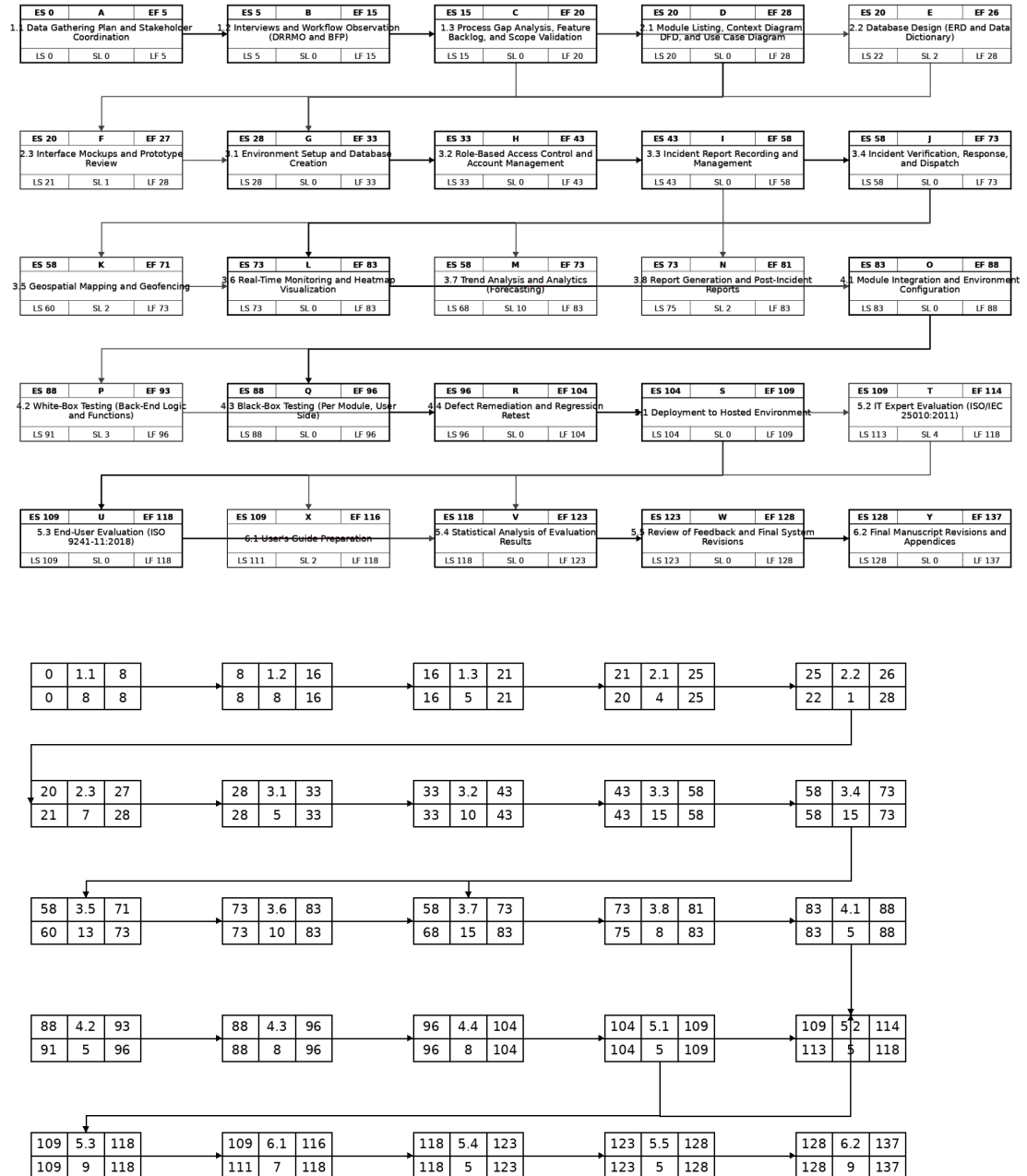


“Appendix I”

PERT – CPM

PERT-CPM

GAOIRS: Geospatial Approach to Optimize Incident Response System with Data Analytics Integration



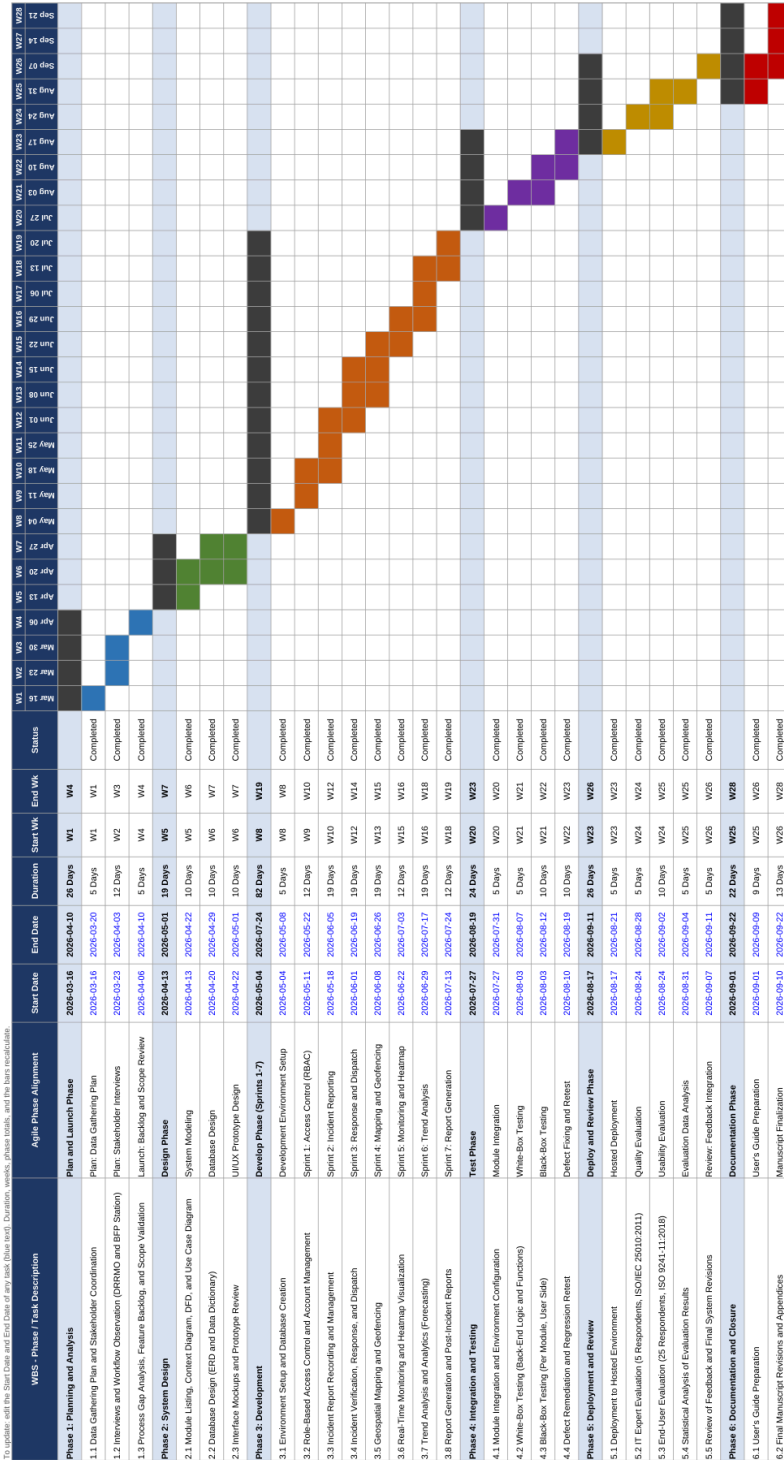


“Appendix J”

GANTT CHART

GAOIRS: GEOSPATIAL APPROACH TO OPTIMIZE INCIDENT RESPONSE SYSTEM WITH DATA ANALYTICS INTEGRATION

Discussion and the Start Date and End Date of each task (blue bar), Duration, Weeks, Sprints, and the task milestones.

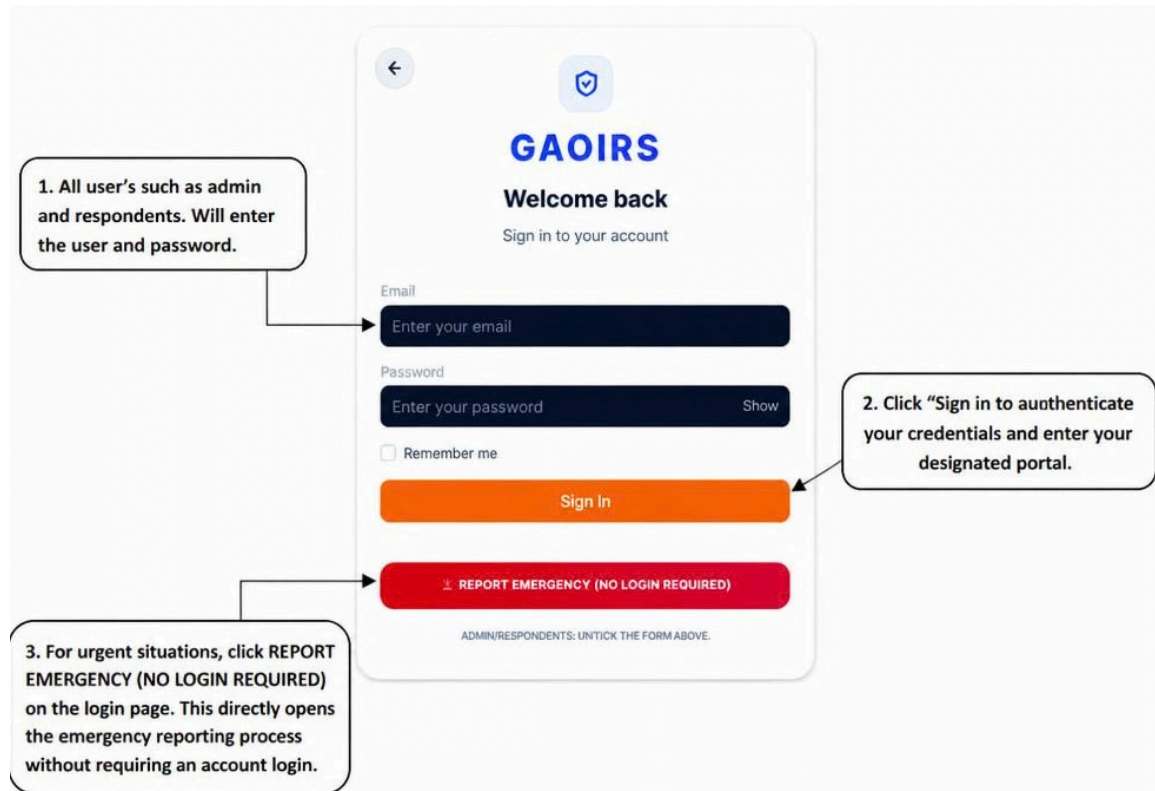


Note: Task dates are proposed placeholders to be replaced with the team's actual schedule. Phases below Section 3.9 (Planning and Analysis, System Design, Development, Testing) and the Agile stages in Figure 1. Blue text = editable input. Dark bar = phase span. Coloured bar = task span.

“Appendix K”

USER’S GUIDE

GAOIRS — Login Options for users



The image shows a mobile app login screen for GAOIRS. At the top, there is a back arrow, a shield icon, and the text "GAOIRS Welcome back Sign in to your account". Below this are two input fields: "Email" with the placeholder "Enter your email" and "Password" with the placeholder "Enter your password" and a "Show" button. There is a "Remember me" checkbox. Below the inputs are two buttons: an orange "Sign In" button and a red "REPORT EMERGENCY (NO LOGIN REQUIRED)" button. At the bottom, there is a small text: "ADMIN/RESPONDENTS: UNTICK THE FORM ABOVE." Three callout boxes provide instructions: 1. Points to the Email field. 2. Points to the Sign In button. 3. Points to the REPORT EMERGENCY button.

1. All user's such as admin and respondents. Will enter the user and password.

2. Click "Sign in to authenticate your credentials and enter your designated portal.

3. For urgent situations, click REPORT EMERGENCY (NO LOGIN REQUIRED) on the login page. This directly opens the emergency reporting process without requiring an account login.

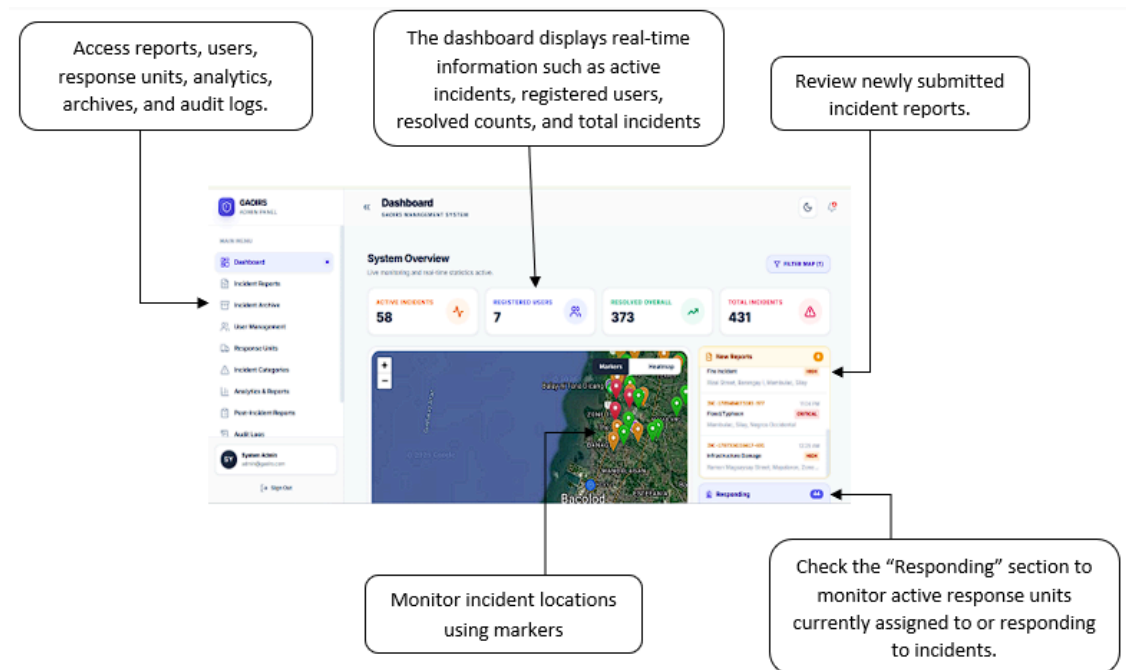
STEPS:

1. Open the <https://gaoirs.vercel.app> using your web browser (Google Chrome or Microsoft Edge).
2. On the GAOIRS Welcome / Login page, enter your registered email address in the Email field.
3. Enter your registered account password in the Password field. Click Show if you want to verify the password you entered.
4. Click Sign In to access your Admin & Respondents account.
5. After successful login, the system redirects you to the Admin Dashboard / Respondents Dashboard, where you can access Administrator Portal and the Respondents Portal.
6. For urgent situations, click REPORT EMERGENCY (NO LOGIN REQUIRED) to submit an emergency



report without signing in.

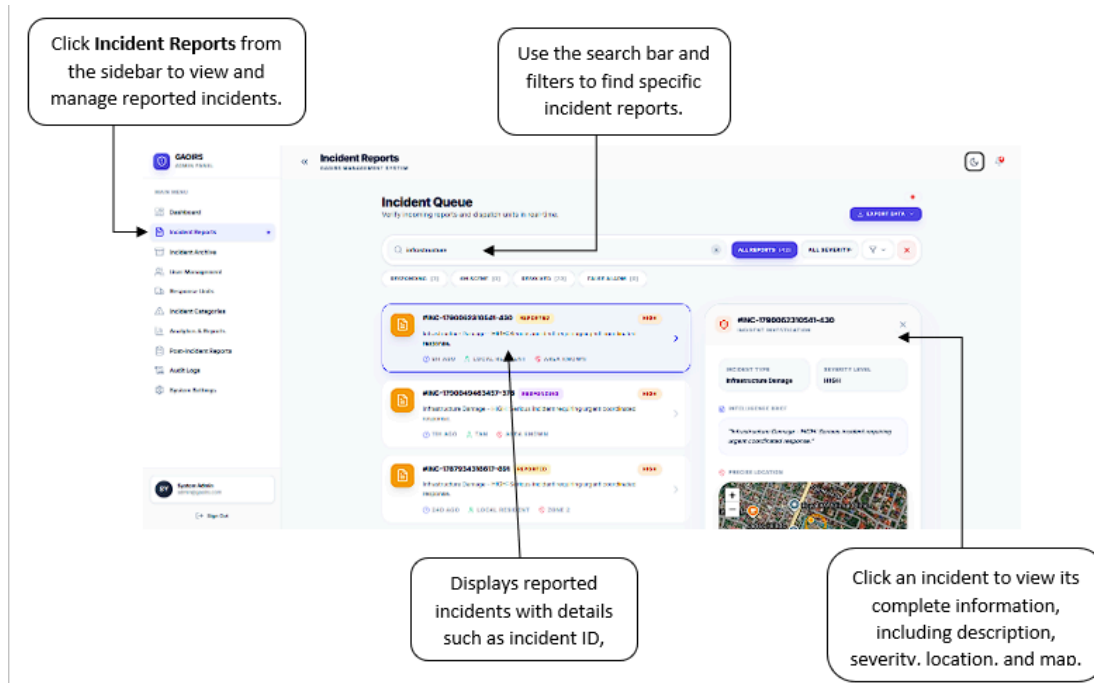
GAOIRS ADMIN — ADMIN Dashboard



STEPS: ADMIN DASHBOARD

1. Access the Administrator Dashboard
After successfully signing in with an Administrator account, the system automatically redirects you to the GAOIRS Administrator Dashboard.
2. View the System Overview
The dashboard displays real-time information such as:
 - Active Incidents – number of currently active incidents.
 - Registered Users – total number of registered system users.
 - Resolved Overall – number of incidents that have been resolved.
 - Total Incidents – total number of recorded incidents.
3. Monitor Incidents on the Map
Use the interactive map to view the locations of reported incidents. You may switch between Markers and Heatmap views and use the + / – buttons to zoom in or out.
4. Check New Reports
The New Reports panel displays recently submitted incidents. Review the incident type, location, priority level, and time reported.
5. Monitor Responding Units
Check the Responding section to monitor response units currently assigned to or responding to incidents.
6. Use the Main Menu
Use the sidebar to access different administrative functions:

GAOIRS ADMIN — INCIDENT REPORTS



The screenshot shows the 'Incident Reports' page in the GAOIRS system. The left sidebar contains a menu with 'Incident Reports' highlighted. The main area displays an 'Incident Queue' with a search bar and filters. A list of incidents is shown, each with a unique ID and status. One incident is expanded to show details like type, severity, description, and location on a map.

Click **Incident Reports** from the sidebar to view and manage reported incidents.

Use the search bar and filters to find specific incident reports.

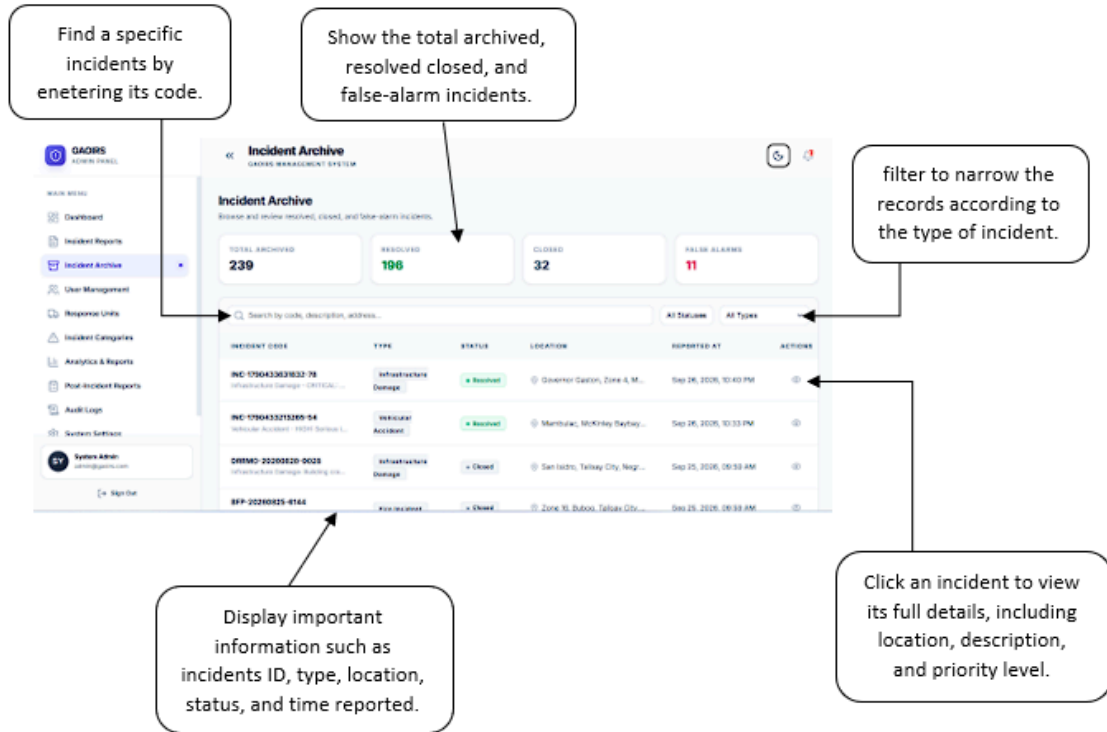
Displays reported incidents with details such as incident ID,

Click an incident to view its complete information, including description, severity, location, and map.

STEPS: INCIDENT REPORTS

1. From the Administrator Dashboard, click Incident Reports in the Main Menu on the left side of the screen.
2. The system will display the Incident Reports page, where incoming and recorded incident reports can be reviewed and managed.
3. Use the Search bar to find a specific incident by entering its incident ID, location, or relevant keyword.
4. Use the available filters to narrow down the incident reports according to their status, severity, or other available criteria.
5. Review the Incident List to see important information such as the incident ID, incident type, location, severity, status, and time reported.
6. Click an incident report from the list to open its detailed information.
7. Review the Incident Details panel, including the incident type, severity level, description, precise location, and map location when available.
8. Use the incident status and details to monitor and track the progress of reported incidents.

GAOIRS Admin — Incident Archive Guide



The screenshot shows the 'Incident Archive' page of the GAOIRS system. The interface includes a sidebar menu on the left with options like Dashboard, Incident Reports, Incident Archive (selected), User Management, Response Units, Incident Categories, Analytics & Reports, Most Recent Reports, Audit Logs, and System Settings. The main content area displays a summary of incident statistics: Total Archived (239), Resolved (196), Closed (32), and False Alarms (11). Below this is a search bar and filters for 'All Statuses' and 'All Types'. A table lists individual incidents with columns for Incident Code, Type, Status, Location, Reported At, and Actions. Callouts provide the following information:

- Find a specific incidents by entering its code.
- Show the total archived, resolved closed, and false-alarm incidents.
- filter to narrow the records according to the type of incident.
- Click an incident to view its full details, including location, description, and priority level.
- Display important information such as incidents ID, type, location, status, and time reported.

Steps: Incident Archive

1. From the Administrator Dashboard, locate the Main Menu on the left side of the screen.
2. Click Incident Archive to open the archived incident records.
3. Review the Archive Summary at the top of the page:
 - Total Archived – displays the total number of archived incidents.
 - Resolved – displays incidents marked as resolved.
 - Closed – displays incidents that have been closed.
 - False Alarms – displays incidents identified as false alarms.
4. Use the Search Bar to find a specific archived incident by entering its incident code, description, or address.
5. Use the All Statuses filter to display incidents according to their current archive status.
6. Use the All Types filter to narrow the records according to the type of incident.
7. Review the Incident Archive List, which displays important information such as:
 - Incident Code
 - Incident Type
 - Status
 - Location
 - Date and Time Reported
8. Click the View (eye) icon under Actions to open and review the complete details of a selected archived incident.



GAOIRS Admin — User Management Guide

The screenshot shows the 'User Management' page in the GAOIRS Admin system. It includes a sidebar menu, a top summary section, a search bar, a filter dropdown, and a table of users. Callouts provide instructions on how to use these features:

- Display all total number of users and response unit in the system.** (Points to the summary cards)
- Use the search bar to find users and the filter to view users by congressional district** (Points to the search bar and filter dropdown)
- Use this button to create a new user account.** (Points to the '+ Create User' button)
- Displays all with their details, including congressional district status, role, date joined, and actions** (Points to the user table)
- Use this icon to edit, activate/deactivate, or delete a user account.** (Points to the action icons in the table)

USER DETAILS	CONGRESSIONAL DISTRICT	STATUS	ROLE	DATE JOINED	ACTIONS
2 2nd District daniel@gaopra.com	2nd District	Active	ADMIN	9/27/2026	[Edit] [Deactivate] [Delete]
3 2nd District daniel@gaopra.com	2nd District	Active	ADMIN	9/27/2026	[Edit] [Deactivate] [Delete]
V Victorias Fire Station victoria@gaopra.com	2nd District	Active	RESPONSE UNIT	9/25/2026	[Edit] [Deactivate] [Delete]
V Victorias ORRMO victoria@gaopra.com	2nd District	Active	RESPONSE UNIT	9/25/2026	[Edit] [Deactivate] [Delete]

STEPS: User Management Guide

1. From the Administrator Dashboard, locate the Main Menu on the left side of the screen.
2. Click User Management to open the System Users page.
3. Review the User Summary to see the total number of registered users and response units.
4. Review the User List to view existing users, their Congressional District, status, system role, date joined, and available actions.
5. Use the Search Bar to find a specific user by name or other available information.
6. Use the Congressional District Filter to display users according to their assigned district.
7. To create a new account, click the + Create User button at the upper-right corner.
8. Enter the required user information, such as the Full Name and Email Address.
9. Select the user's Congressional District based on where the user is located.
10. Select the appropriate System Role, such as Response unit or System Administrator.
11. Review the entered information and click Save User to create the account.
12. For existing accounts, use the Action icons to edit, activate/deactivate, or delete a user account, depending on the administrator's permissions.



GAOIRS Admin — Response Units Guide

The screenshot shows the 'Response Units' management interface. It includes a sidebar menu with 'Response Units' highlighted. The main content area displays a summary of units (Total: 10, Available: 10) and a table of registered units. Callouts provide the following information:

- Shows the total number of registered unit and the currently available**: Points to the summary cards at the top.
- Find specific unit by entering its name, type, or contact number.**: Points to the search bar.
- Click this button to register new response unit.**: Points to the '+ Register Unit' button.
- Displays all registered units with their name, type, status, pin, location, contact number, and actions.**: Points to the unit list table.
- View location, Edit information, or delete the unit.**: Points to the action icons in the table.

UNIT IDENTIFIER	TYPE	STATUS	PINNED	CONTACT	ACTIONS
Talibay Fire Station	FIRE	Available	Pinned	-	[View] [Edit] [Delete]
Silay Fire Station	FIRE	Available	Pinned	0957400000	[View] [Edit] [Delete]
Silay DRRMO	DRRMO	Available	Pinned	0954043700	[View] [Edit] [Delete]
Victorias Fire Station	FIRE	Available	Pinned	-	[View] [Edit] [Delete]
Victorias DRRMO	DRRMO	Available	Pinned	-	[View] [Edit] [Delete]

STEP: Response Units

1. From the Administrator Dashboard, locate the Main Menu on the left side of the screen.
2. Click Response Units to open the Response Units management page.
3. Review the Unit Summary to view the total number of registered response units and the number currently available.
4. Review the Response Unit List to see each unit's name, type, status, pinned location, and contact number.
5. Use the Search Bar to find a specific response unit.
6. To register a new response unit, click the + Register Unit button at the upper-right corner.
7. Enter the Unit Name of the response team or station.
8. Enter the Contact Number of the response unit so it can be reached when necessary.
9. Select the appropriate Unit Type from the available options:
 - o Fire Engine
 - o Medical / EMS
 - o Barangay Responders
 - o DRRMO Team
10. Provide or configure the required location/pin information for the response unit, if applicable.
11. Review the entered information and save the new unit to add it to the Response Units list.



12. Use the Action icons beside an existing unit to view its location, edit its information, or delete the unit.
13. Monitor the Available status of response units to determine which units can be dispatched when responding to incidents.

GAOIRS Admin — Incident Categories Guide

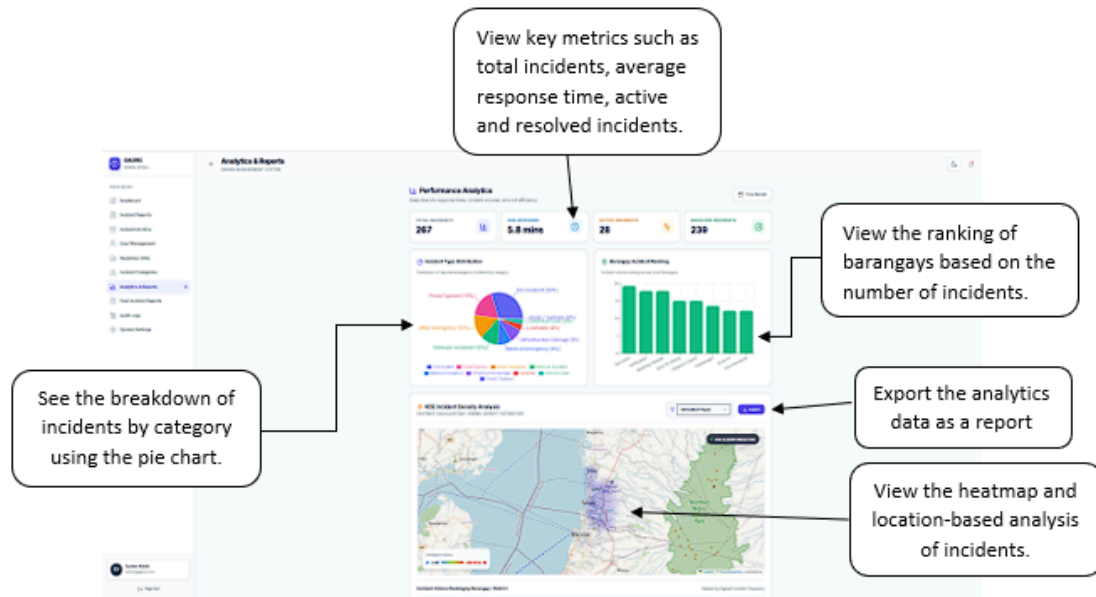
The screenshot shows the 'Incident Categories' management page in the GAOIRS Admin system. The page includes a sidebar menu on the left with options like Dashboard, Incident Reports, Incident Archive, User Management, Response Units, Incident Categories (highlighted), Analytics & Reports, Post-Incident Reports, Audit Logs, and System Settings. The main content area is titled 'Incident Categories' and contains a search bar, a table of existing categories, and a '+ New Category' button. The table has columns for Category Name, Default Unit, Color Theme, Description, and Actions. Callouts provide instructions: 'Find a specific Category by entering its name.' points to the search bar; 'Click this button to create new category' points to the '+ New Category' button; 'Displays all incident categories with their default unit, color theme, description, and actions.' points to the table; and 'Click the icon to update category or delete icon to remove it.' points to the edit and delete icons in the Actions column.

CATEGORY NAME	DEFAULT UNIT	COLOR THEME	DESCRIPTION	ACTIONS
Fire Incident	FIRE	#F08080	Structural, chemical, or grass fire emergency	[Edit] [Delete]
Flood / Typhoon	DRINK	#8000FF	Severe flooding, storm surge, or typhoon etc...	[Edit] [Delete]
Landslide	DRINK	#FFA500	Landslide or soil erosion hazard	[Edit] [Delete]
Medical Emergency	DRINK	#800080	Emergency medical response and trauma etc...	[Edit] [Delete]
Vehicular Accident	DRINK	#FF0000	Road crash or transportation accident	[Edit] [Delete]
Infrastructure Damage	DRINK	#800080	Bridge, road, power line, or utility failure	[Edit] [Delete]

STEPS: Incident Categories

1. From the Administrator Dashboard, locate the Main Menu on the left side of the screen.
2. Click Incident Categories to open the Incident Categories management page.
3. Review the list of existing incident categories, including their Category Name, Default Unit, Color Theme, and Description.
4. Use the Search Bar to find a specific incident category by entering its name.
5. Use the All Types filter to narrow down the displayed categories according to their type.
6. To create a new incident category, click the + New Category button at the upper-right corner.
7. Enter the required category information, including the Category Name, Default Unit, Color Theme, and Description.
8. Review the entered information and save the new category to add it to the incident category list.
9. To modify an existing category, click the Edit (pencil) icon under the Actions column.
10. To remove an existing category, click the Delete (trash) icon under the Actions column.
11. Review the updated category list to confirm that the changes have been applied.

GAOIRS Admin — Analytics & Reports Guide



STEPS: Analytics & Reports

1. From the Administrator Dashboard, locate the Main Menu on the left side of the screen.
2. Click Analytics & Reports to open the analytics dashboard.
3. Review the Performance Analytics summary, including:
 - Total Incidents
 - Average Response Time
 - Active Incidents
 - Resolved Incidents
4. Use the Time Filter at the upper-right corner to select the period of data you want to view.
5. Review the Incident Type Distribution chart to see the breakdown of reported incidents according to their category.
6. Review the Barangay Incident Ranking chart to identify the number of incidents recorded in each barangay.
7. Examine the KDE Incident Density Analysis map to visualize the geographic concentration and density of reported incidents.
8. Use the Incident Type Filter on the map to display density information for a specific incident type.
9. Click Export to generate or download the available analytics report/data.
10. Review the displayed analytics and reports for monitoring and planning purposes.



GAOIRS Admin — Post-Incident Reports Guide

The screenshot shows the 'Post-Incident Reports' page in the GAOIRS Admin system. The interface includes a sidebar menu on the left with options like Dashboard, Incident Reports, Incident Archive, User Management, Response Logs, Incident Categories, Analytics & Reports, Audit Logs, and System Settings. The main content area is titled 'Post-Incident Reports' and includes a subtitle 'Review and acknowledge reports submitted after incident resolution'. It features a 'TOTAL REPORTS' counter showing 5 reports and an 'Export' button. Below this is a 'Filters' section with a search bar for incident codes, a dropdown for incident types (currently set to 'All Types'), and date range pickers for start and end dates. The main part of the page is a table of submitted reports with columns for Incident Code, Type, Submitted By, Status, Date Submitted, and Action. The table lists five reports, all with a status of 'APPROVED'. Callouts provide instructions: 'Use the search bar to find specific reports you are finished' points to the search bar; 'Select a specific incident type from the dropdown' points to the incident type dropdown; 'Choose the date range for the submission period' points to the date range pickers; 'Click to export the available reports data' points to the 'Export' button; 'Shows the list of submitted reports with their details' points to the table; 'Shows the current status of each report' points to the 'Status' column; and 'Click review to inspect a post-incident report' points to the 'Review' link in the 'Action' column.

Use the search bar to find specific reports you are finished

Select a specific incident type from the dropdown

Choose the date range for the submission period

Click to export the available reports data

Shows the list of submitted reports with their details

Shows the current status of each report

Click review to inspect a post-incident report

STEPS: Post-Incident Reports

1. From the Administrator Dashboard, click Post-Incident Reports in the Main Menu.
2. Review the Total Reports displayed at the top-right of the page.
3. Use the Search Bar to find a specific post-incident report by entering its incident code.
4. Use the All Types filter to display reports according to the incident type.
5. Select the Start Date and End Date to filter reports according to the submission period.
6. Review the Post-Incident Report Table, which displays the:
 - Incident Code
 - Incident Type
 - Submitted By
 - Status
 - Date Submitted
 - Action
7. Check the Status of each report, such as Approved.
8. Click Review under the Action column to inspect the details of a post-incident report.
9. Click Export to download or export the available post-incident report data.



GAOIRS Admin — Audit Logs Guide

The screenshot shows the 'Audit Logs' page in the GAOIRS Admin system. The left sidebar contains a 'Main Menu' with options like Dashboard, Incident Reports, Incident Archive, User Management, Response Units, Incident Categories, Analytics & Reports, Prior Incident Reports, Audit Logs (highlighted), and System Settings. The main content area is titled 'Audit Logs' and 'System Audit Logs'. It features a search bar with the placeholder 'Search by user, action, resource...' and a filter dropdown set to 'All Actions'. Below these is a table of audit logs with columns: Timestamp, User, Action, Resource, and Details. The table lists several login attempts by users like 'Talway DRRMO', 'System Admin', and 'Victorlas DRRMO'. Callouts point to the search bar, the filter dropdown, and a row in the table.

Search by user, action, or resource

Filter logs by system action

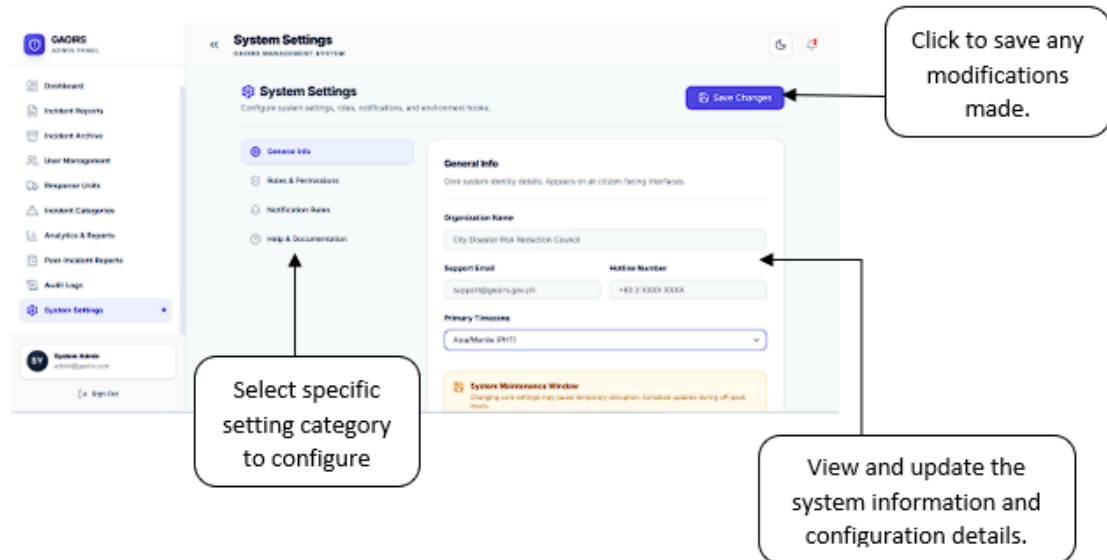
View timestamp, actions, resources and details of each activity.

STEPS: Audit Logs

1. From the Administrator Dashboard, locate the Main Menu on the left side of the screen.
2. Click Audit Logs to open the System Audit Logs page.
3. Review the Total Entries displayed at the upper-right corner to see the number of recorded system activities.
4. Use the Search Bar to find a specific audit record by entering a user, action, or resource.
5. Use the All Actions filter to narrow the audit records according to the type of system action performed.
6. Review the Audit Log Table, which displays:
 - Timestamp – date and time the activity occurred.
 - User – account that performed the action.
 - Action – activity performed, such as USER_LOGIN.
 - Resource – system area or resource affected.
 - Details – additional information about the activity.
7. Review the recorded activities to track system actions for accountability and compliance.



GAOIRS Admin — System Settings Guide

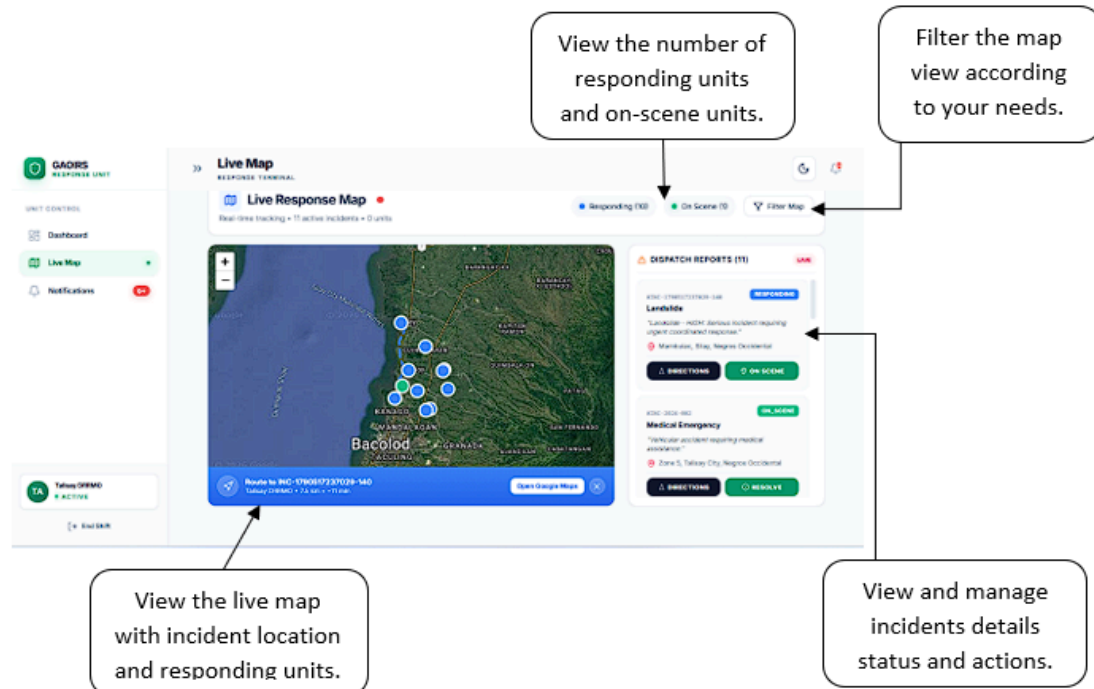


Steps: System Settings

1. From the Administrator Dashboard, locate the Main Menu and click System Settings.
2. The System Settings section will be displayed.
3. Use the Sub Menu to select a settings category, such as General Info, Roles & Permissions, Notification Rules, or Help & Documentation.
4. Navigate through the tabs to access the different configuration sections.
5. Make the necessary changes in the settings form, such as Organization Name, Support Email, Hotline Number, and Primary Timezone.
6. Click Save Changes to apply your modifications.
7. Review the updated information to ensure the changes were saved correctly.
8. When finished, click Sign Out from the Administrator Account.



GAOIRS Response Unit — Live Map Guide

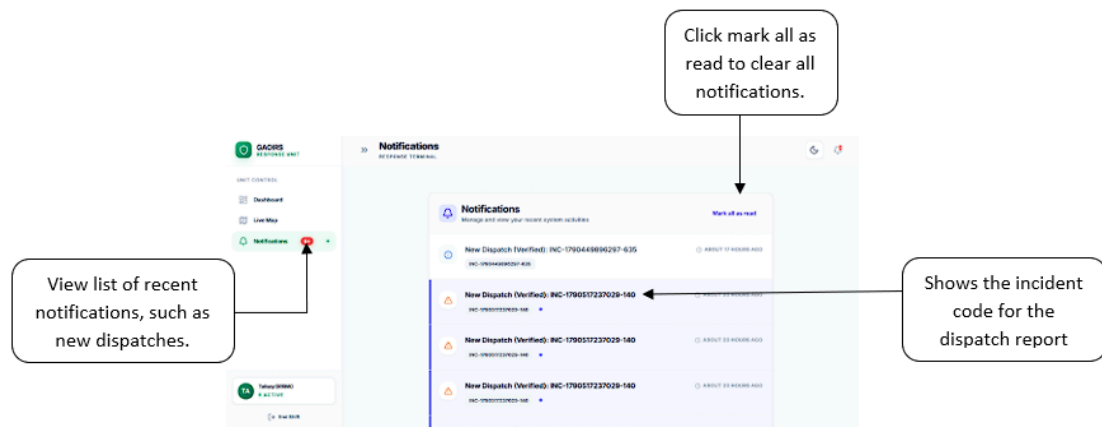


STEPS: Live Map

1. Log in to the GAOIRS Response Unit Portal using your registered response-unit account.
2. From the Unit Control menu, click Live Map to open the real-time response dashboard.
3. Review the Live Response Map to view the locations of active incidents and response units.
4. Check the status indicators at the top of the map, such as Responding and On Scene, to monitor current response activities.
5. Use the Filter Map button to filter the incidents or map information displayed.
6. Review the Dispatch Reports panel on the right side to see incoming incidents assigned to response units.
7. Select a dispatch report to view important information such as the Incident Code, Incident Type, Severity, Location, and incident description.
8. Click Directions to view the route to the reported incident location.
9. Use Open Google Maps when external navigation is needed to reach the incident location.
10. After arriving at the incident location, click On Scene to update the response status.
11. Once the incident has been handled, click Resolve to indicate that the response has been completed.
12. Monitor the Notifications section for new dispatch or incident updates.



GAOIRS Response Unit — Notifications Guide

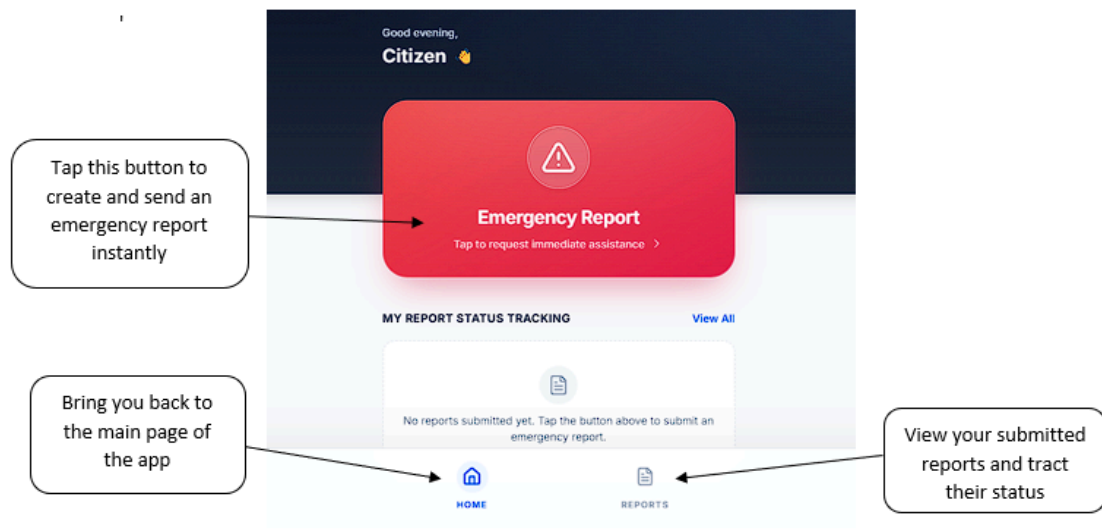


STEPS: Notifications

1. From the Response Unit Dashboard, locate the Unit Control menu on the left side.
2. Click Notifications to open the Notifications page.
3. Review the Notifications page to see recent system activities and dispatch updates.
4. Check each notification marked “New Dispatch (Verified)” to identify newly verified incidents.
5. Review the Incident Code shown in each notification to identify the corresponding dispatch report.
6. Check the timestamp to determine when the notification was received.
7. Look for the blue dot/unread indicator to identify notifications that have not yet been viewed.
8. Click Mark all as read after reviewing the notifications to clear the unread indicators.
9. Continue checking the Notifications page regularly for new dispatch assignments and system updates.
10. Review your Response Unit Account and Status at the bottom-left to confirm that your unit is active.
11. When your duty or response shift is finished, click End Shift at the bottom-left of the screen.



GAOIRS Citizen — Emergency Dashboard Guide

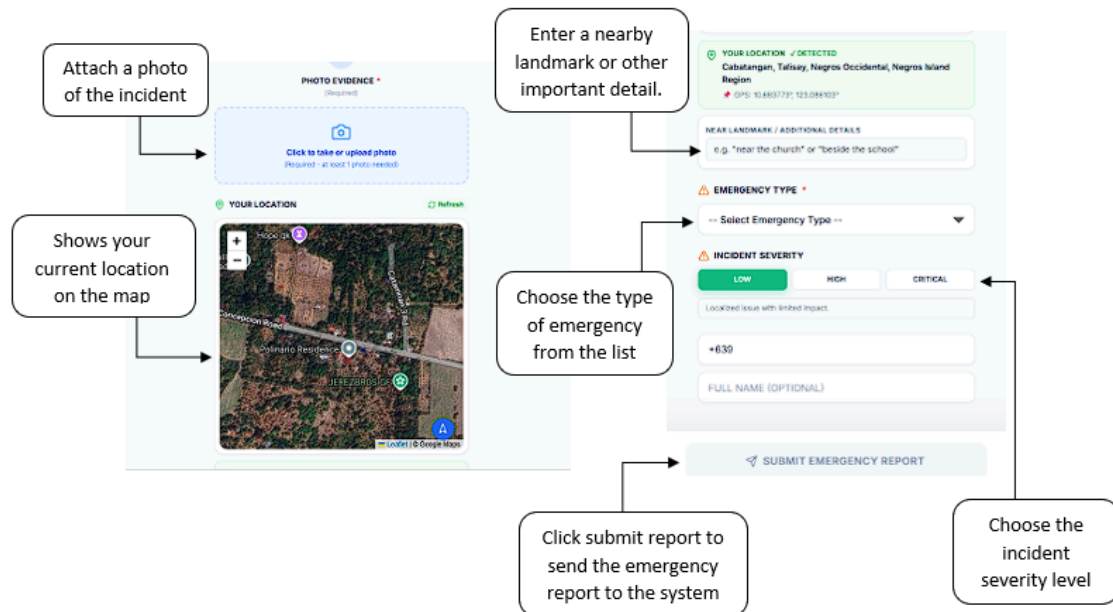


STEPS: Emergency Dashboard

1. Click the Emergency Report button to create and submit an emergency report and request immediate assistance.
2. Check the My Report Status Tracking section to view your submitted reports and monitor their current status.
3. Click View All beside the report status section to see the complete list of your submitted reports.
4. Click the Home icon at the bottom navigation bar to return to the main page.
5. Click the Reports icon to view your submitted reports and track their status.



GAOIRS Citizen — Emergency Report Guide



STEPS: Citizen — Emergency Report

1. From the GAOIRS Login Page, click REPORT EMERGENCY (NO LOGIN REQUIRED).
2. The Emergency Report form will open without requiring the citizen to log in.
3. Click Take or Upload Photo and attach a photo of the emergency. At least one photo is required.
4. The system will automatically detect your current location when the emergency report page is opened.
5. Check Your Location / Detected section to verify the automatically detected address and GPS coordinates.
6. If necessary, click Refresh to update the detected location.
7. Enter a Nearby Landmark / Additional Details, such as *"near the church"* or *"beside the school,"* to help responders locate the incident.
8. Select the appropriate Emergency Type from the dropdown list.
9. Select the Incident Severity:
 - Low
 - High
 - Critical
10. Review all the information you entered, especially the photo, automatically detected location, GPS coordinates, emergency type, and severity.
11. Click Submit Report to send the emergency report to GAOIRS.



12. The report will be received by the appropriate authorities/response units for verification and response.



“Appendix L”

CURRICULUM VITAE

Tristan Zane C. Lim

PROGRAMMER

☎ +639511490082

✉ tzlim.chmsu@gmail.com

📍 Zone 2 Barangay Mambulac, Silay City Negros Occidental



Passionate and detail-oriented Programmer with experience in building web applications, backend APIs, geospatial data integration, and database management. Dedicated to writing clean, maintainable code and solving complex technical challenges.

EDUCATION

- Carlos Hilado Memorial State University
Bachelor of Science in Information System (2023 - 2027)

SKILLS

- Programming Languages: JavaScript, Node.js, Python, HTML/CSS
- Frameworks & Libraries: React, React Native, Express, Vite, Tailwind CSS, Leaflet
- Tools: Git, Socket.io, Prisma, Cloudinary, Firebase, Postman
- Databases: PostgreSQL, Supabase, PostGIS

EXPERIENCE

- Analyzed requirements & designed geospatial analytics incident response workflow
- Integrated mapping data with analytics to track locations and optimize response
- Defined metrics, validated design, and prepared system specifications

CERTIFICATES

- Trust and Security with Google Cloud - Simplilearn

Issued Oct 2025

- A Practical Introduction to Cloud Computing - EC-Council

Issued Nov 2025

- Selenium Automation Testing for Beginners - EC-Council

Issued Nov 2025

- Cybersecurity for Businesses - The Fundamental Edition - EC-Council

Issued Nov 2025



Rubelyn Margaha

PROJECT MANAGER

CONTACT:

-  09056189675
-  rubelynmargaha4@gmail.com
-  Sagay City, Negros Occ.
-  Rubelyn Margaha
-  <https://www.linkedin.com/in/rubelyn-margaha-1b6800361>

SKILLS:

- Web Design
- Design Thinking
- Problem-Solving
- Strong Communication

LANGUAGE:

- English
- Filipino

PROFILE:

I am an average learner seeking for opportunities that will help me gain valuable experiences to learn from. I am someone that has the ability to work well with others and is cooperative.

ROLE STRENGTHS:

- Planned and scheduled project activities and monitored progress across all project phases.
- In charge of the capstone manuscript: managed its formatting, structure, and consistency across chapters.
- Revised and refined manuscript sections
- Organized sprint meetings and documented outcomes for each project phase.
- Coordinated tasks among team members to keep deliverables on schedule.

EDUCATION:

SAGAY NATIONAL HIGH SCHOOL **2021- 2023**

Science Technology Engineering and Mathematics

CARLOS HILADO MEMORIAL STATE UNIVERSITY **2023 - 2027**

Bachelor of Science in Information Systems



JUSTINE M. BIBANCO

QUALITY ASSURANCE

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 justinebibanco11@gmail.com

 Brgy. Cabatangan, Talisay City, Negros Occidental 6100

QUALITY ASSURANCE

About me

Detail-oriented and motivated Quality Assurance professional with a passion for ensuring software quality, identifying issues, and improving systems through effective testing and documentation.



EDUCATION

- **Carlos Hilado Memorial State University**
Bachelor of Science in Information System
2023 - 2027

SKILLS

- **Software Testing:**
Black-Box Testing, Alpha Testing, Functional Testing
- **QA Tool:**
Qase.io, Playwright
- **Documentation:**
Test Cases, Bug reports, Test results, User manuals

EXPERIENCES

- Analyzed requirements & designed geospatial analytics incident response workflow
- Integrated mapping data with analytics to track locations and optimize response
- Defined metrics, validated design, and prepared system specifications

CERTIFICATES

- **Introduction to CISSP Security Assessment & Testing and Security Operations**
April 10, 2025
- **Cyber Incident Response and Digital Forensics**
May 24, 2025